

POLYMER CHARACTERIZATION FACTUAL REPORT

Test Report for the National Transportation Safety Board
NTSB No. HWY06MH024

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1 COMPONENTS EXAMINED

Thirty-eight epoxy samples were received from the National Transportation Safety Board (NTSB) for polymer characterization. Of the samples received, 17 were removed from anchors collected at the failure area, 14 were removed from anchors not from the failure area, 5 were removed from anchors involved in 2006 testing by the Federal Highway Administration (FHWA), and 2 were cured samples of Powers Fast Set and Standard Set provided by NTSB for reference.

2 BACKGROUND

In January 2007, NTSB requested the characterization of epoxy samples from the adhesive anchoring system used to secure concrete ceiling panels that formed part of the ventilation ducting to the concrete tunnel roof. These samples, categorically labeled NTSB HWY06MH024, were delivered by a Massachusetts state trooper in plastic specimen bags. Samples typically consisted of epoxy pieces removed from anchor threads, small pieces of fractured epoxy, and epoxy crumbles.

Of the samples received, 17 were removed from failed anchors from the M1 row of module 081 of the eastbound tunnel. These samples are labeled 1-2, 1-3, 1-4, 2-1, 2-3, 3-1, 3-2, 3-3, 3-4, 4-2, 5-1, 5-2, 5-3, 6-1, 6-2, 6-3, and E1-1. Three of these samples (1-2, 1-4, and 3-3) are yellowish in appearance. Additionally, samples (8) of typical appearance removed from anchors not from the failure area were received. Also received were 4 samples of epoxy removed from anchors not from the failure area that appear brown and exhibited adhesive failure, as well as 2 samples which failed adhesively but otherwise appear typical. These samples are labeled based on the tunnel—eastbound (EB) or westbound (WB)—module number (e.g. module 081), anchor row (e.g. row M1), the hanger(s) associated with the anchor, and the position of the anchor in the fitting attached to the tunnel roof. Labels also include the location on the anchor (top – deepest into concrete, middle, bottom) from which the epoxy samples were taken.

Samples received further included one bag of cured Powers Fast Set epoxy shavings and one bag of Powers Standard Set epoxy shavings, as well as 5 epoxy samples taken from FHWA anchor testing. The Fast Set reference sample was from a 22-ounce package with Lot Number 062080I2LC and Use By date of 08/2008. The Standard Set reference sample was from a 22-ounce package with Lot Number 062280H1LC and Use By date of 08/2008. Both of the reference samples were cast on August 29, 2006 using the Powers manual injection tool and static mixing nozzle. Upon arrival at MIT, all sample bags were placed in a locked drawer and stored there while not in use.

3 FTIR CHARACTERIZATION

Fourier Transform Infrared Spectroscopy (FTIR) is a method that can be used to chemically characterize samples by using infrared radiation (IR). The absorption spectrum of a sample is indicative of the chemical composition of the irradiated sample.¹ The chemical composition of samples can be compared by observing similarities and differences in absorption peaks, particularly those in the so called "fingerprint region" (1900 cm^{-1} to 600 cm^{-1}). As per NTSB request, FTIR analysis was performed on epoxy samples in order to identify any significant anomalies, as well as to ascertain whether epoxy used in the tunnel was Powers Fast Set or Standard Set.

A Thermo Nicolet Nexus 870 FTIR with a DTGS TEC detector and KBr beam splitter was used in Attenuated Total Reflection (ATR) mode to scan from 5500 cm^{-1} to 525 cm^{-1} . A resolution of 4 wavenumbers was used, and 64 scans were performed and averaged per sample. An aperture of 90 and mirror velocity of 0.6329 cm/s was used. Crumbled samples were placed on the diamond window of the ATR accessory and pressure was applied to ensure flatness. Larger or non-flat epoxy pieces were first flattened between weigh paper with a hammer before being placed on the window. A background was captured before every run and was subtracted from raw data.

FTIR analysis shows that the IR spectra of the Powers Fast Set and Standard Set reference samples differ essentially in two peaks. The Fast Set reference displays a pronounced absorption peak at 755 cm^{-1} that is not present in the Standard Set reference. Additionally, an absorption peak at 735 cm^{-1} can be seen in the Standard Set reference but is not present in the Fast Set reference. These peaks are marked in Figure 3.1. All epoxy samples removed from the tunnel exhibit FTIR spectra that correspond closely to that from the Powers Fast Set including the absorption peak at 755 cm^{-1} . The epoxy samples removed from the tunnel do not exhibit a peak at 735 cm^{-1} . Thus, FTIR indicates that tunnel epoxy samples correspond to Powers Fast Set epoxy rather than Standard Set epoxy.

¹ Cheremisinoff, N.P. Polymer Characterization – Laboratory Techniques and Analysis. (1996) 64 – 73.

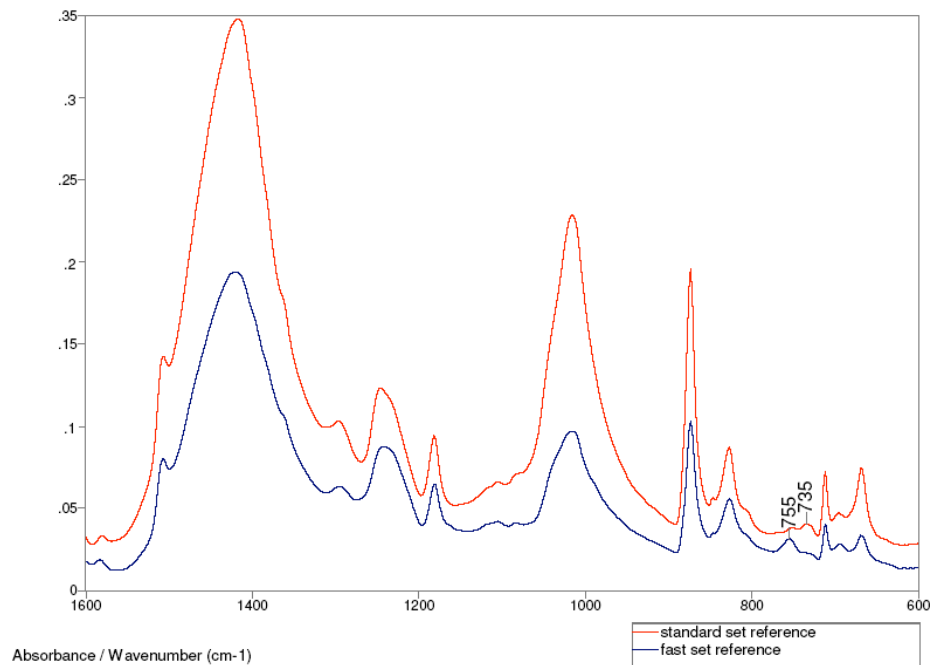


Figure 3.1 Portion of fingerprint region of Fast Set and Standard Set reference samples with relevant peaks marked at 735 and 755 wavenumbers.

All samples examined, including Fast Set and Standard Set references, exhibit an amine absorption peak at 3675 cm^{-1} . Furthermore, all tunnel epoxy samples, the Fast Set reference sample, and all FHWA samples (with the exception of FHWA unclean 1) display very similar absorption spectra. FTIR spectra of the 3 yellowed samples do not appear significantly different from other samples.

Some epoxy samples from both in and outside of the failure area, all the brown epoxy samples, some FHWA epoxy samples, and all the adhesive failure epoxy samples show a small absorption peak at 2850 cm^{-1} which is not present in the Fast Set reference sample. This peak is marked in Figure 3.2.

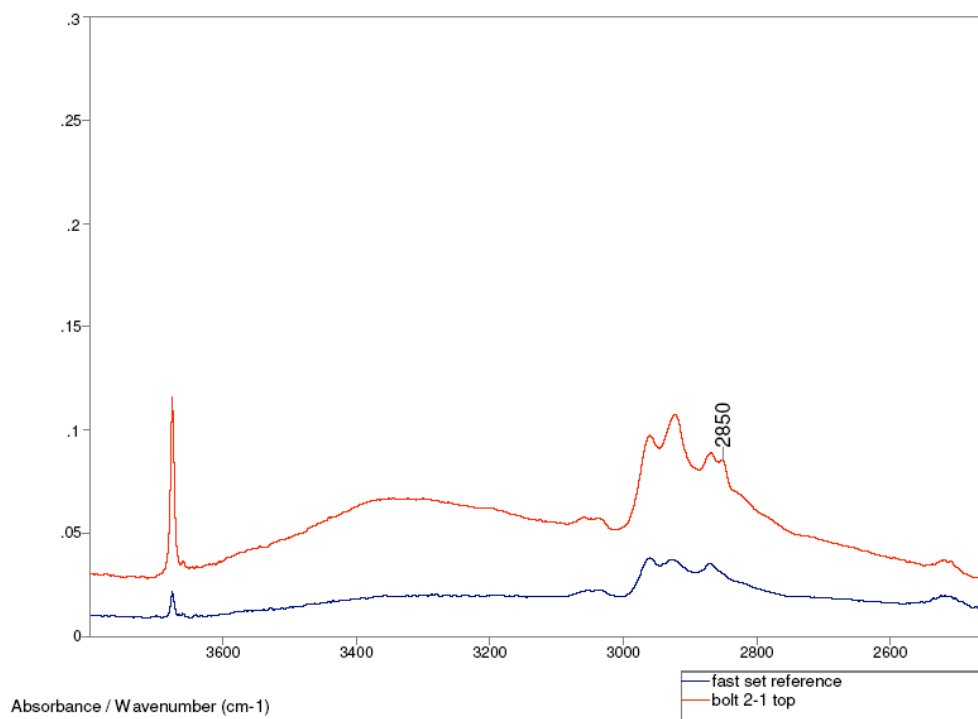


Figure 3.2 Example of weak peak at 2850 cm^{-1} that is not found in the Fast Set reference but is found in most samples from the failure area, one typical epoxy sample not from the failure area, all brown epoxy samples not from the failure area, all adhesive failure epoxy samples not from the failure area, and most FHWA epoxy samples.

The sample labeled FHWA unclean 1 (see Figure A36.1 and A36.2) was observed to have a brown-colored surface and exhibits a notably different peak configuration in the 3000 cm^{-1} to 2800 cm^{-1} region as well as attenuated peak intensity in the fingerprint region (1900 cm^{-1} to 600 cm^{-1}). This finding is consistent with the presence of a contamination layer on the surface.

Three of the 4 brown samples notably show a small, broad peak at 1647 cm^{-1} that is not seen in the Fast Set reference or most other epoxy samples. This peak is marked in Figure 3.3.

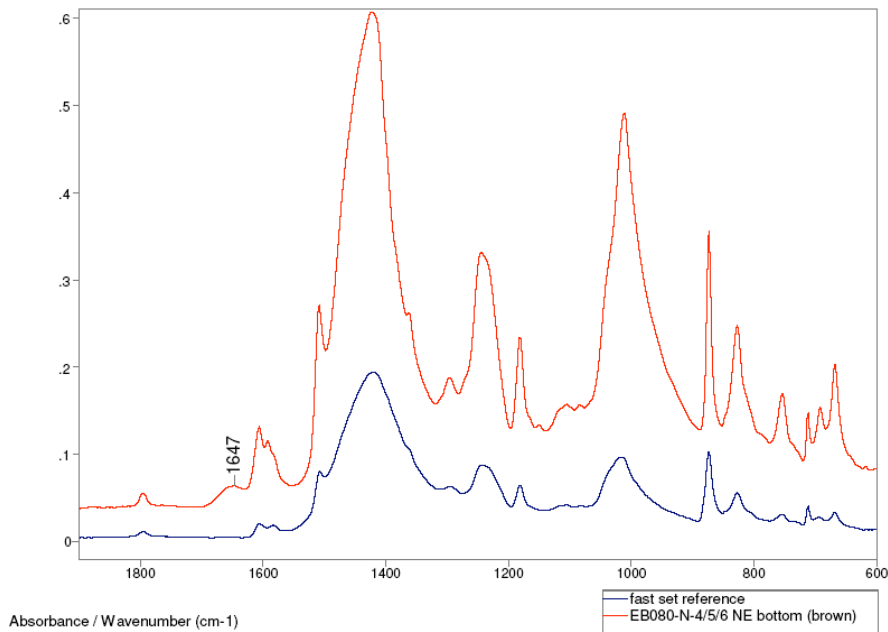


Figure 3.3 Example of weak, broad peak at 1647 cm^{-1} present in most brown samples but not present in the Fast Set reference or other epoxy samples. Spectra are vertically offset for clarity.

Most peaks in the fingerprint region have absorbance in the range of 0.1 – 0.3. One brown sample (EB080-N-4/5/6 NE bottom) had significantly higher absorbance than the other brown samples. However, precise quantitative analysis is not possible with these measurements, as peak intensities are highly dependent on sample thickness and flatness. Also, while classes of compounds, such as amines, can be identified in the FTIR spectra, it is not straightforward to draw more specific and certain conclusions about the chemical composition/bonding of samples from FTIR analysis. This is primarily due to the number of different amine components (reacted and unreacted) in the cured epoxy and the lack of reference spectra for individual components in their pure (unreacted) state and for the many possible types of pairs of components that have undergone reaction.

Comments on FTIR spectra from all the individual samples can be found in Section 5. Images of FTIR spectra from 4000 cm^{-1} to 2450 cm^{-1} and the fingerprint region (1900 cm^{-1} to 600 cm^{-1}) are included in Appendix A. Peaks between 2450 cm^{-1} and 1900 cm^{-1} are due to environmental influences and no peaks were seen above 4000 cm^{-1} . Measurements below 600 cm^{-1} are beyond the limits of the ATR crystal.

4 DSC CHARACTERIZATION

Differential Scanning Calorimetry (DSC) is an analytical method used to characterize thermal transitions within a material. DSC can be used gather information about transformations such as glass transition, melting and physical aging in a glassy polymer. Additionally, the DSC scan can indicate the occurrence of bond-forming reactions such as crosslinking, which are seen as exothermic (i.e. – heat flow out of the specimen) heat

flow.² As requested by NTSB, DSC analysis was performed on samples in order to document glass transition temperatures, evidence of further crosslinking upon exposure to high temperature, and to observe any anomalies.

Epoxy samples $5 \text{ mg} \pm 1 \text{ mg}$ in mass were loaded into standard aluminum DSC pans, which were then crimped. A TA Instruments DSC Q1000 was used. Samples were equilibrated at $-50 \text{ }^{\circ}\text{C}$ for 3 minutes, ramped at $20 \text{ }^{\circ}\text{C}/\text{min}$ to $150 \text{ }^{\circ}\text{C}$, held isothermally for 5 minutes at $150 \text{ }^{\circ}\text{C}$, ramped at $-20 \text{ }^{\circ}\text{C}/\text{min}$ to $-50 \text{ }^{\circ}\text{C}$, held isothermally for 3 minutes at $-50 \text{ }^{\circ}\text{C}$, and finally ramped at $20 \text{ }^{\circ}\text{C}/\text{min}$ to $150 \text{ }^{\circ}\text{C}$ again. An empty pan background was run for at least every 3 sample runs and was subtracted from the raw data.

For epoxy samples removed from the tunnel, with the exception of most of the brown epoxy samples and sample anchor 5-1, measurable glass transition temperatures (T_g) on the first heating ramp to $150 \text{ }^{\circ}\text{C}$ range from $42 \text{ }^{\circ}\text{C}$ to $52 \text{ }^{\circ}\text{C}$ (see Table 4.1). Sample Anchor 5-1 shows a T_g of $39 \text{ }^{\circ}\text{C}$ on the first ramp. Two of the brown epoxy samples (EB083-S-1-S top and EB080-N-4/5/6-NE bottom) exhibit a T_g of about $11 \text{ }^{\circ}\text{C}$ on the first ramp, while a third brown sample (EB081-S-8-S top) does not exhibit a recognizable T_g . The T_g of the Fast Set reference, two of the typical epoxy samples not from the failure area, and most of the FHWA samples could not be precisely determined on the first heating scan due to interference from an adjacent endothermic peak.

Endothermic peaks are seen in most of the epoxy samples from the failure area, all of the typical epoxy samples not from the failure area, all of the FHWA epoxy samples, one of the adhesive failure samples not from the failure area, and both Fast Set and Standard Set references. These peaks typically occur either at approximately $55 \text{ }^{\circ}\text{C}$ and/or $70 \text{ }^{\circ}\text{C}$ for samples removed from the tunnel and at approximately $55 \text{ }^{\circ}\text{C}$ for FHWA samples (see Table 4.1).

In addition to an endothermic peak, samples 3-1, 6-3, and two of the brown epoxy samples exhibit a shallow but notable exothermic peak at approximately $135 \text{ }^{\circ}\text{C}$ on the first temperature ramp. FHWA samples also display a small, broad exothermic peak centered at approximately $115 \text{ }^{\circ}\text{C}$ on the first temperature ramp. It should be noted that on the first temperature ramp, almost all samples examined display at least some exothermic heat flow either in the form of a notable exothermic peak or gradual exothermic curvature in the heat flow curve between $75 \text{ }^{\circ}\text{C}$ and $150 \text{ }^{\circ}\text{C}$.

No endothermic peaks are seen on the second ramp to $150 \text{ }^{\circ}\text{C}$ for any of the samples examined. With the exception of EB081-S-8-S-top (a brown sample which did not show a T_g on the first ramp), all samples examined show a readily recognizable T_g on the second ramp. Most samples from the failure area and the typical samples not from the failure area have a second heating run T_g in the range of $49 \text{ }^{\circ}\text{C}$ to $58 \text{ }^{\circ}\text{C}$. The average T_g increase is $6.1 \text{ }^{\circ}\text{C}$ for samples from the failure area and $3.2 \text{ }^{\circ}\text{C}$ for typical samples not from the failure area. Unlike these samples, the brown epoxy samples do not show an increase in T_g on the second ramp. The FHWA samples which did not show a clearly measurable T_g on the first

² Sichina, W.J. Perkin Elmer Instruments Applications Note: Measurement of T_g by DSC.
http://las.perkinelmer.com/content/ApplicationNotes/APP_ThermalTgByDSC.pdf

ramp due to the presence of an adjacent endothermic peak show a distinct T_g in the range of 36 °C to 41 °C on the second heating ramp.

Table 4.1 lists first and second heating ramp T_g values, T_g change, and normalized size of any endothermic peaks present. Analysis of DSC results for all individual epoxy samples can be found in Section 5. Appendix B includes plots of the DSC data.

Failure area samples	1st ramp T_g (°C)	2nd ramp T_g (°C)	T_g change (°C)	Endo peak at 55°C (J/g)	Endo peak at 70°C (J/g)
1-2 middle	46	55	9	–	0.67
1-3 top	45	54	9	–	1.52
1-4 middle	48	54	6	–	1.58
2-1 top	46	52	6	0.11	0.22
2-3 top	47	56	9	–	0.37
3-1 bottom	45	49	4	0.7	–
3-2 top	48	54	6	0.4	0.14
3-3 middle	47	54	7	–	1.29
3-4 bottom	48	51	3	0.63	–
4-2 middle	45	54	9	–	1.31
5-1 middle	39	42	3	–	–
5-2 bottom	44	46	2	–	–
5-3 bottom	44	49	5	1	–
6-1 top	52	56	4	0.05	0.1
6-2 middle	43	52	9	–	2.67
6-3 bottom	45	51	6	0.54	–
E1-1 top	52	59	7	0.18	–
Average	46.12	52.24	6.12	0.45	0.99
Typical epoxy samples not from failure site	1st ramp T_g (°C)	2nd ramp T_g (°C)	T_g change (°C)	peak at 55°C (J/g)	peak at 70°C (J/g)
EB080-M1-3/4/5-SW middle	46	46	0	0.46	–
EB080-N-4/5/6-NW bottom	47	52	5	0.74	–
WB084-N-4/5/6-NW bottom	47	50	3	–	0.22
WB082-S-2-S middle	*	54	N/A	–	1.76
EB081-M2-2/3/4-SE top	51	56	5	–	1.26
EB082-M1-8/9/10-SW top	51	56	5	–	0.91

– No endothermic peak at this temperature

* T_g could not be determined due to interference by endothermic peak or lack of distinct glass transition

EB081-S-9-S bottom	49	50	1	0.19	0.13
EB083-N-3-S middle	*	56	N/A	–	2.56
Average	48.50	52.50	3.17	0.46	1.14
Brown epoxy samples	1st ramp T_g (°C)	2nd ramp T_g (°C)	T_g change (°C)	peak at 55°C (J/g)	peak at 70°C (J/g)
EB081-S-8-S top	*	*	N/A	–	–
EB083-S-1-S top	11	11	0	–	–
EB081-S-10-S top	43	44	1	0.44	–
EB080-N-4/5/6-NE-bottom	9	12	3	–	–
average	21	22.33	1.33	0.44	N/A
Adhesive failure samples	1st ramp T_g (°C)	2nd ramp T_g (°C)	T_g change (°C)	peak at 55°C (J/g)	peak at 70°C (J/g)
EB083-M2-11-N bottom	45	47	2	–	–
EB083-M2-7/8/9-SW bottom	45	49	4	–	0.5
average	45	48	3	N/A	0.5
FHWA samples	1st ramp T_g (°C)	2nd ramp T_g (°C)	T_g change (°C)	peak at 55°C (J/g)	peak at 70°C (J/g)
Best practice 6 inside cap	50	51	1	0.62	–
bubble w/o twist 9 middle	*	41	N/A	1.5	–
bubble w/o twist 1 top	*	39	N/A	1.54	–
bubble w/twist 4bottom	*	38	N/A	0.61	–
unclean 1 below cap	*	36	N/A	1.44	–
average	50	41	1	1.14	N/A
Reference samples	1st ramp T_g (°C)	2nd ramp T_g (°C)	T_g change (°C)	peak at 55°C (J/g)	peak at 70°C (J/g)
Fast Set 50/50	*	48	N/A	–	0.89
Standard Set 50/50	*	56	N/A	–	0.68

Table 4.1 Summary of first and second ramp T_g, T_g change, and magnitude of endothermic peaks.

5 EPOXY SAMPLE DETAIL

This section addresses the analysis of DSC and FTIR results for each sample examined. Condition and visual appearance of samples are also discussed. Any significant anomalies are noted.

Samples from the Failure Area

Anchor 1-2 middle: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. Compared to other samples received, the epoxy appears to have a yellow tint that is especially noticeable in the rough, fractured surface. DSC results show a first (scan) T_g at 46 °C, a second (scan) T_g at 55 °C, and an endothermic peak at 73 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Anchor 1-3 top: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 45 °C, a second T_g at 54 °C, and an endothermic peak at 67 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Anchor 1-4 middle: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. Compared to other samples received, the epoxy appears to have a yellow tint that is especially noticeable in the rough, fractured surface. DSC results show a first T_g at 48 °C, a second T_g at 54 °C, and an endothermic peak at 68 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Anchor 2-1 top: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 46 °C, a second T_g at 52 °C, and small endothermic peaks at 56 °C and 70 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Anchor 2-3 top: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 47 °C, a second T_g at 56 °C, and a small endothermic peak 70 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Anchor 3-1 bottom (cap area): The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth, as does the material on the opposite side (towards the cap) though with some minor evidence of scratching. DSC results show a first T_g at 45 °C, a second T_g at 49 °C, and an endothermic peak 54 °C. A small, broad exothermic peak is also seen at 135 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Anchor 3-2 top: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 48 °C, a second T_g at 54 °C, and small endothermic peaks at 56 °C and 70 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Anchor 3-3 middle: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. Compared to other samples received, the epoxy appears to have a yellow tint that is especially noticeable in the rough, fractured surface. DSC results show a first T_g at 47 °C, a second T_g at 54 °C, and an endothermic peak at 70 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Anchor 3-4 bottom: The sample received consists primarily of epoxy pieces removed from anchor threads as well as epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth, while for the small epoxy pieces the material on the opposite side (towards the concrete) appears rough and fractured. For the larger epoxy chunks, the material on the concrete side appears smooth. DSC results show a first T_g at 48 °C, a second T_g at 51 °C, and an endothermic peak at 56 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Anchor 4-2 middle: The sample received consists primarily of epoxy pieces removed from anchor threads as well as one epoxy chunk removed from anchor threads. For both the pieces and the chunk, material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 45 °C, a second T_g at 54 °C, and an endothermic peak at 69 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Anchor 5-1 middle: The sample received consists primarily of small epoxy bits removed from anchor threads. The material appears smooth on all sides and very few rough, fractured surfaces are seen. DSC results show a first T_g at 39 °C and a second T_g at 42 °C, which is lower than other samples from the failure area. No endothermic peak is seen. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Anchor 5-2 bottom (cap area): The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth, as does the

material on the opposite side (towards the cap) though with some minor evidence of scratching. DSC results show a first T_g at 44 °C and a second T_g at 46 °C, which is slightly lower than most other samples from the failure area. No endothermic peak is seen. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Anchor 5-3 bottom: The sample received consists primarily of epoxy pieces removed from anchor threads as well as epoxy chunks removed from anchor threads. For both pieces and chunks, the material appears smooth on all sides and very few rough, fractured surfaces are seen. DSC results show a first T_g at 44 °C, a second T_g at 49 °C, and an endothermic peak at 57 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Anchor 6-1 top: The sample received consists primarily of epoxy pieces removed from anchor threads as well as fractured bits of epoxy. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 52 °C, a second T_g at 56 °C, and endothermic peaks at 58 °C and 72 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Anchor 6-2 middle: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 43 °C, a second T_g at 52 °C, and an endothermic peak at 69 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Anchor 6-3 bottom (cap area): The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth, as does the material on the opposite side (towards the cap) though with some minor evidence of scratching. DSC results show a first T_g at 45 °C, a second T_g at 51 °C, and an endothermic peak at 55 °C. A small, broad exothermic peak is also seen at 132 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Anchor E1-1 top: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 52 °C, a second T_g at 59 °C, and an endothermic peak at 60 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

Typical Samples Not from the Failure Area

EB080-m1-3/4/5-SW middle: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 46 °C, a second T_g at 46 °C, and a small endothermic peak at 59 °C. Unlike other samples examined, there was no change in T_g on the second ramp. The FTIR spectrum is consistent with that of the Fast Set reference.

EB080-N-4/5/6-NW bottom: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 47 °C, a second T_g at 52 °C, and an endothermic peak at 56 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

WB084-N-4/5/6-NW bottom: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 47 °C, a second T_g at 50 °C, and an endothermic peak at 69 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

WB082-S-2-S middle: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a second ramp T_g at 54 °C and an endothermic peak at 68 °C. No T_g could be measured on the first ramp due to interference from the endothermic peak located at 68 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

EB081-M2-2/3/4-SE top: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 51 °C, a second T_g at 56 °C, and an endothermic peak at 70 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

EB082-M1-8/9/10-SW top: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 51 °C, a second T_g at 56 °C, and an endothermic peak at 69 °C. The FTIR spectrum is consistent with that of the Fast Set reference.

EB081-S-9-S bottom: The sample received consists primarily of epoxy pieces removed from anchor threads as well as bits of fractured epoxy. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 49 °C, a second T_g at 50 °C, and small endothermic peaks at 56 °C and 69 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

EB083-N-3-S middle: The sample received consists primarily of epoxy pieces removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a second ramp T_g at 56 °C and an endothermic peak at 69 °C. T_g on the first ramp could not be measured due to interference from the endothermic peak at 69 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Brown Epoxy Samples not from the Failure Area

EB081-S-8-S top: The sample received consists primarily of bits of fractured epoxy. Some are normal while others have a distinct brown appearance. Most of the brown pieces are soft and pliable, but have hardened to the touch somewhat since arrival at the end of January. DSC results do not show a recognizable T_g on the first ramp or second ramp and no endothermic peaks are seen. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

EB083-S-1-S top: The sample received consists primarily of bits of fractured epoxy. Some appear normal while others have a distinct brown appearance. Some of the brown pieces are soft and pliable and have somewhat hardened since arrival, while a few of the darker brown pieces were brittle at the time of arrival. DSC results show a first T_g at $11\text{ }^{\circ}\text{C}$ and a second T_g at $11\text{ }^{\circ}\text{C}$, both of which are very low compared to other samples examined. No endothermic peaks are seen, but an exothermic peak is seen at $139\text{ }^{\circ}\text{C}$ on the first ramp. The FTIR spectrum displays a small peak at 2850 cm^{-1} and a small, broad peak at 1647 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

EB081-S-10-S top: The sample received consists primarily of epoxy chunks removed from anchor threads as well as smaller bits of epoxy. The smaller pieces are brown in appearance and are soft and pliable, though they have hardened somewhat since arrival at MIT. For some of the larger chunks brown epoxy is seen on one side but not on the other, while for other large chunks brown epoxy can be seen on both sides (against anchor and towards concrete). Also included is a large chunk of dark brown epoxy which was brittle at time of arrival. DSC results show a first T_g at $43\text{ }^{\circ}\text{C}$ and a second T_g at $44\text{ }^{\circ}\text{C}$. An endothermic peak is seen at $59\text{ }^{\circ}\text{C}$. The FTIR spectrum displays a small peak at 2850 cm^{-1} and a small, broad peak at 1647 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

EB080-N-4/5/6-NE bottom: The sample received consists primarily of epoxy chunks removed from anchor threads as well as smaller bits of epoxy removed from anchor threads. Some of the small pieces are brown in appearance and are soft and pliable. For the large epoxy chunks, the material that had been against the anchor appears smooth and typical while material on the opposite side (towards the concrete) appears brownish and uneven. DSC results show a first T_g at $9\text{ }^{\circ}\text{C}$ and a second T_g at $12\text{ }^{\circ}\text{C}$, both of which are very low compared to other samples examined. No endothermic peaks are seen, but an exothermic peak is seen at $137\text{ }^{\circ}\text{C}$ on the first ramp. The FTIR spectrum displays a small peak at 2850 cm^{-1} and a small, broad peak at 1647 cm^{-1} but is otherwise consistent with that of the Fast Set reference. Peak intensities are high compared to other samples.

Adhesive Failure Samples not from the Failure Area

EB083-M2-11-N bottom: The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at $45\text{ }^{\circ}\text{C}$ and a second T_g at $47\text{ }^{\circ}\text{C}$. No endothermic peak could be seen. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

EB083-M2-7/8/9-SW bottom: The sample received consists of epoxy chunks removed from anchor threads as well as some pieces of fractured epoxy. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a first T_g at 45 °C, a second T_g at 49 °C, and an endothermic peak at 69 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

FHWA Samples (cured in 2006)

Best practice 6 inside cap: The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth, as does material on the other side (towards the cap). DSC results show a first T_g at 50 °C, a second T_g at 51 °C, and an endothermic peak at 60 °C. Additionally, a broad exothermic peak at 115 °C is seen on the first ramp. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Bubble w/o twist 9 middle: The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a second ramp T_g at 41 °C and an endothermic peak at 58 °C. T_g on the first ramp could not be determined due to interference from the endothermic peak at 58 °C. Additionally, a broad exothermic peak is seen at 117 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Bubble w/o twist 1 top: The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a second ramp T_g at 39 °C and an endothermic peak at 58 °C. T_g on the first ramp could not be determined due to interference from the endothermic peak at 58 °C. Additionally, a broad exothermic peak is seen at 123 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Bubble w/ twist 4 bottom: The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side (towards the concrete) appears rough and fractured. DSC results show a second ramp T_g at 38 °C and an endothermic peak at 57 °C. T_g on the first ramp could not be determined due to the endothermic peak at 57 °C. Additionally, a broad exothermic peak is seen at 117 °C. The FTIR spectrum displays a small peak at 2850 cm^{-1} but is otherwise consistent with that of the Fast Set reference.

Unclean 1 below cap: The sample received consists of epoxy chunks removed from anchor threads. Material that had been against the anchor appears smooth while material on the opposite side appears dark brown and flat. DSC results show a second ramp T_g at 36 °C and an endothermic peak at 58 °C. T_g on the first ramp could not be determined due to interference from the endothermic peak at 58 °C. Additionally, a broad exothermic peak is seen at 114 °C. The FTIR spectrum is mostly similar to that of the Fast Set reference, but the signal in the fingerprint region is greatly attenuated. The peak configuration between 3000 cm^{-1} and 2800 cm^{-1} is also slightly different.

Reference Samples

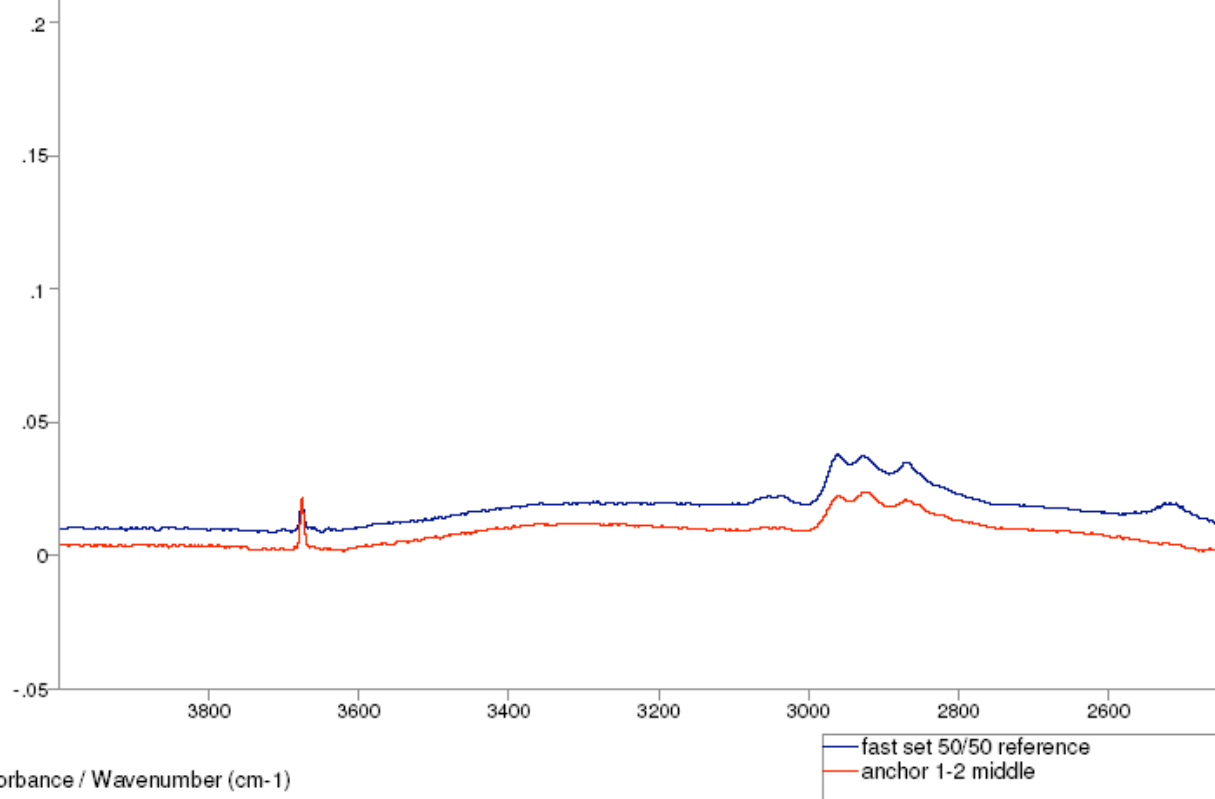
Fast Set 50/50: The sample received primarily consists of shavings from a cured epoxy sample. A large cylindrical chunk of epoxy was also included. DSC results show a second ramp T_g at 48 °C, and an endothermic peak at 68 °C. T_g on the first ramp could not be determined due to interference from the endothermic peak at 68 °C.

Standard Set 50/50: The sample received primarily consists of shavings from a cured epoxy sample. A large chunk of epoxy was also included. DSC results show a second ramp T_g at 56 °C, and an endothermic peak at 65 °C. T_g on the first ramp could not be determined due to interference from the endothermic peak at 65 °C. A large endothermic peak at 150 °C is also seen on the first ramp. The FTIR spectrum is similar to that of the Fast Set reference except for the presence of a peak at 735 cm^{-1} and the lack of a peak at 755 cm^{-1} .

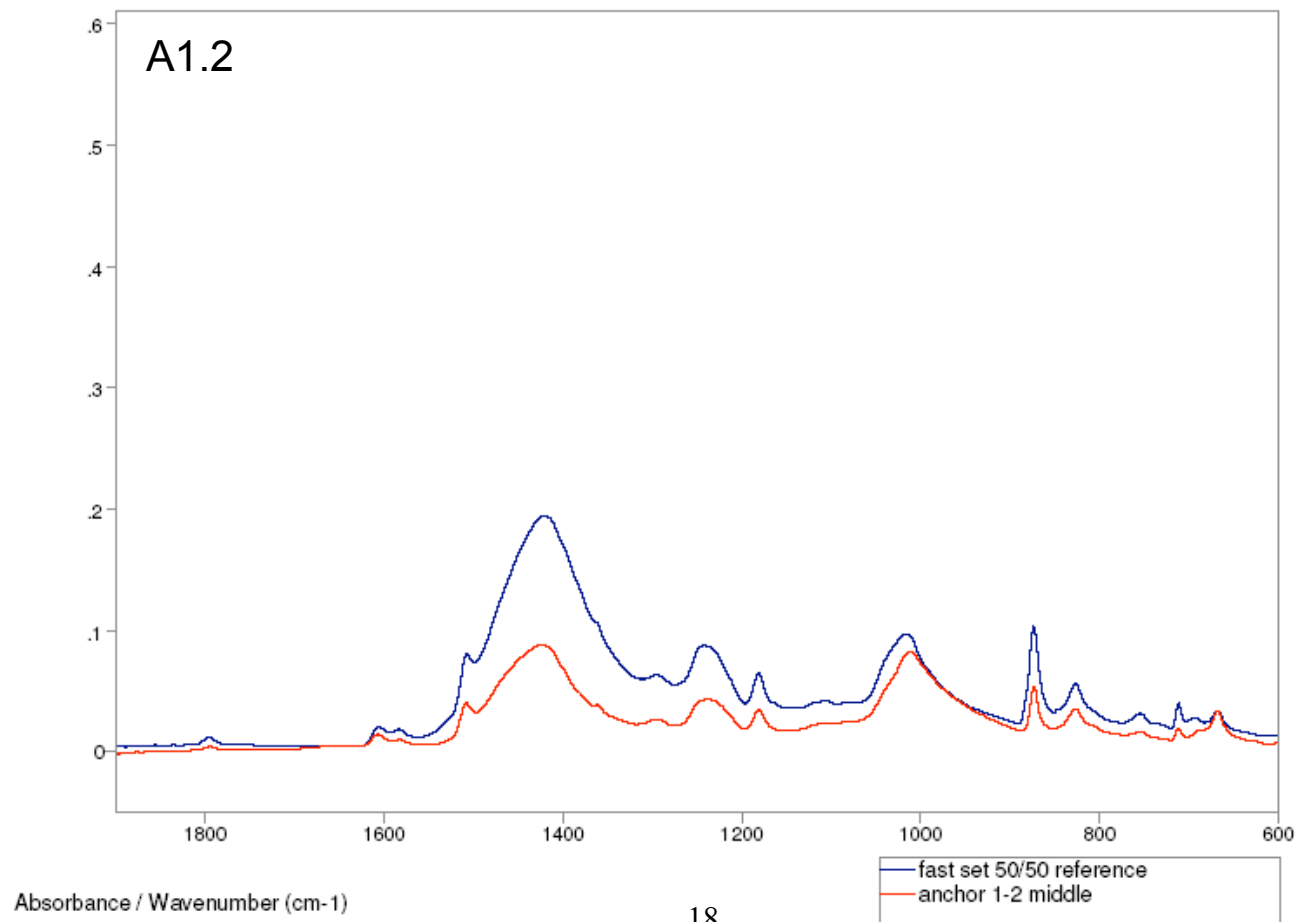
APPENDIX A: FTIR SPECTRA

FTIR spectra of all samples examined are presented in this section. For each sample (red), a Fast Set reference spectrum is presented (navy). Wavelength ranges presented are 4000 cm^{-1} to 2450 cm^{-1} and 1900 cm^{-1} to 600 cm^{-1} .

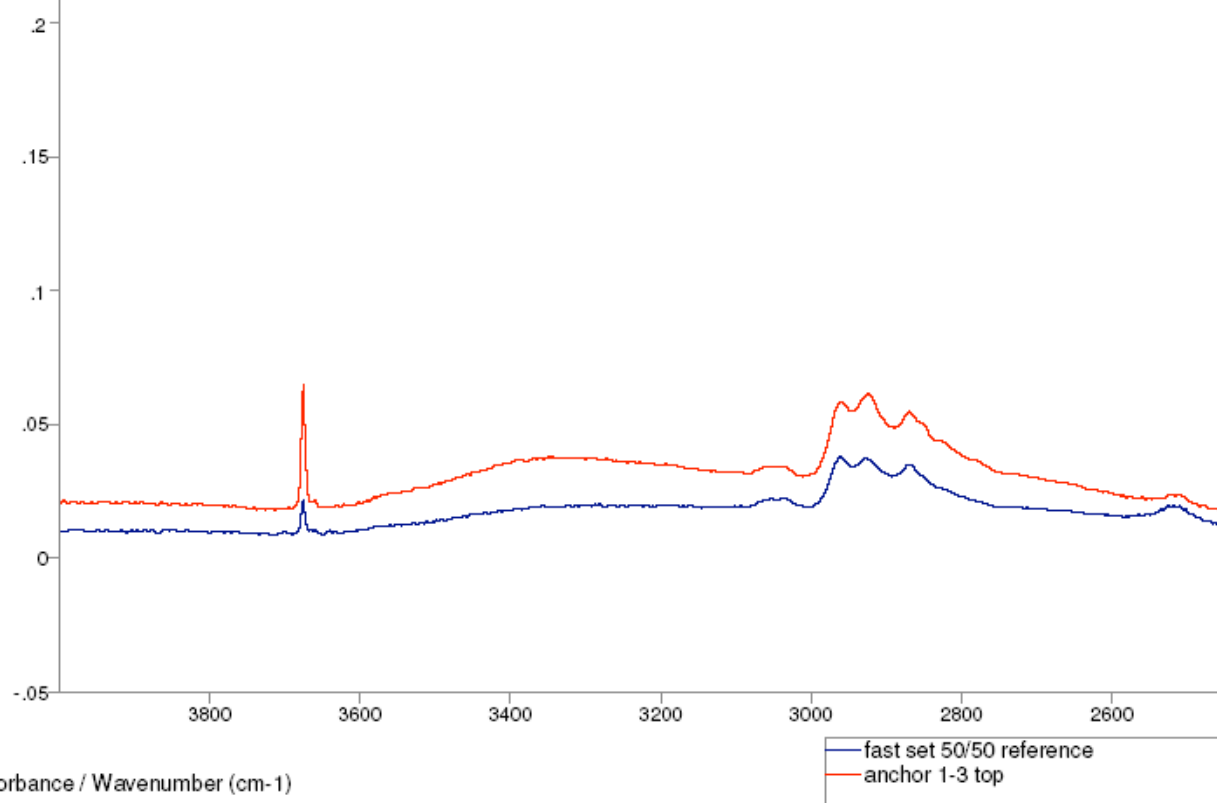
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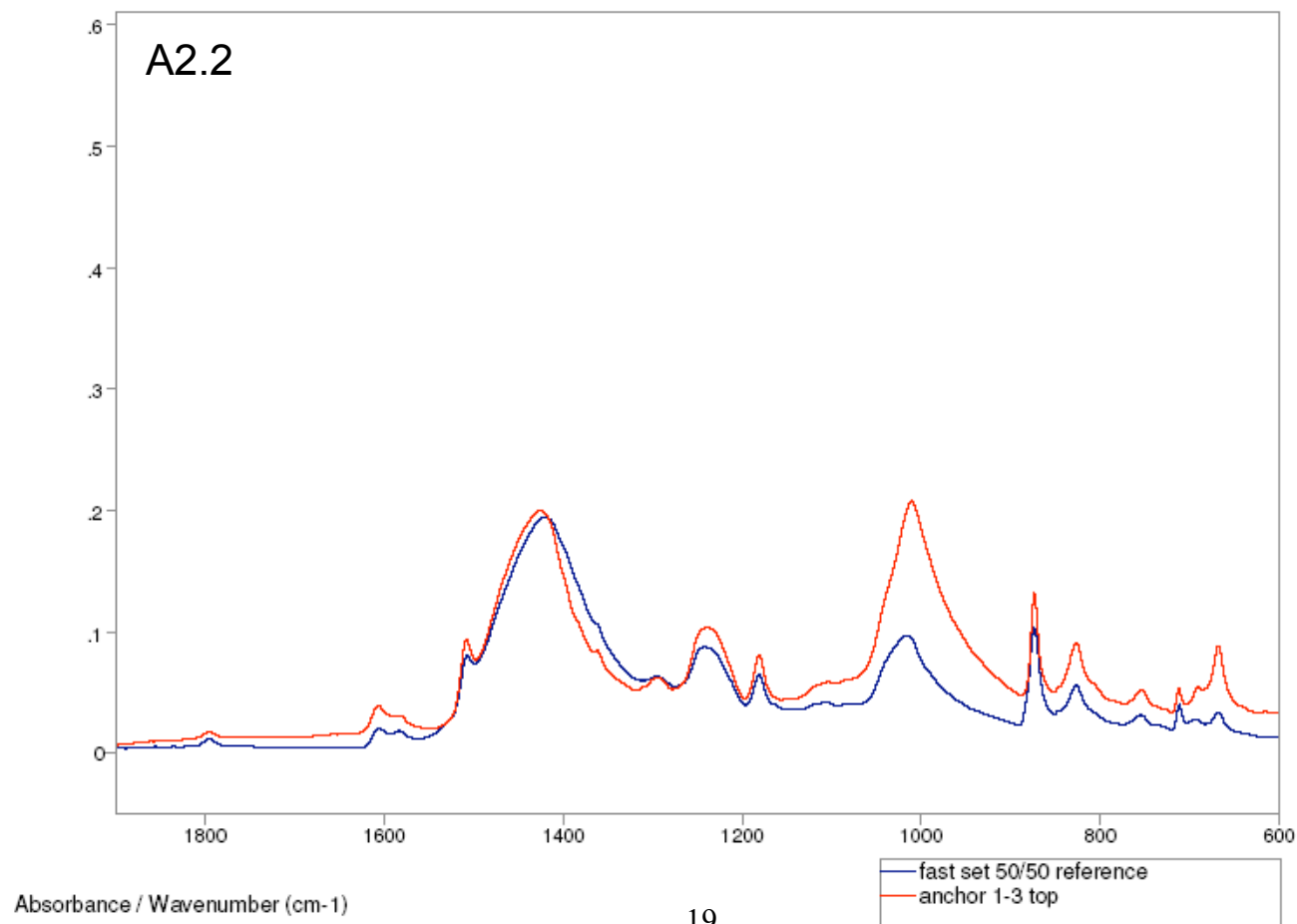
A1.2



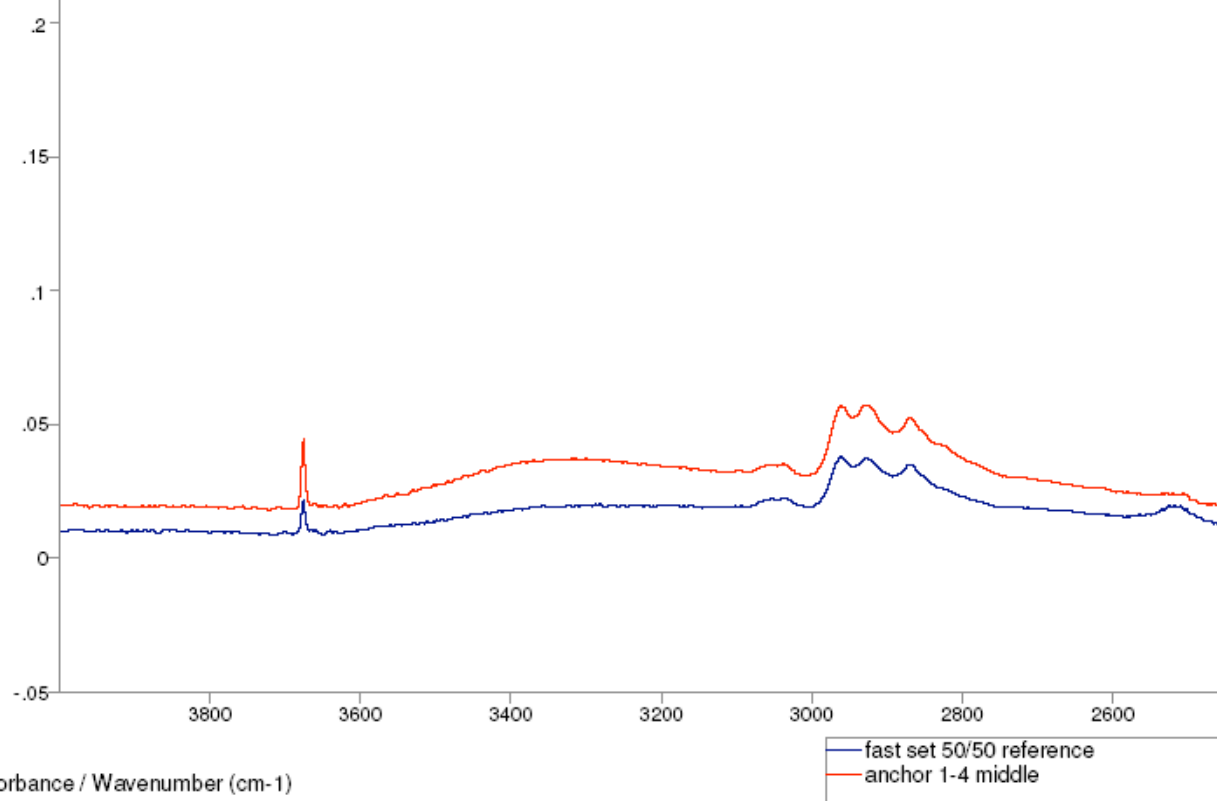
A2.1



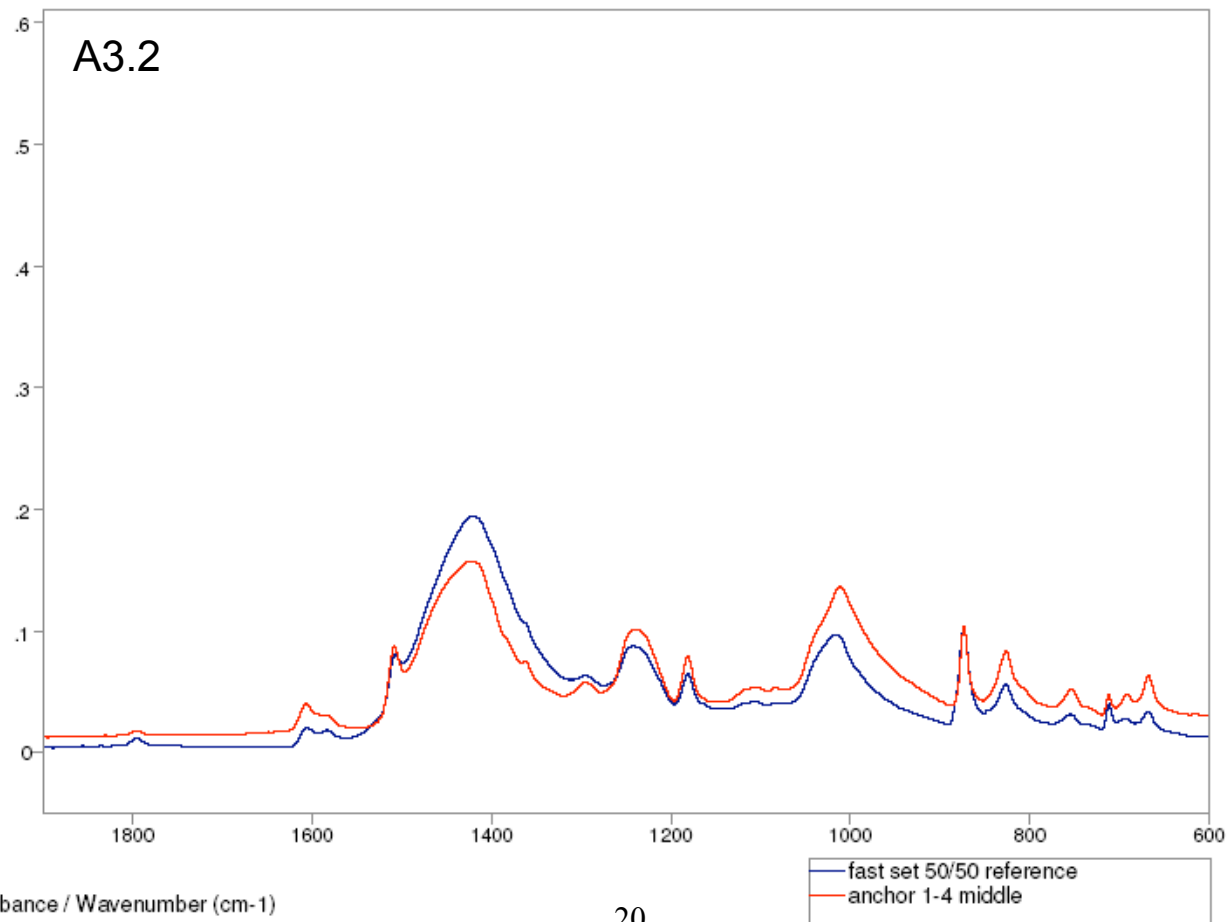
A2.2



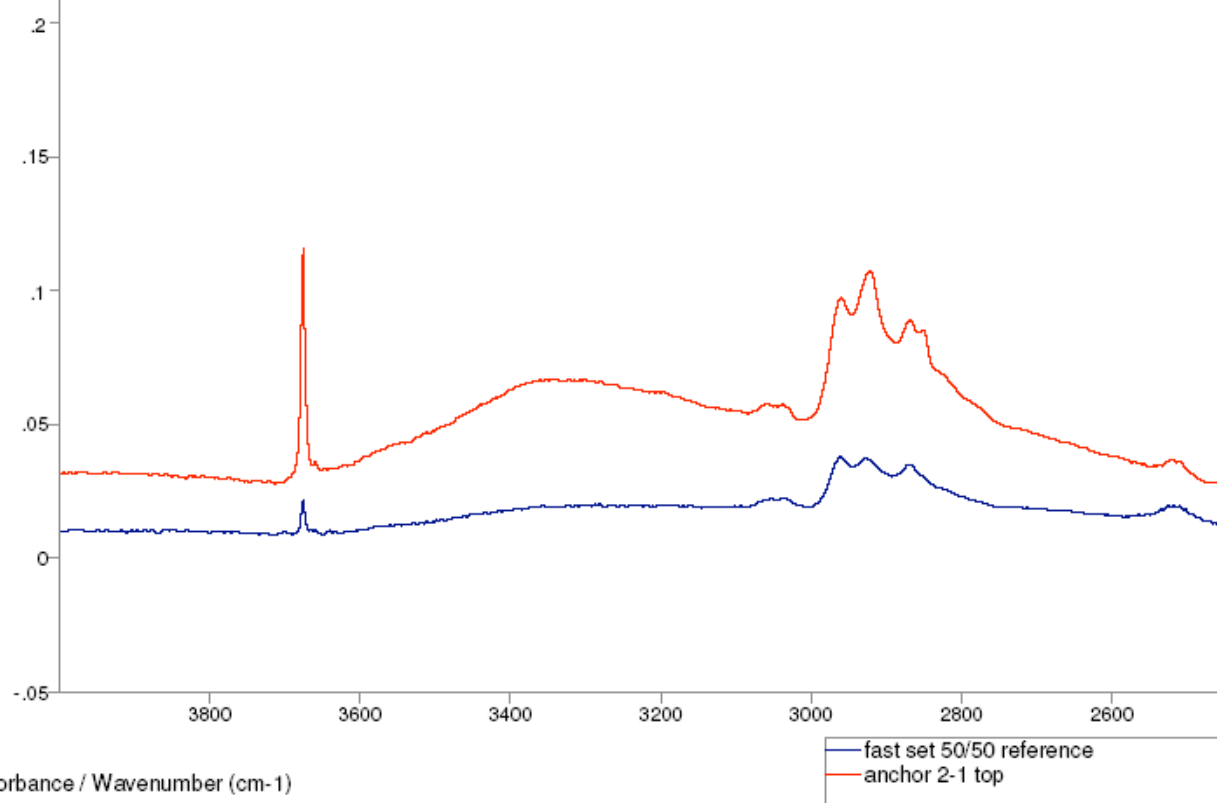
A3.1



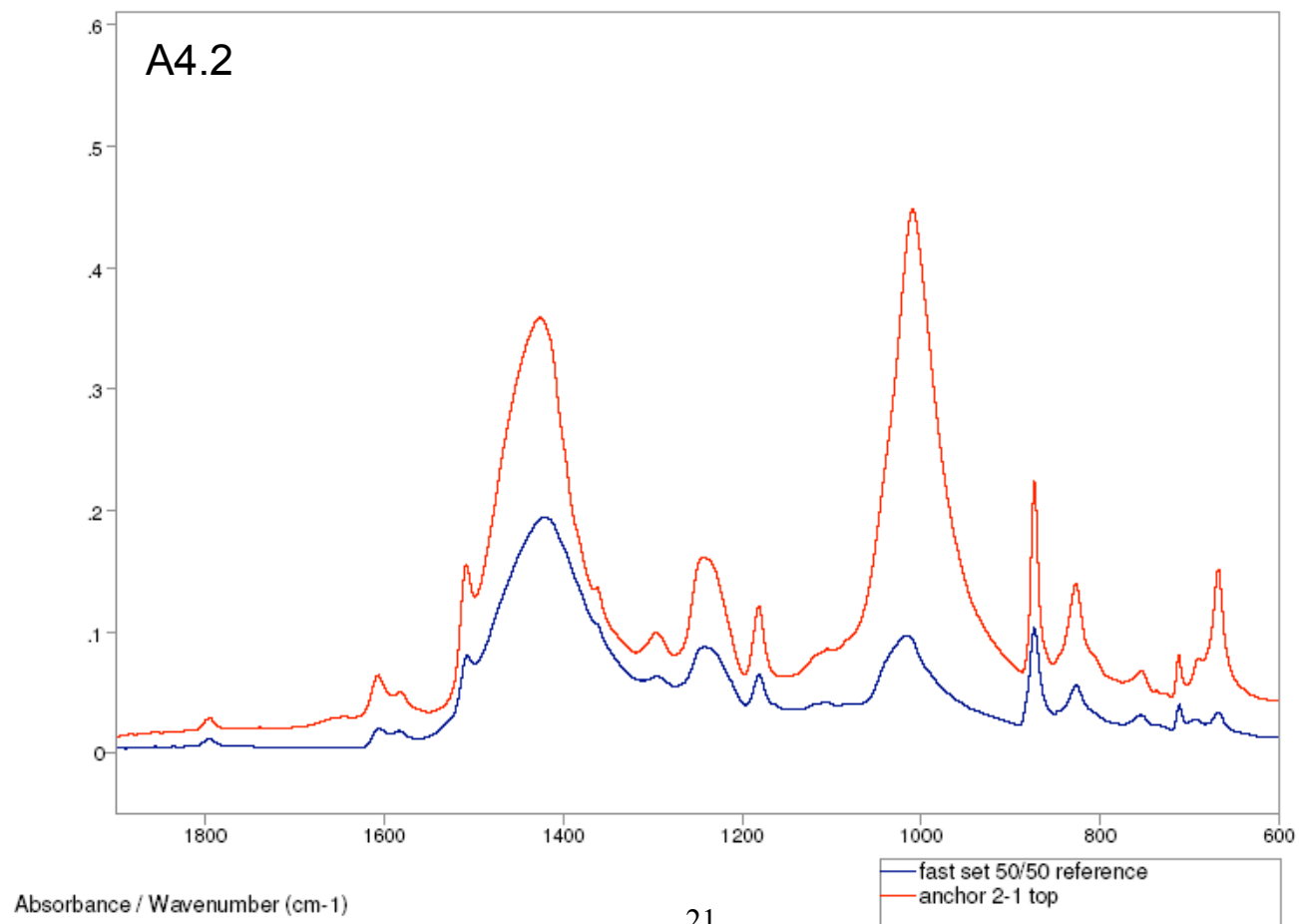
A3.2



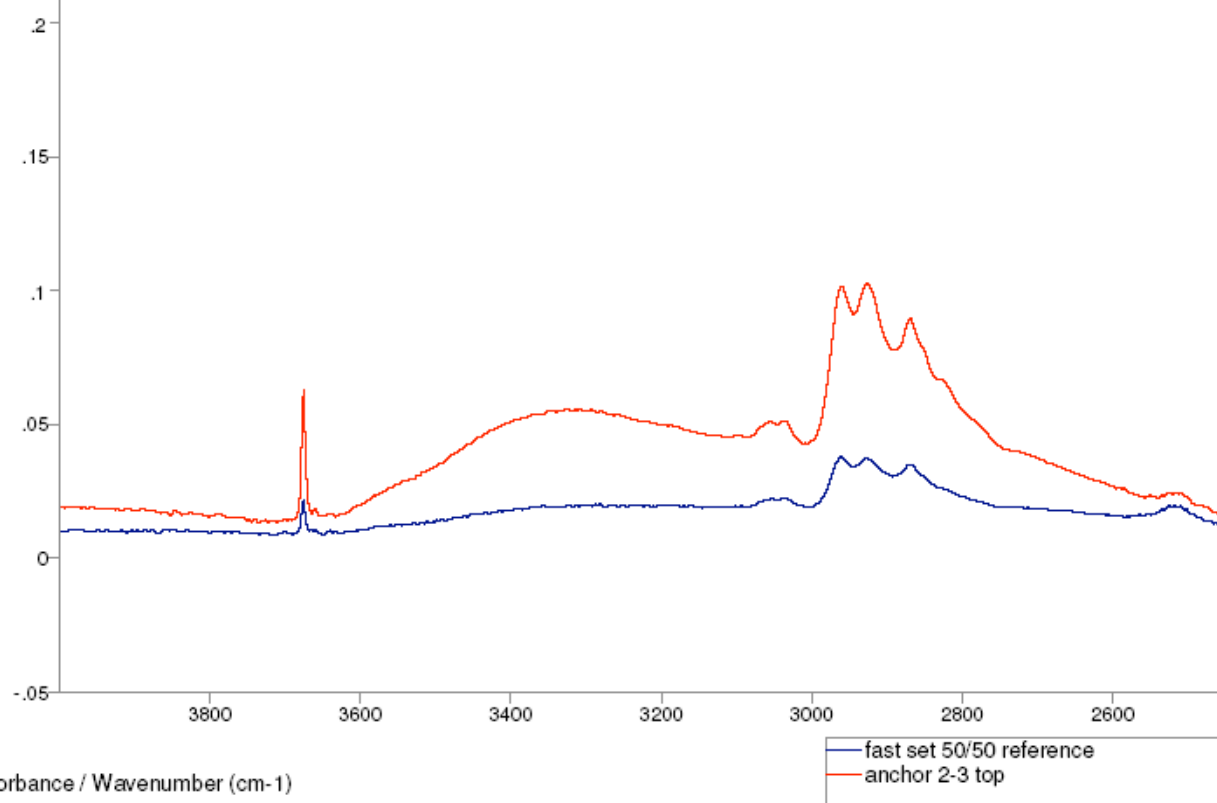
A4.1



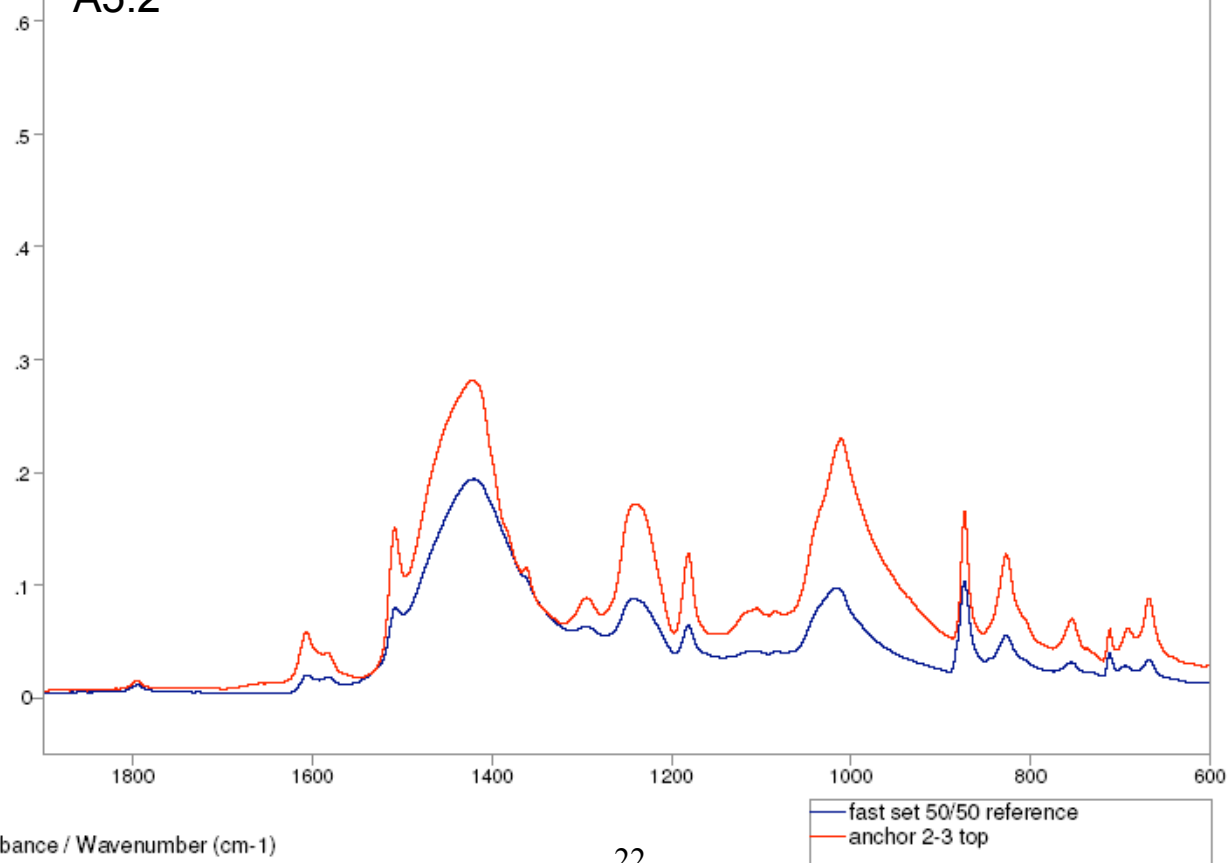
A4.2



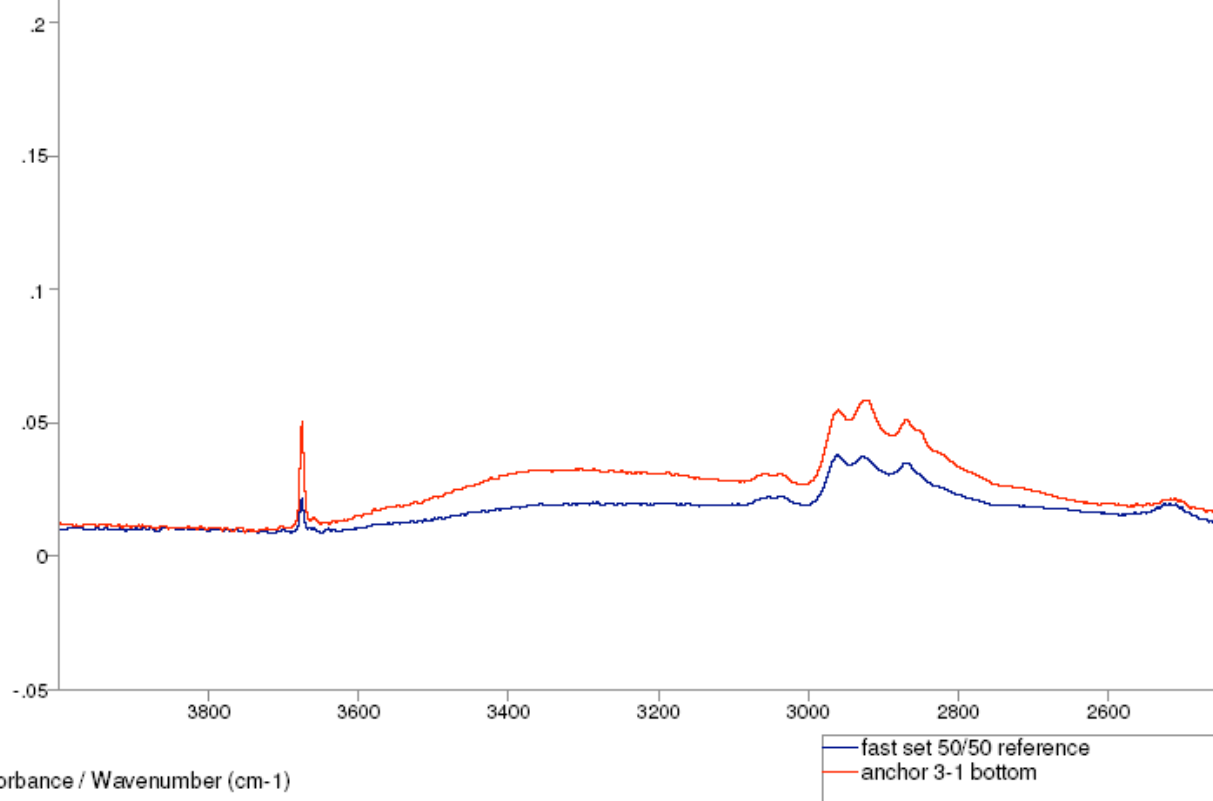
A5.1



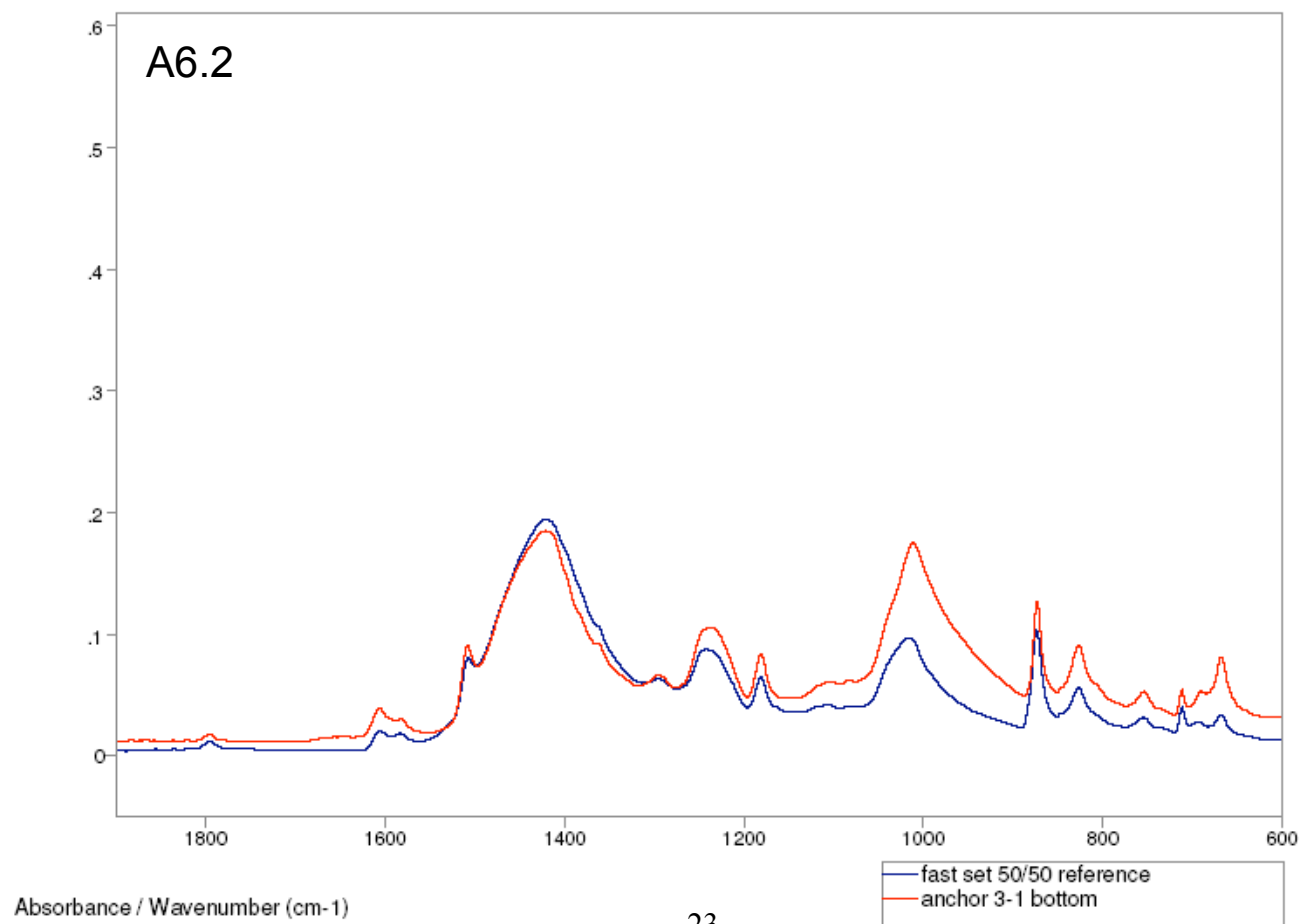
A5.2



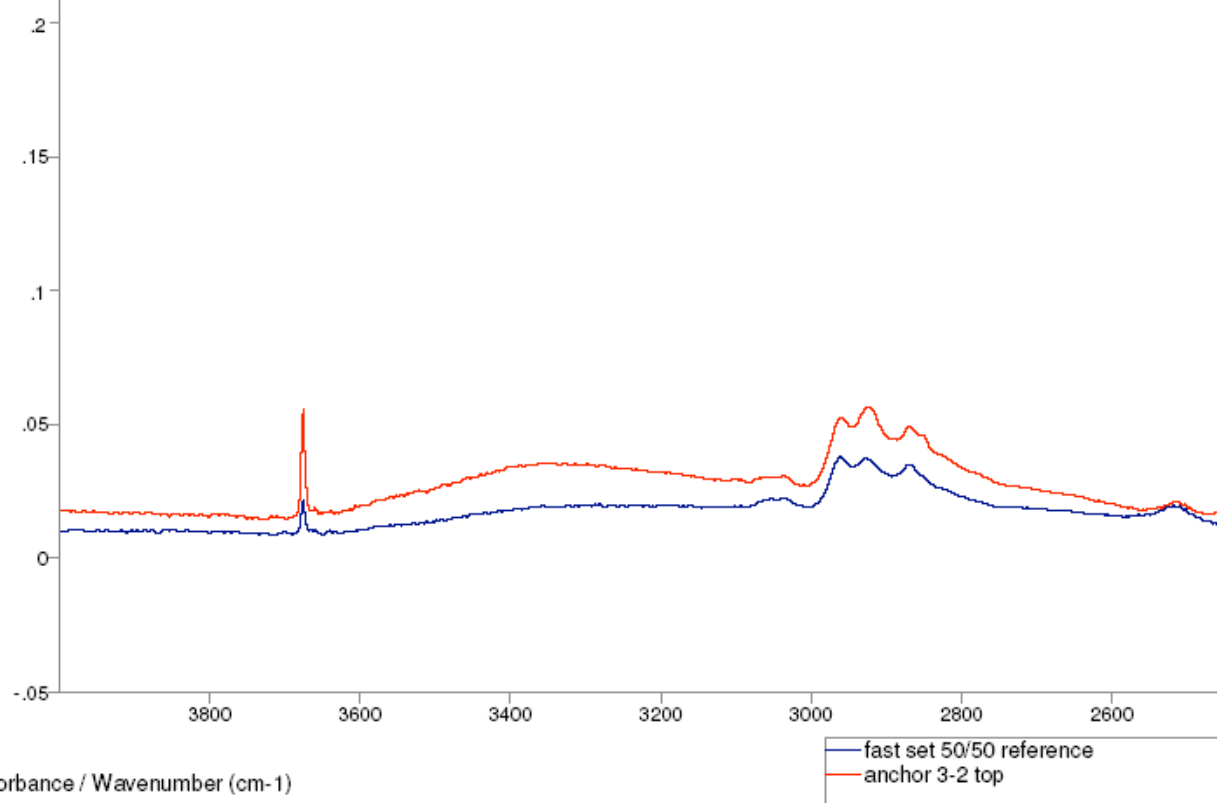
A6.1



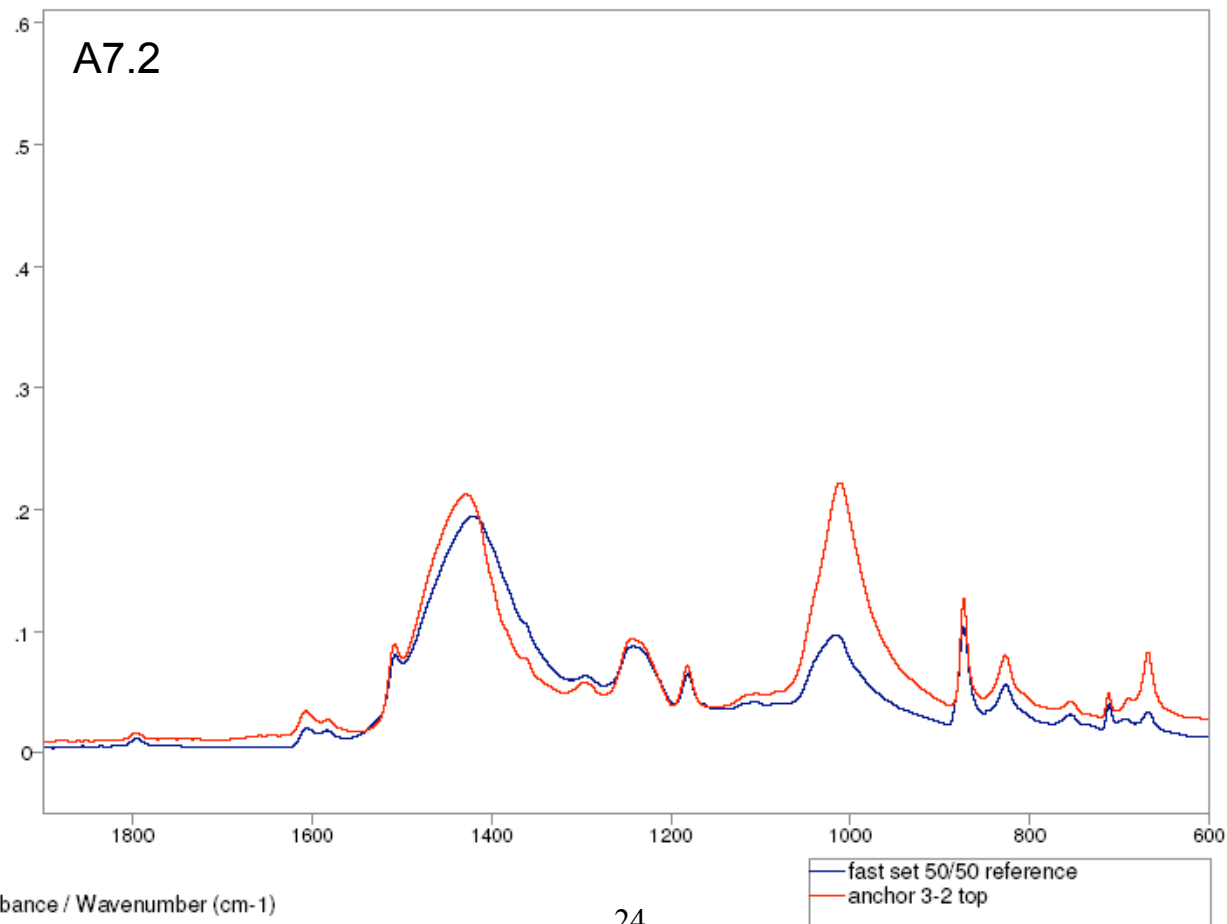
A6.2



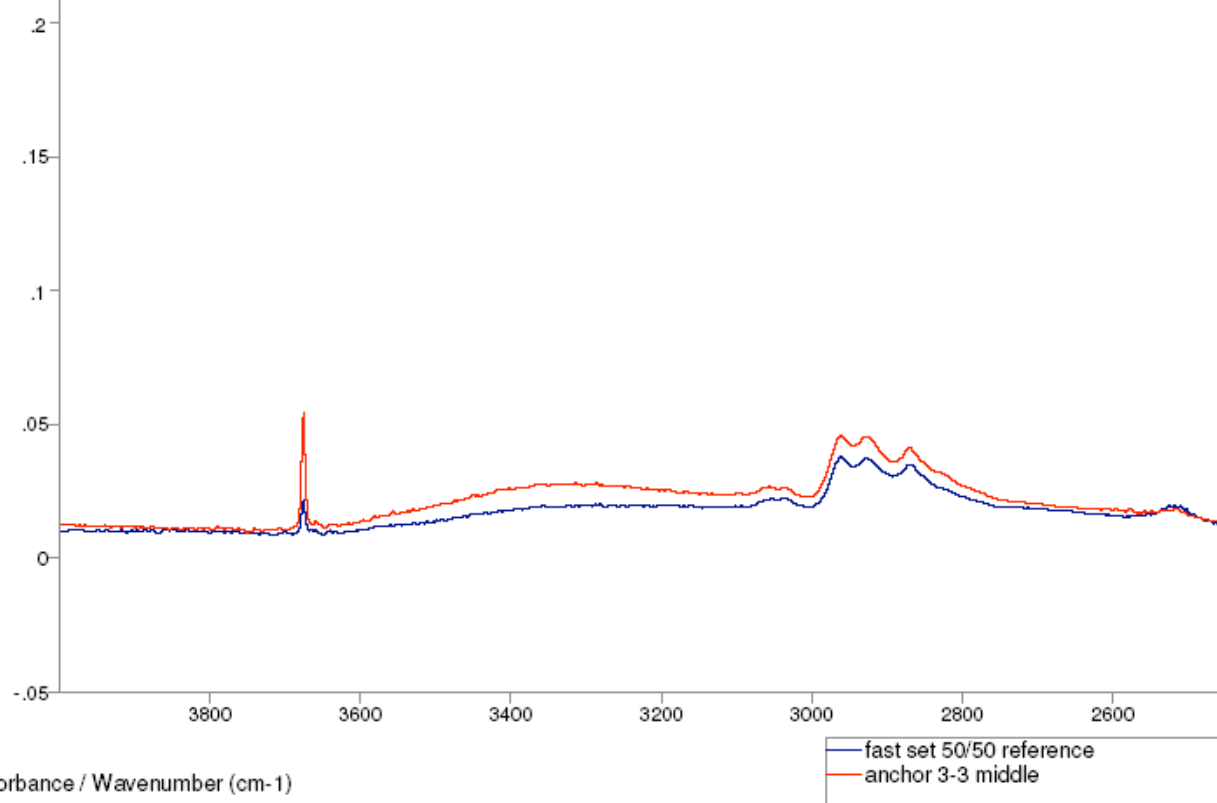
A7.1



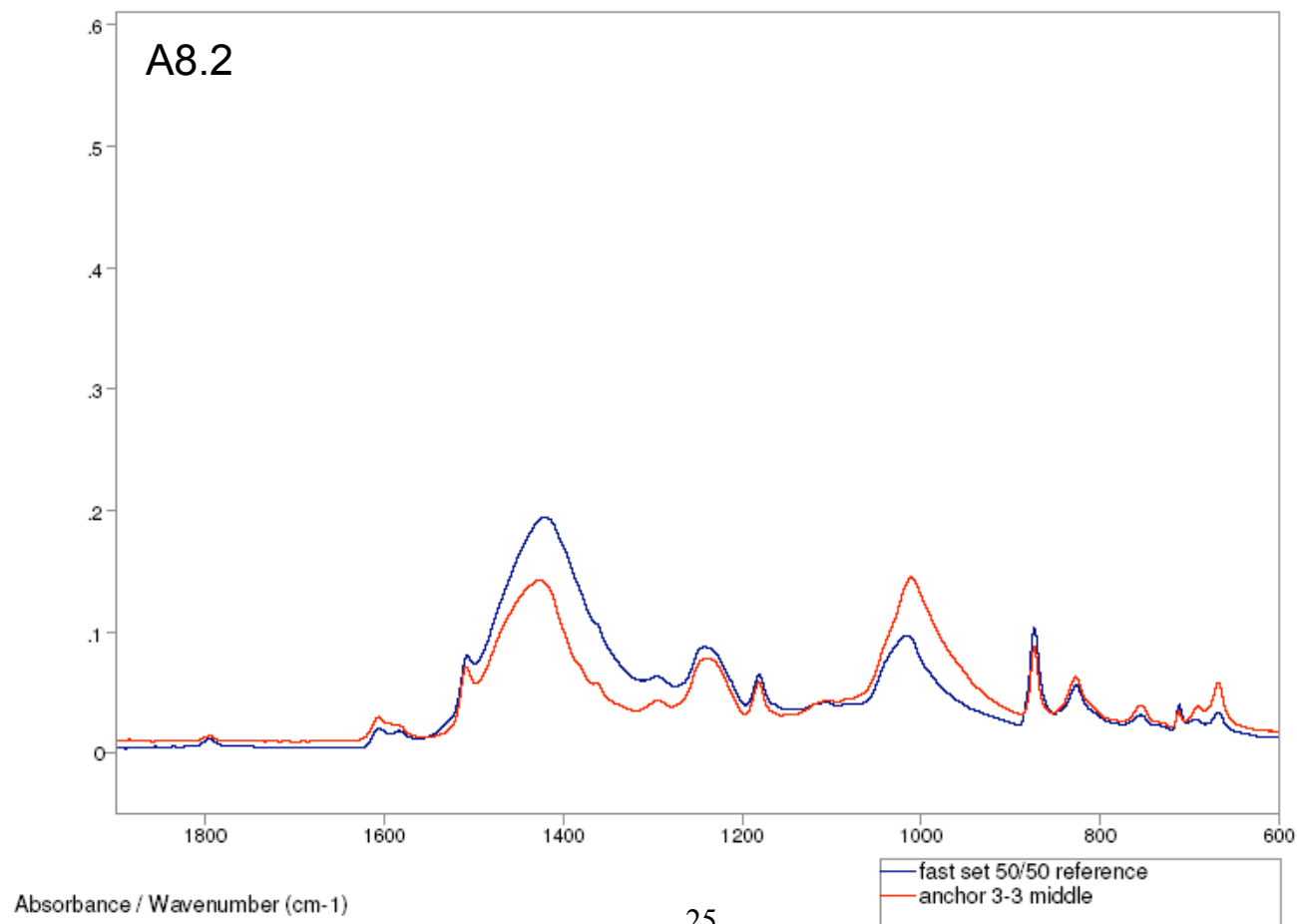
A7.2



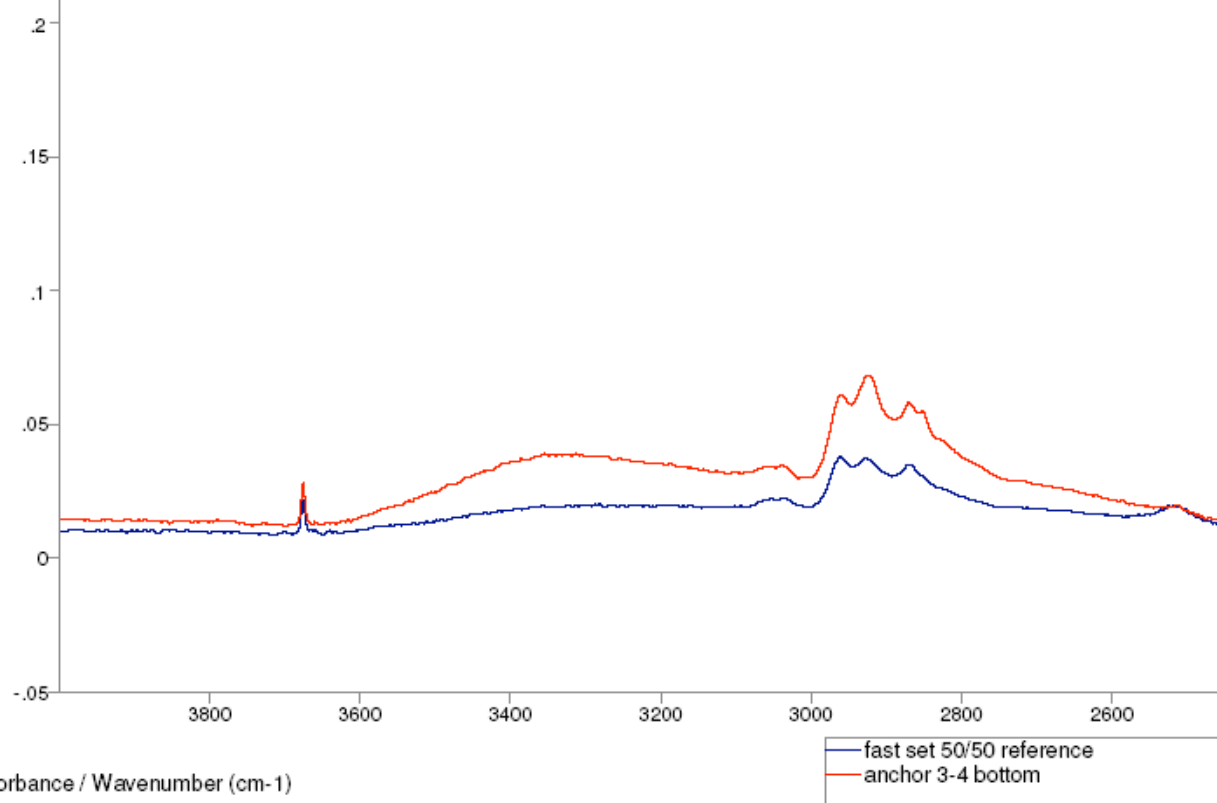
A8.1



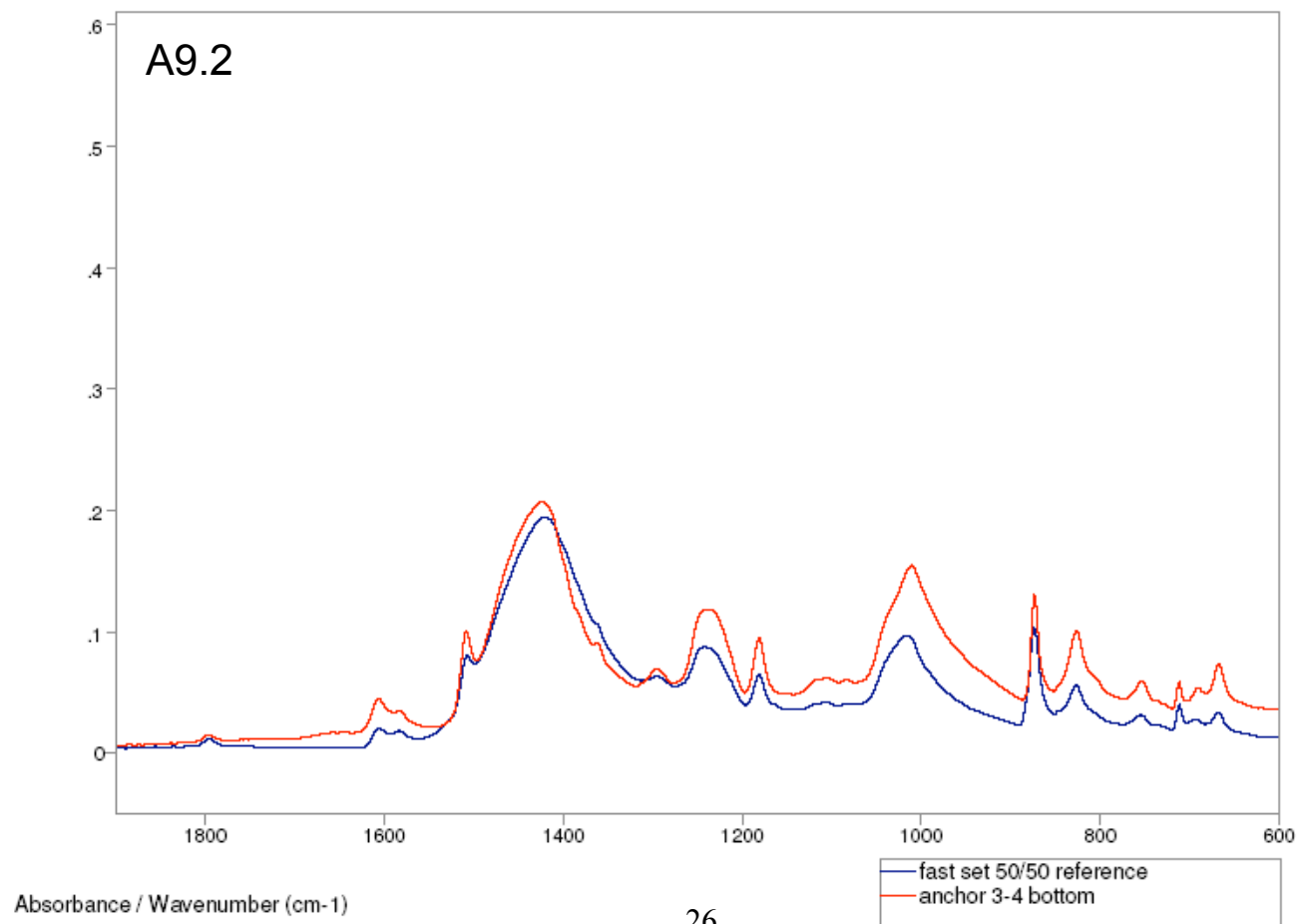
A8.2



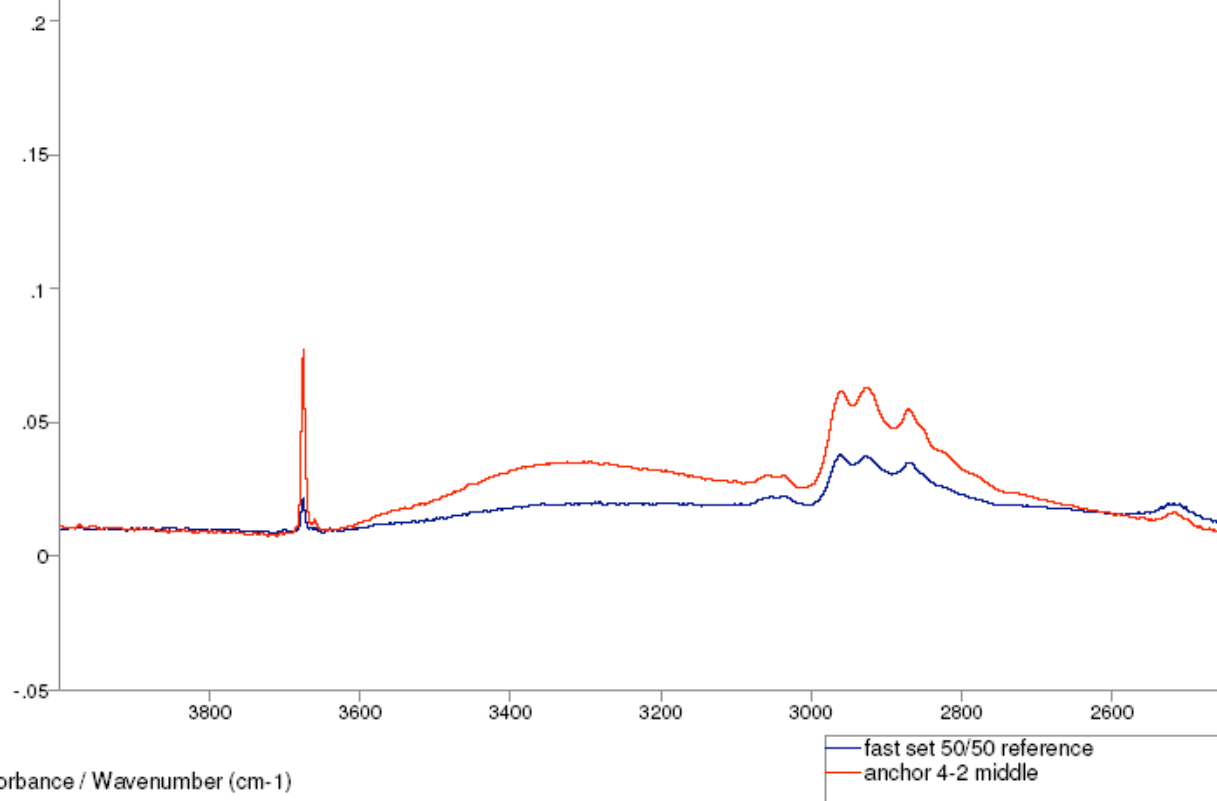
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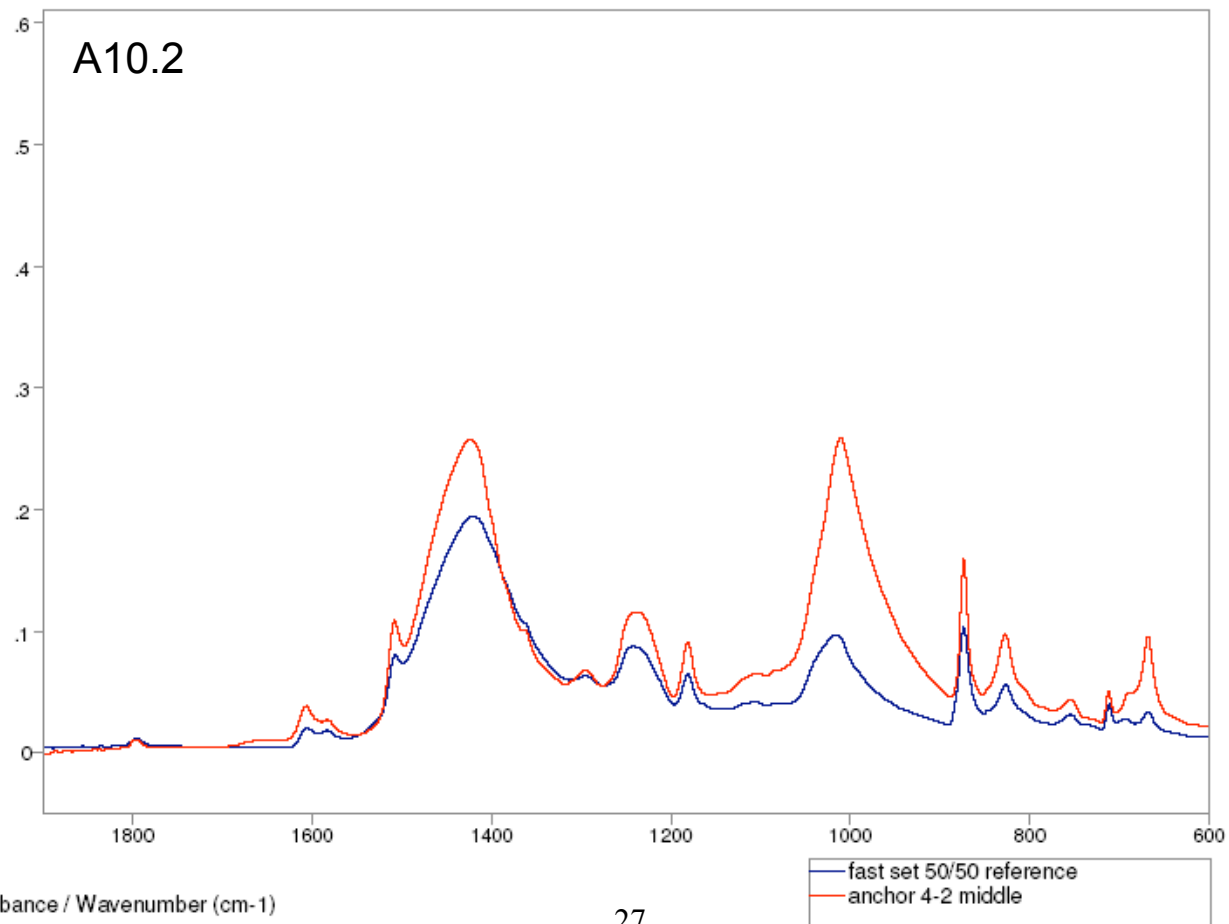
A9.2



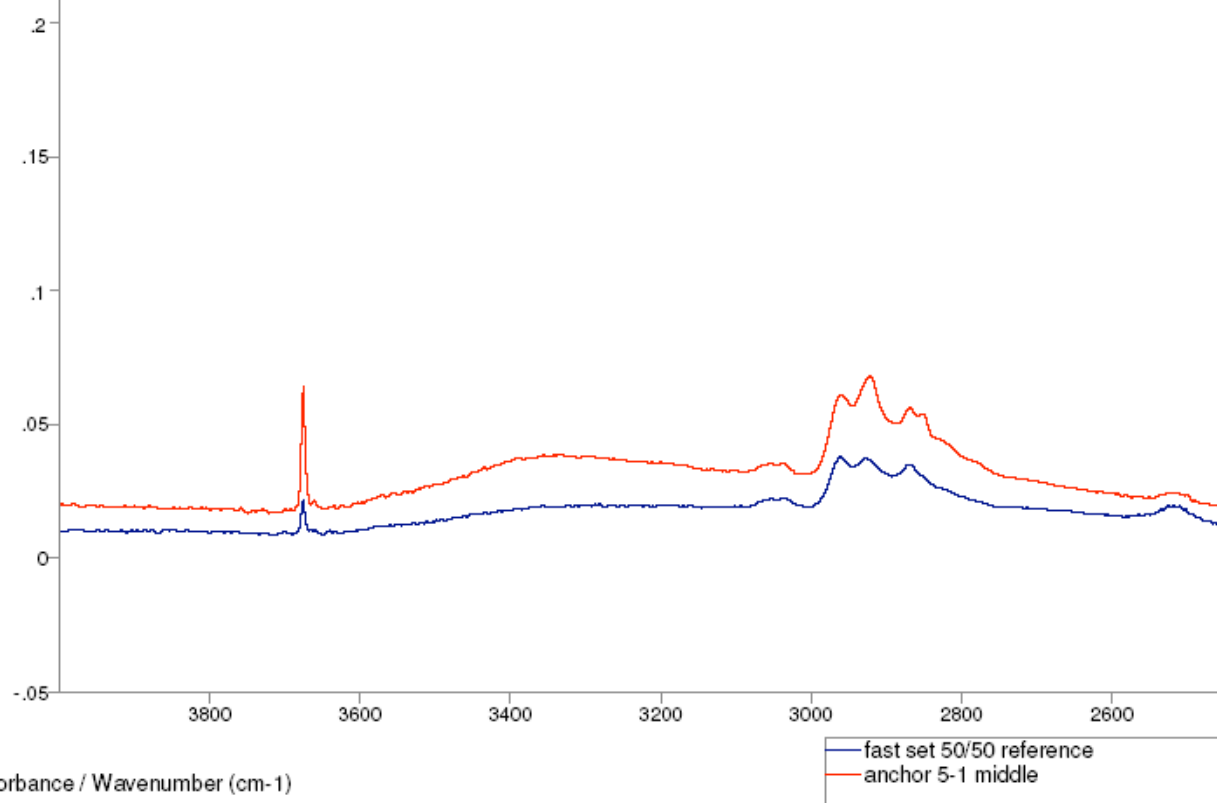
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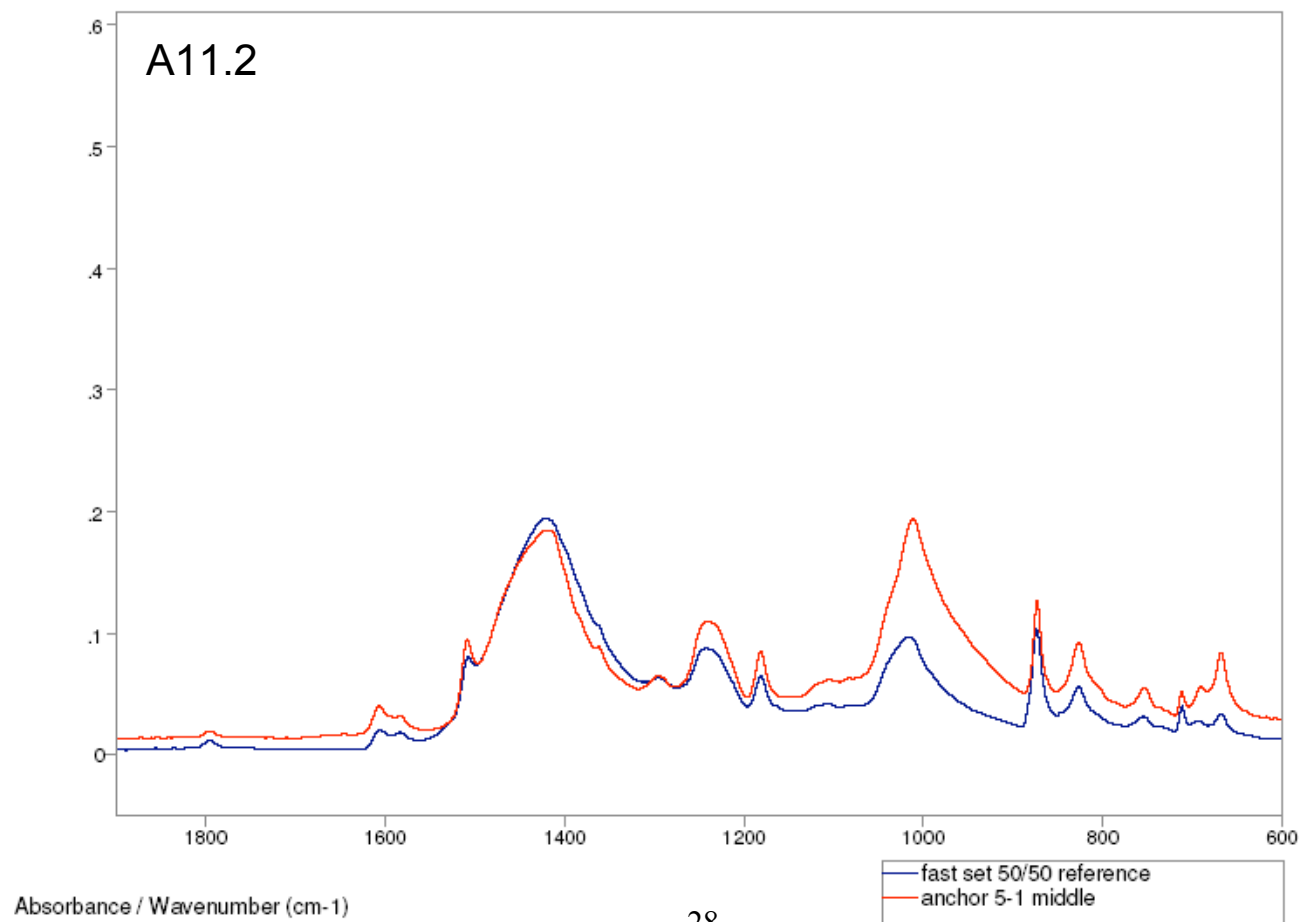
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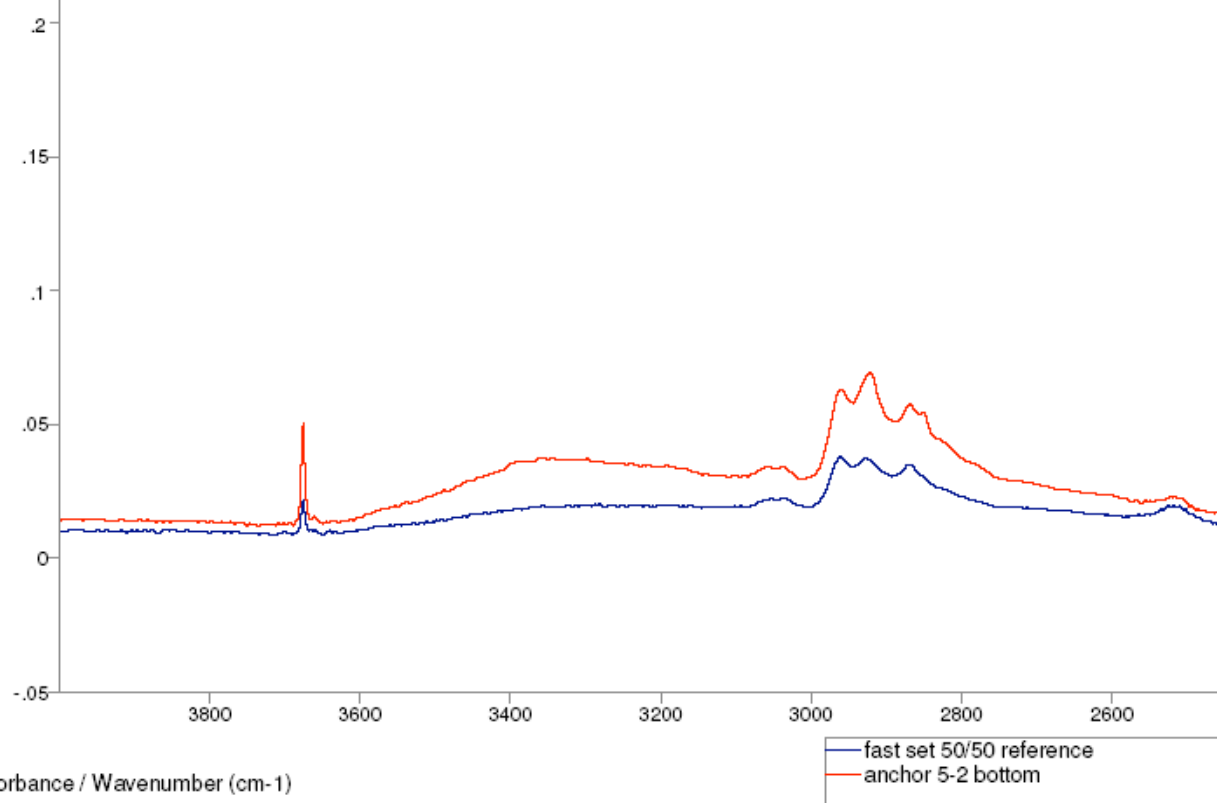
A11.1



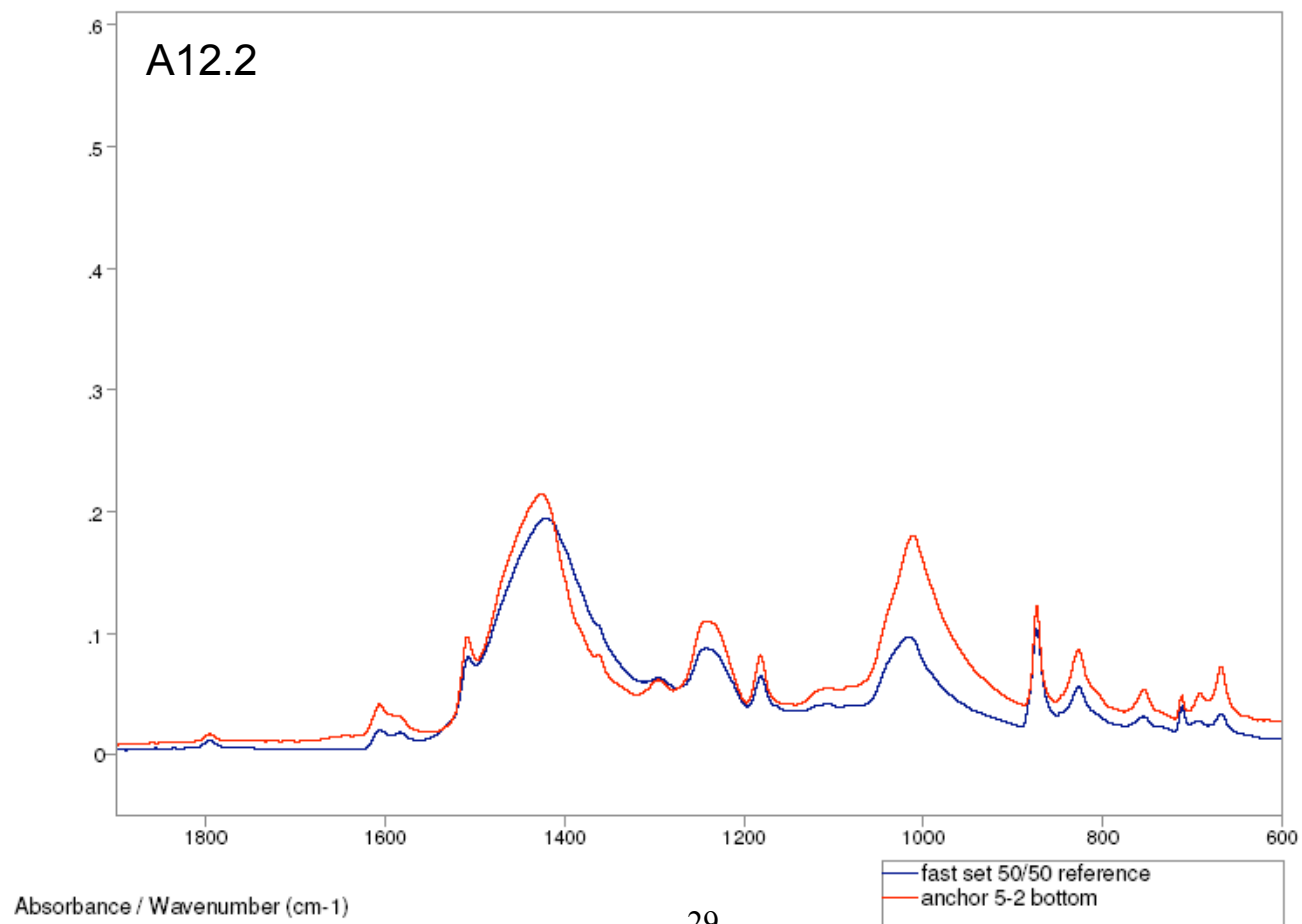
A11.2



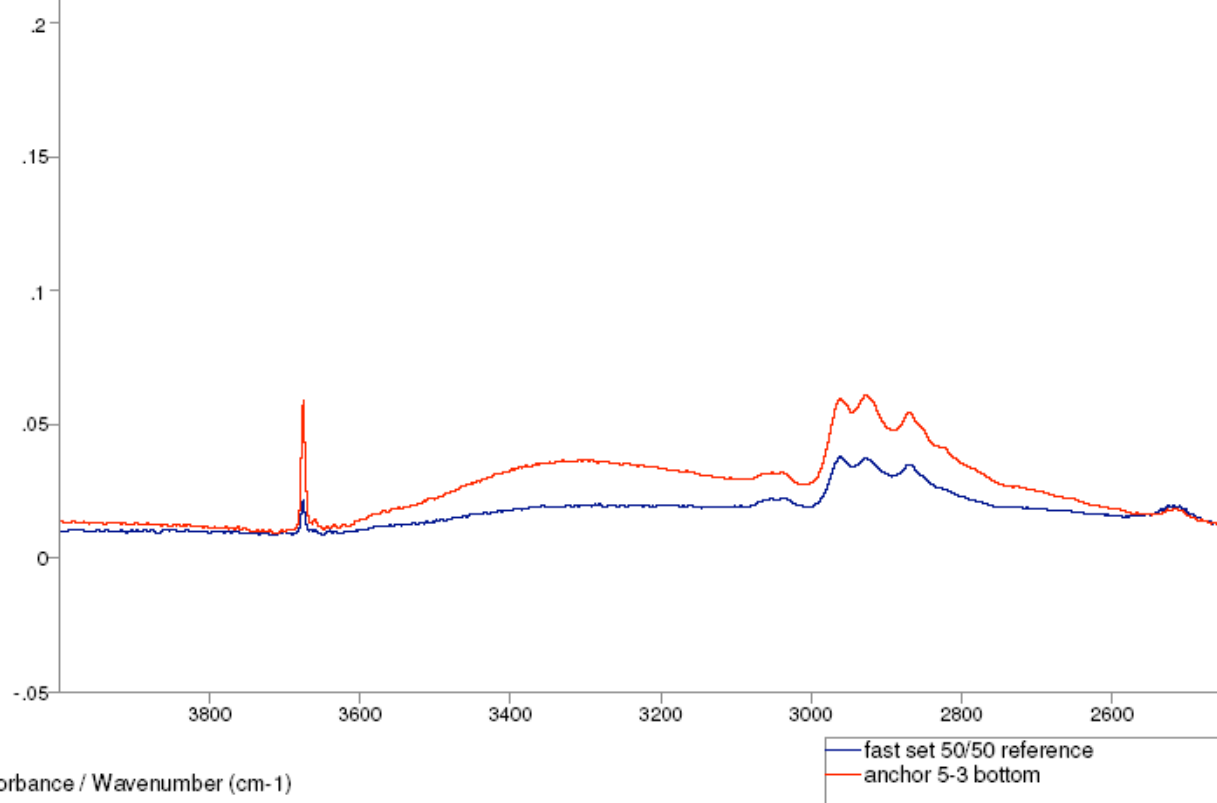
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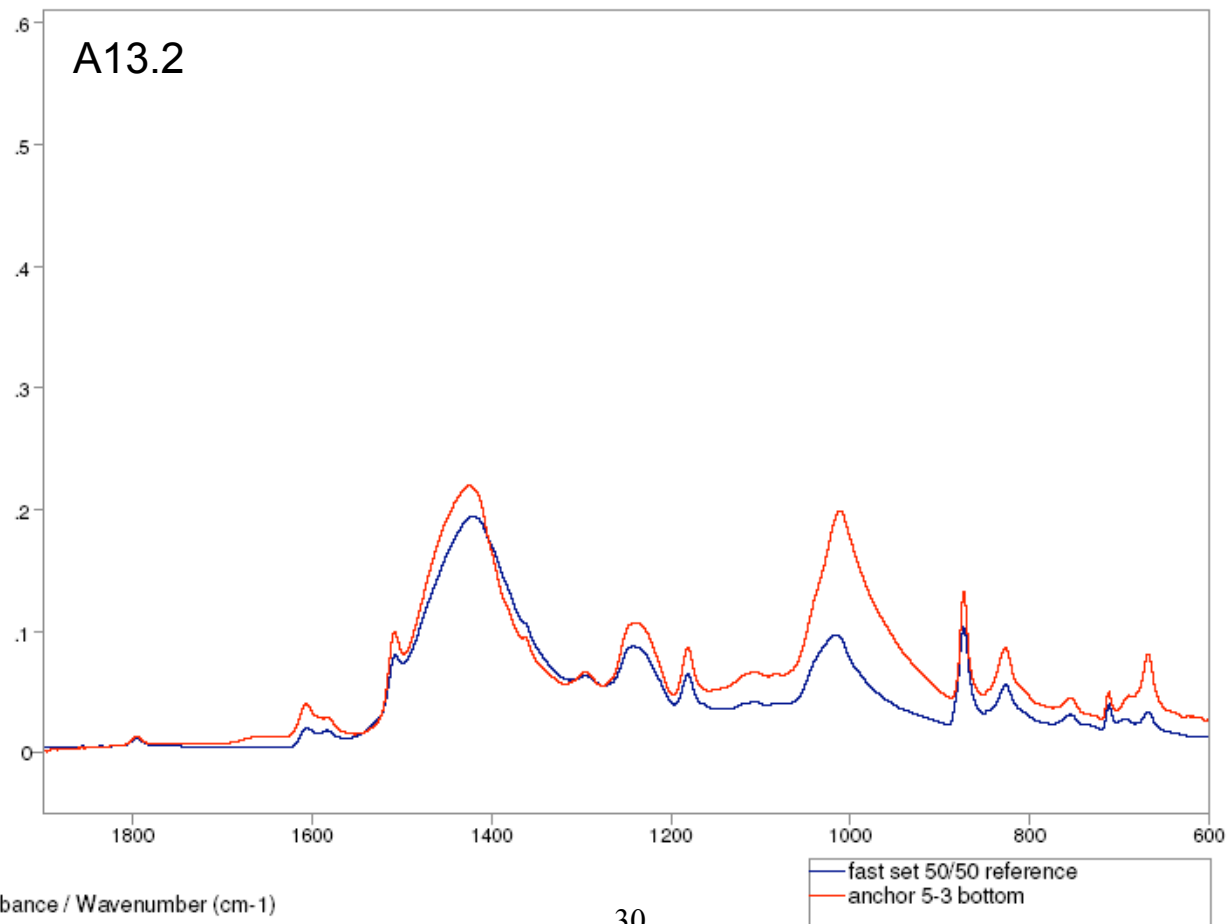
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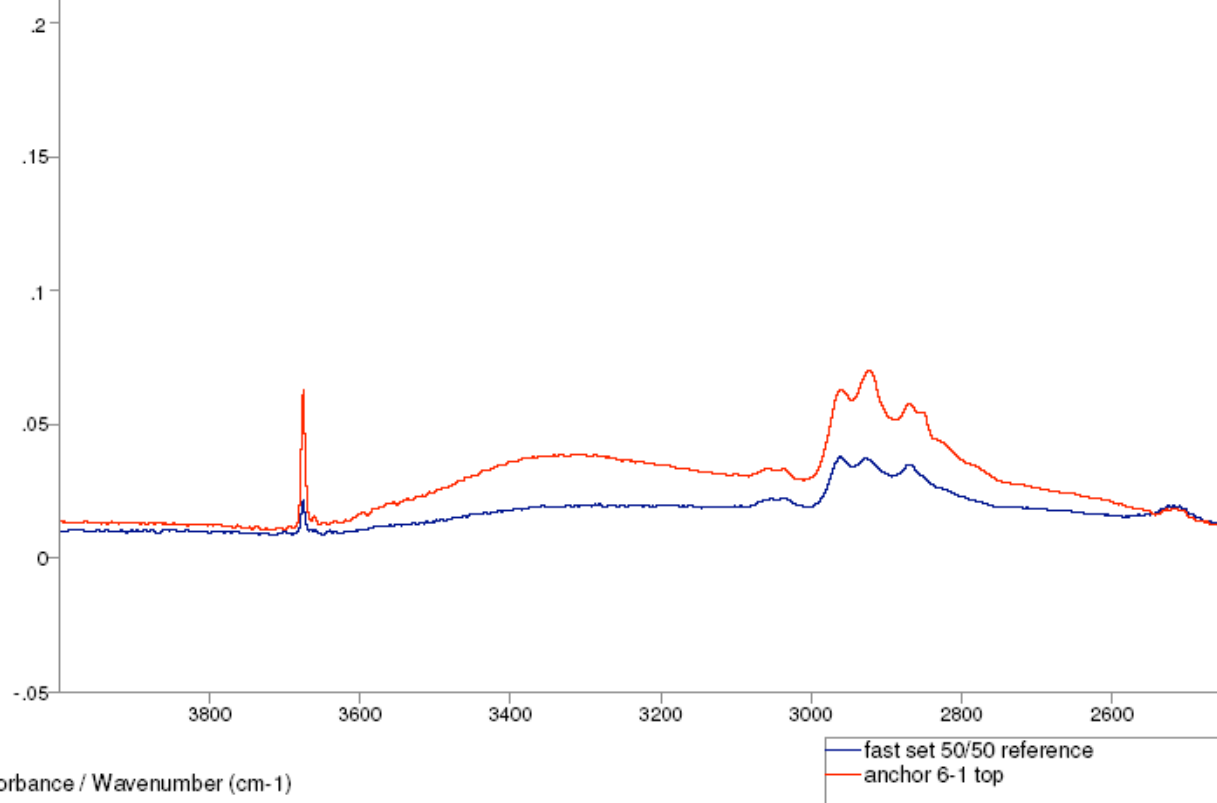
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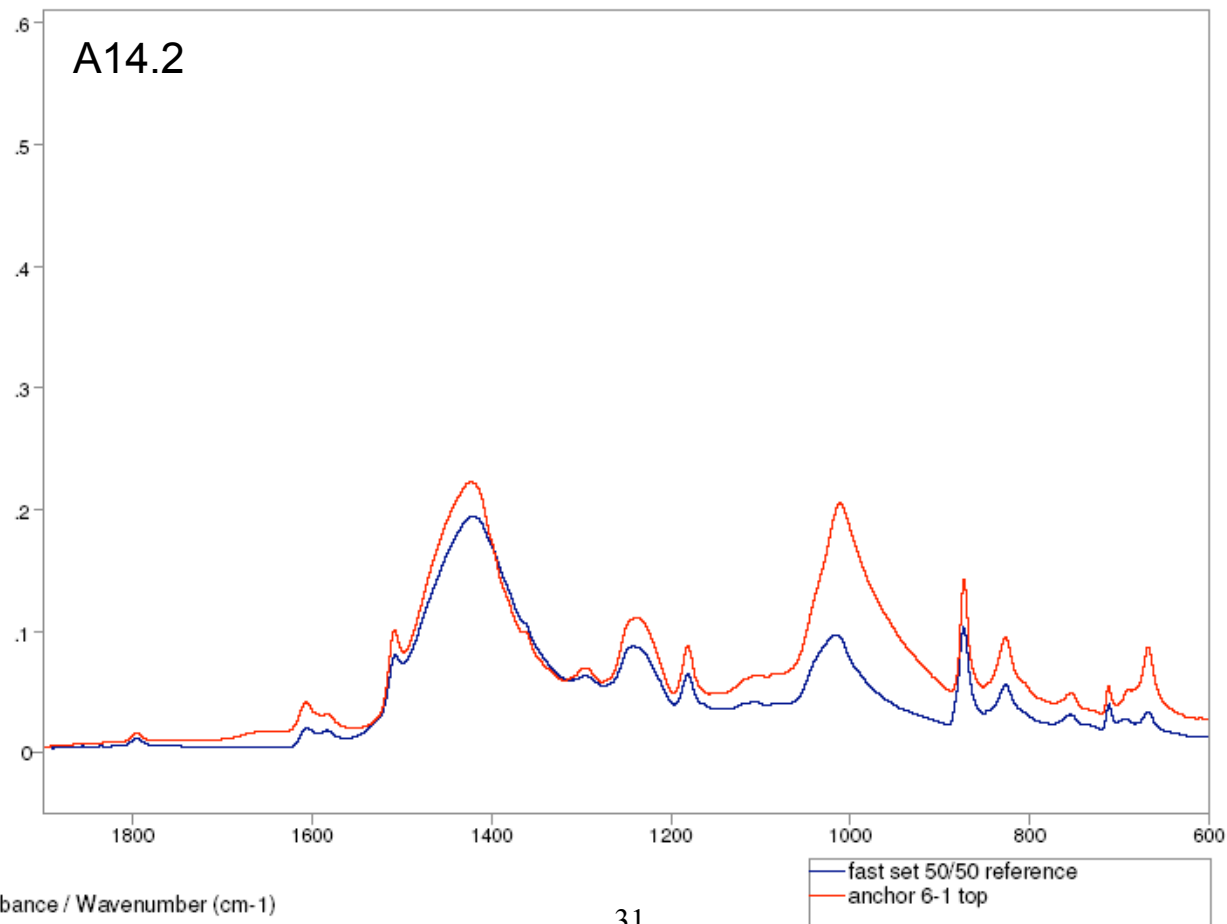
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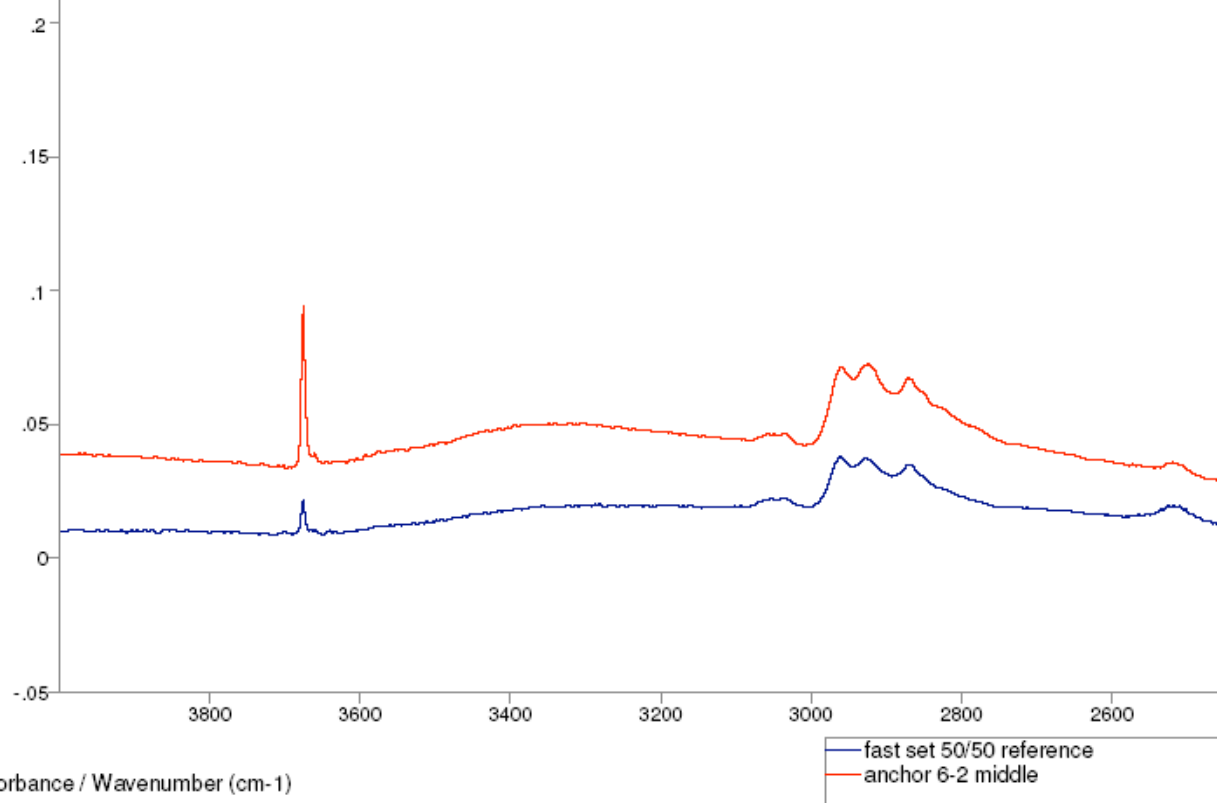
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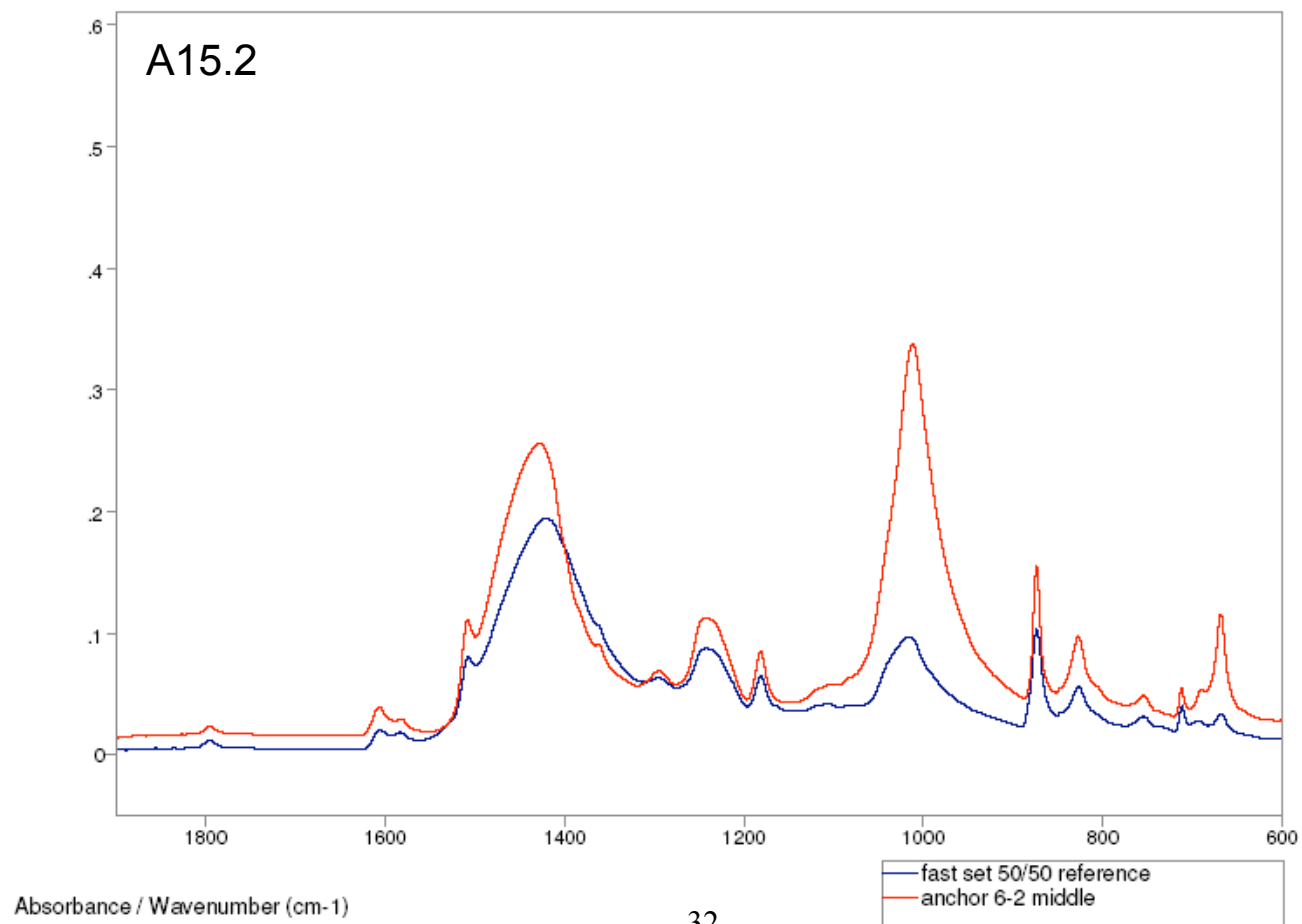
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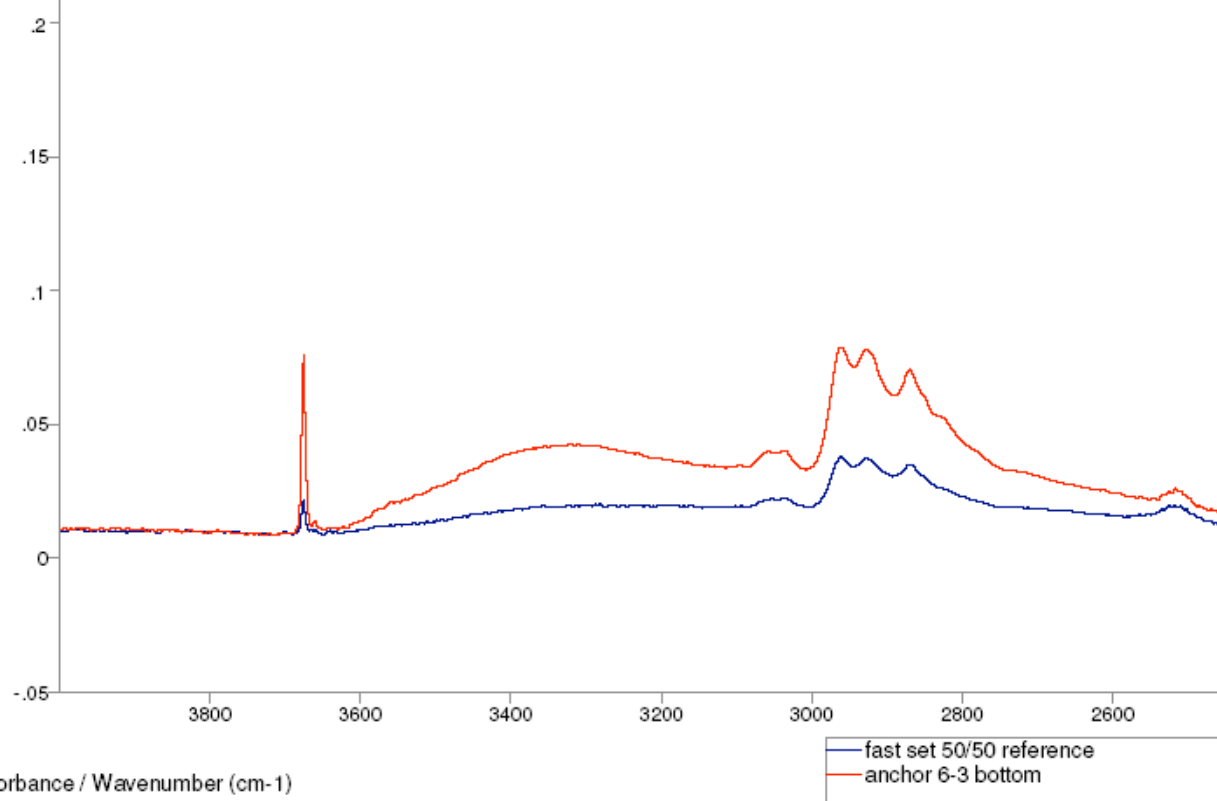
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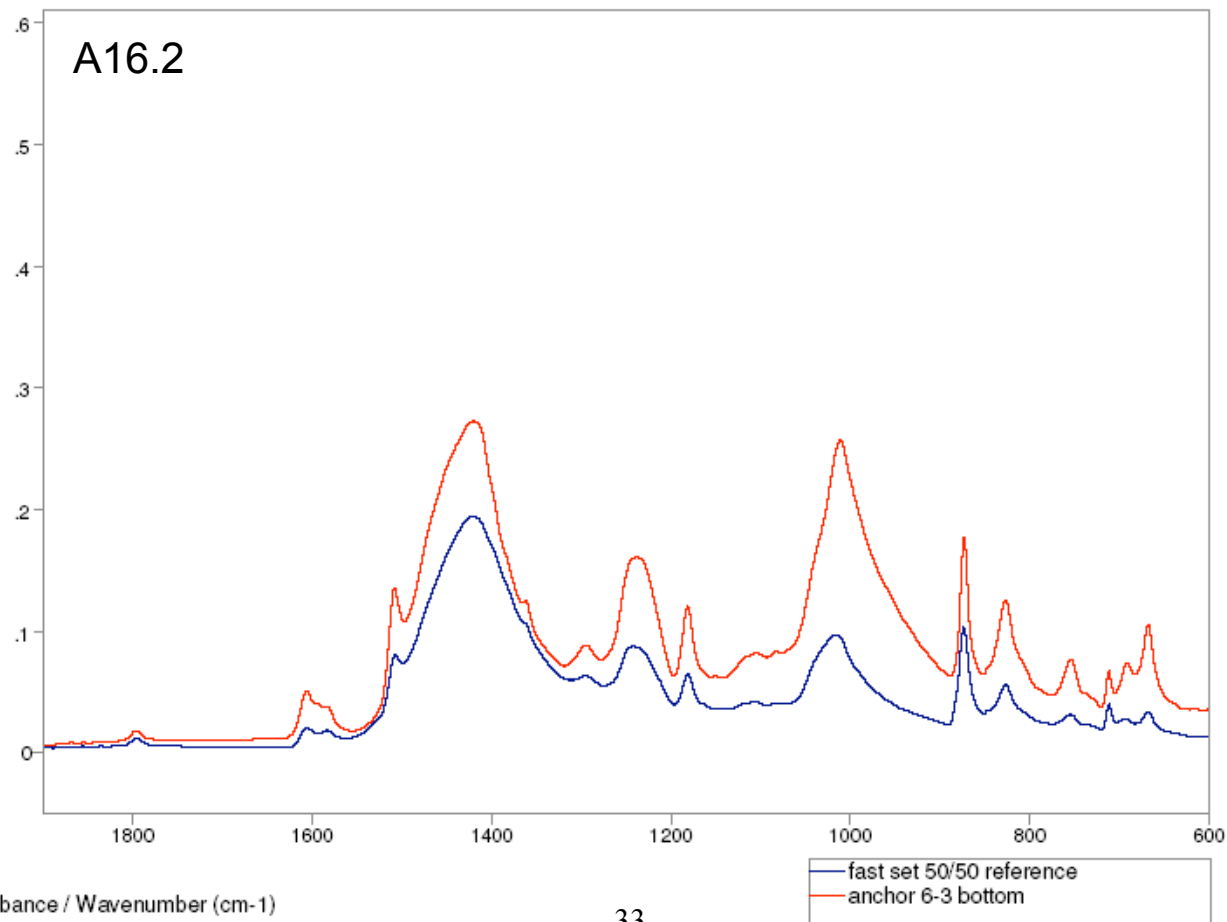
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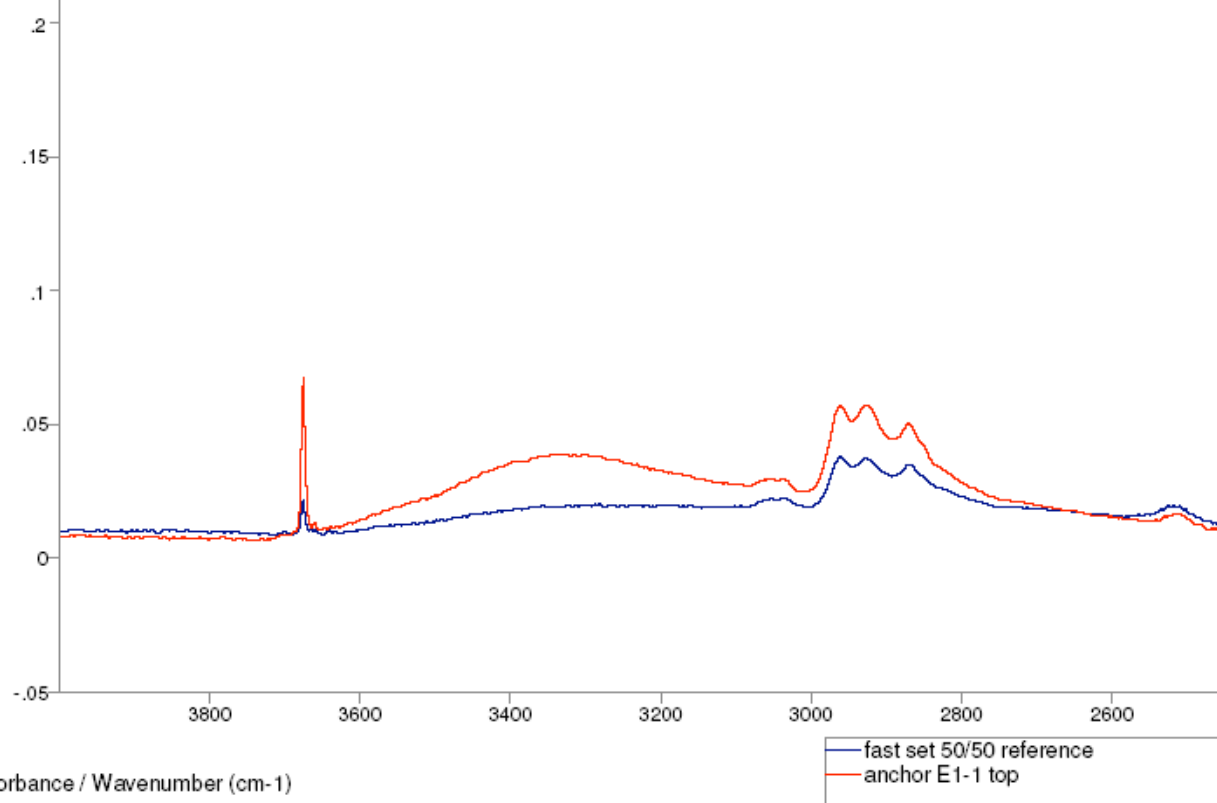
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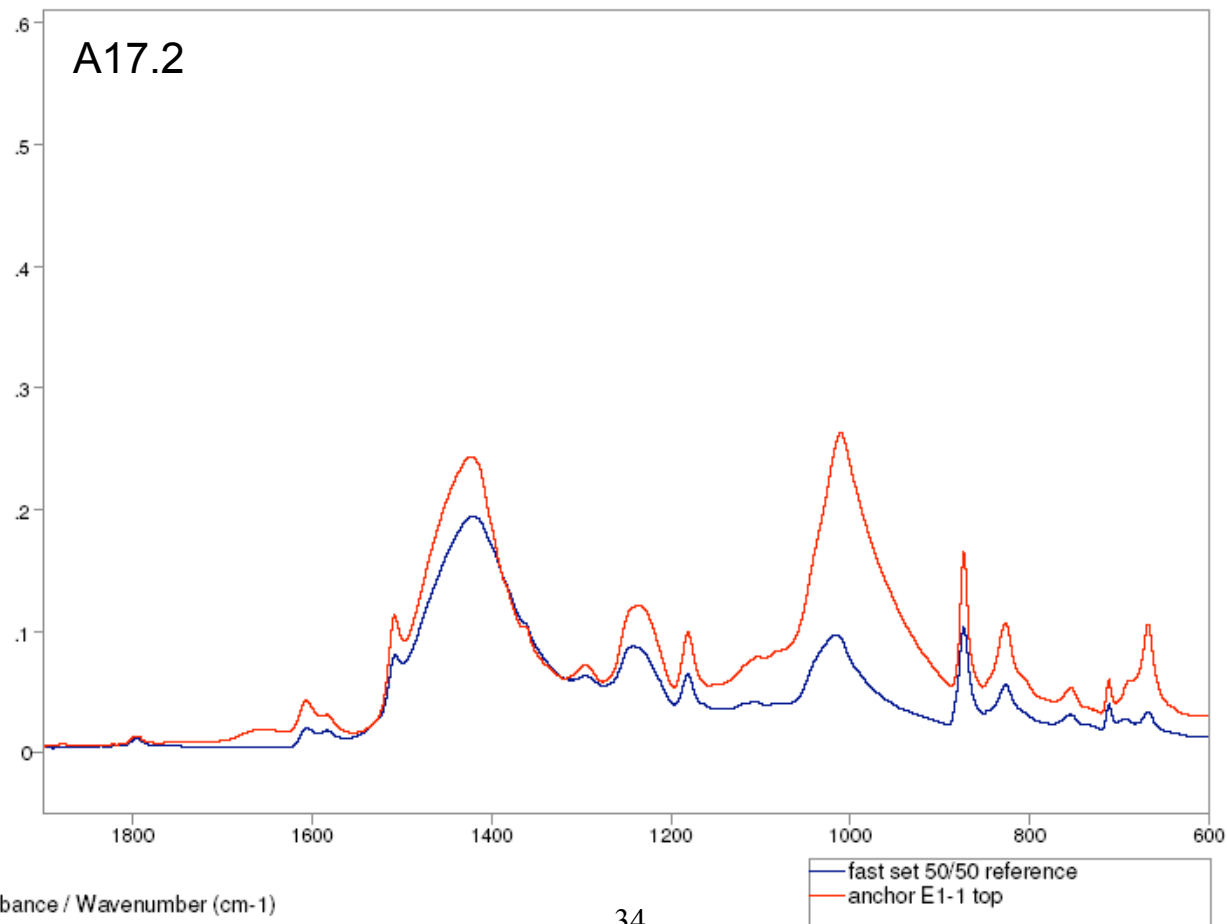
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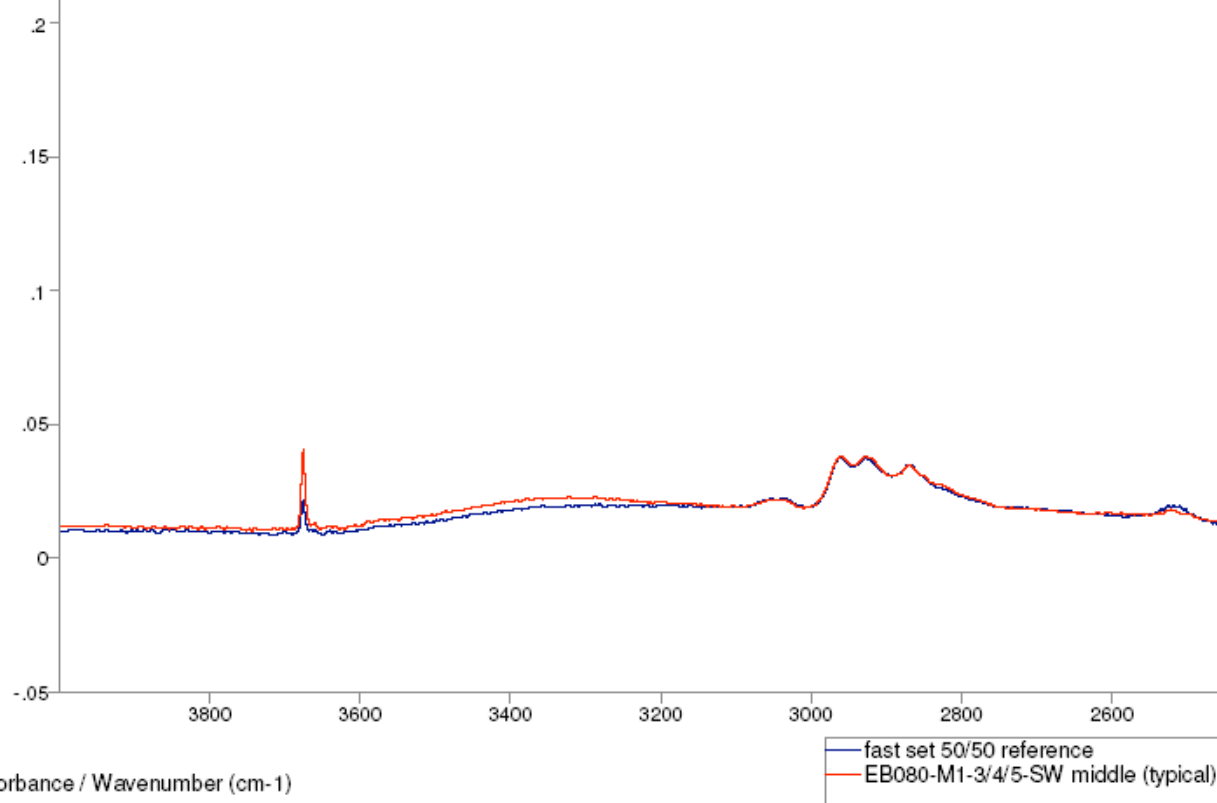
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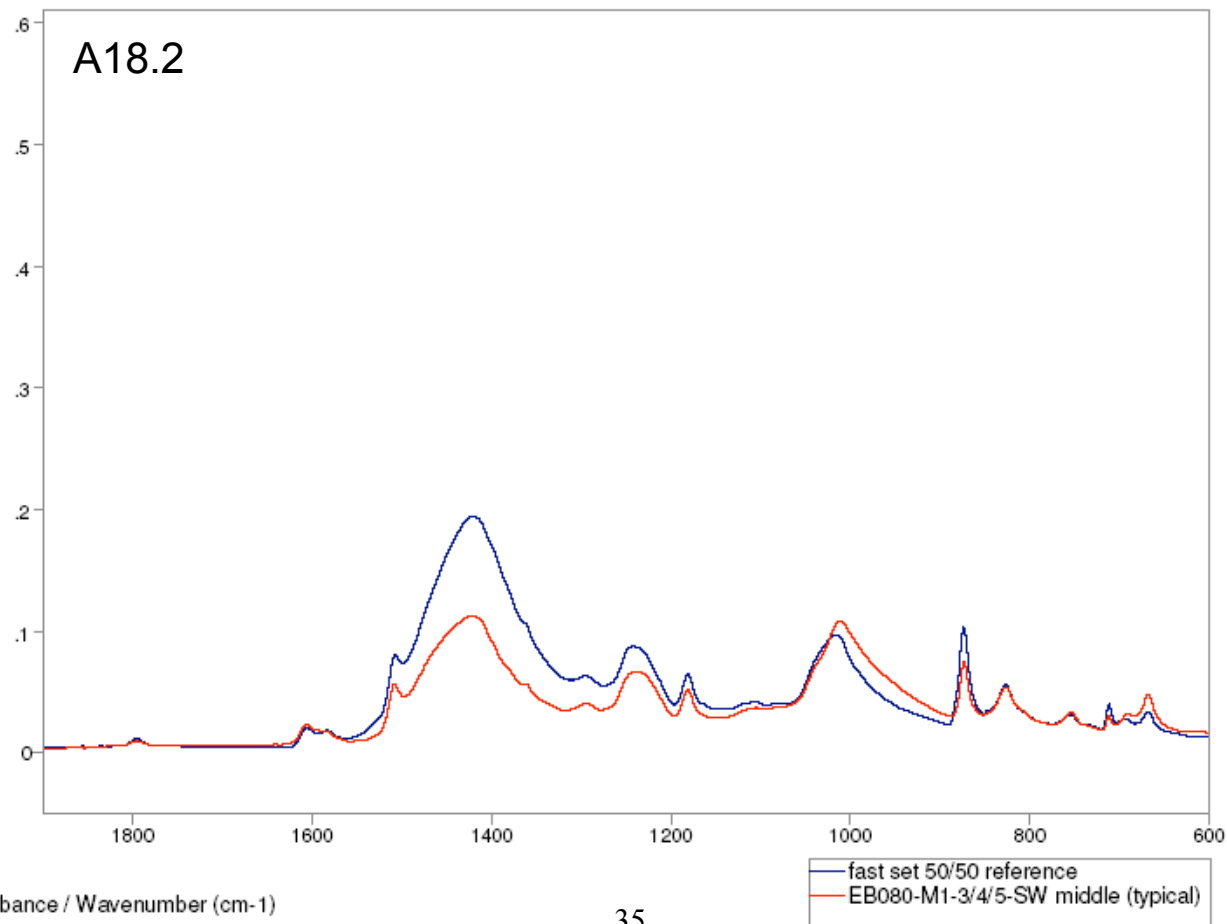
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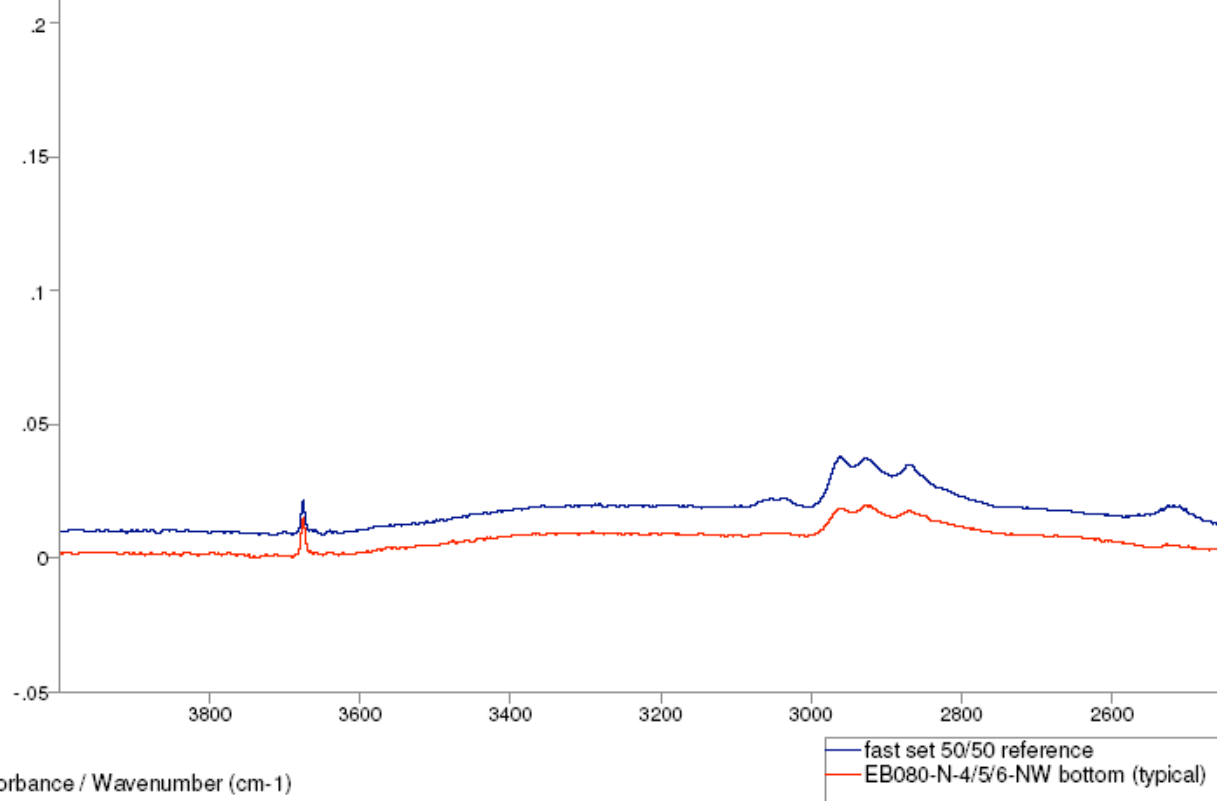
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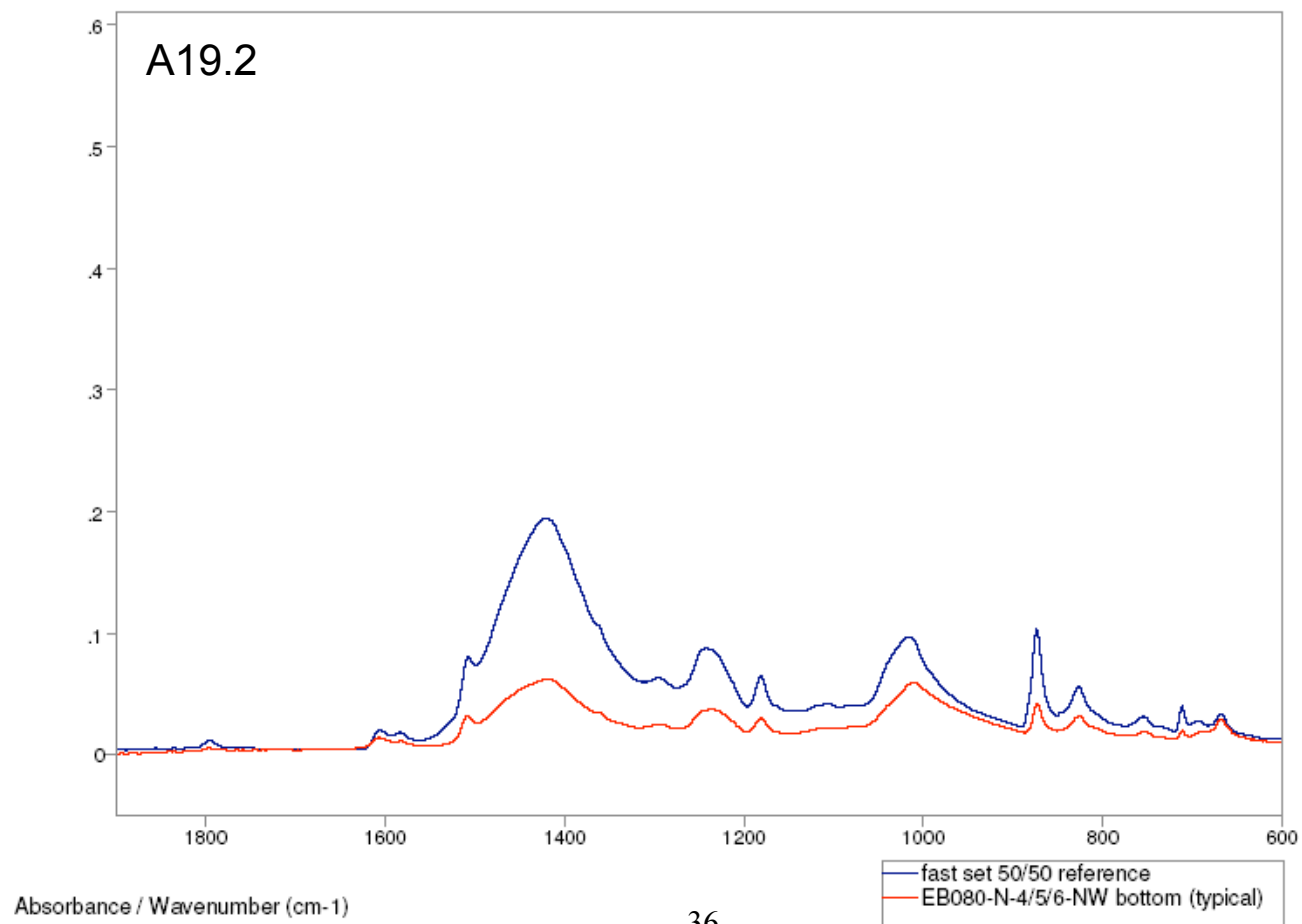
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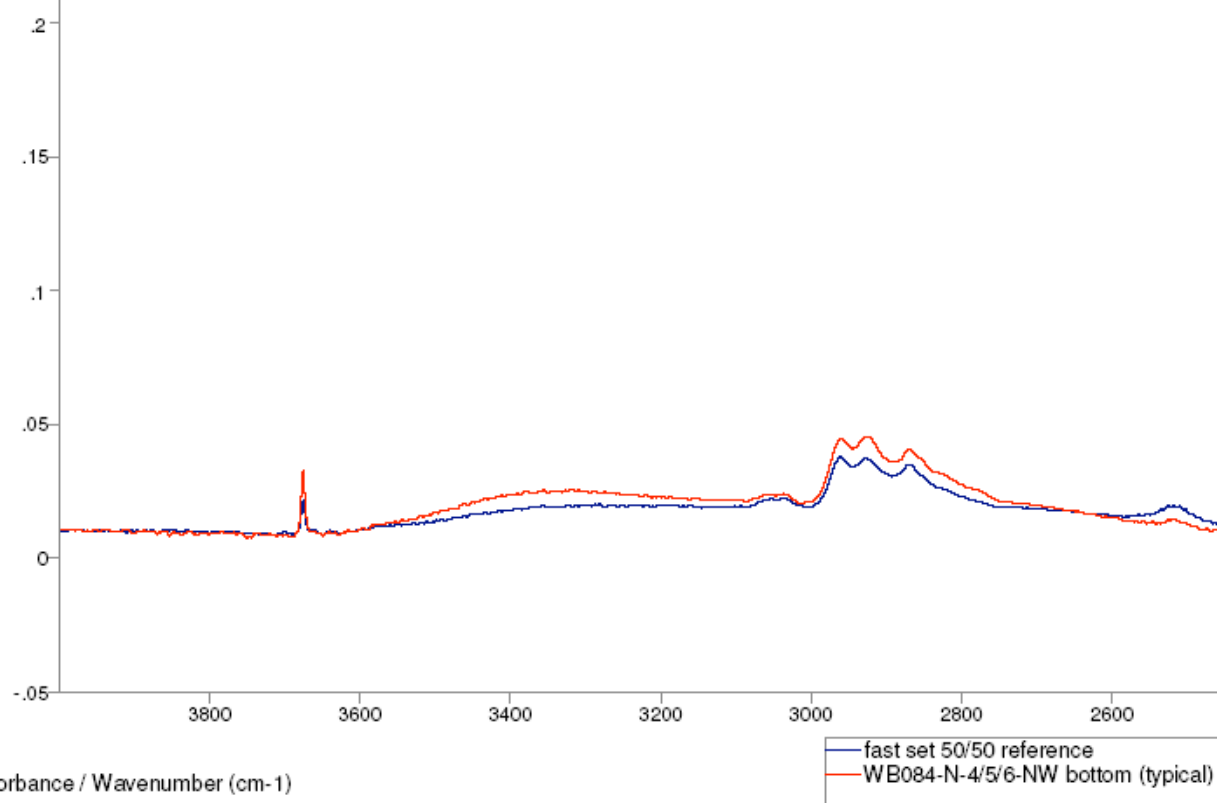
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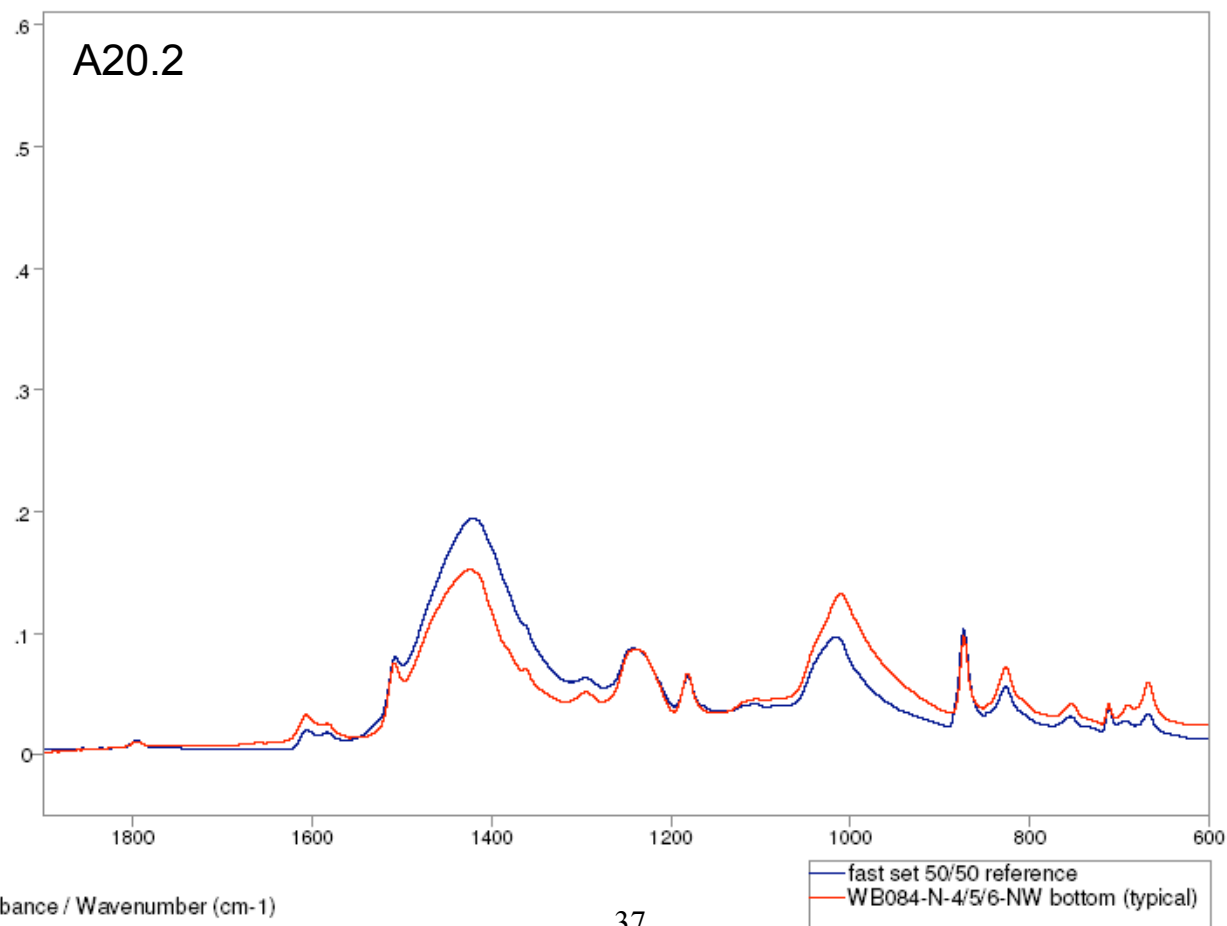
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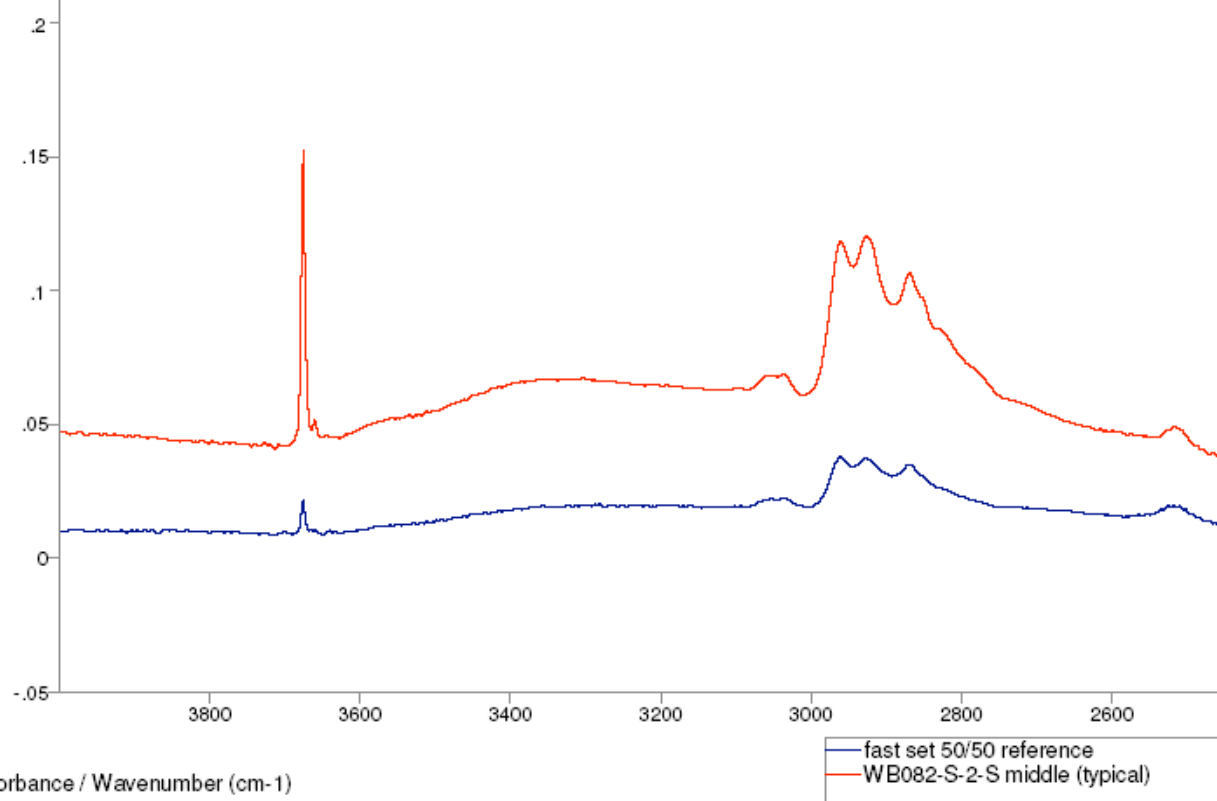
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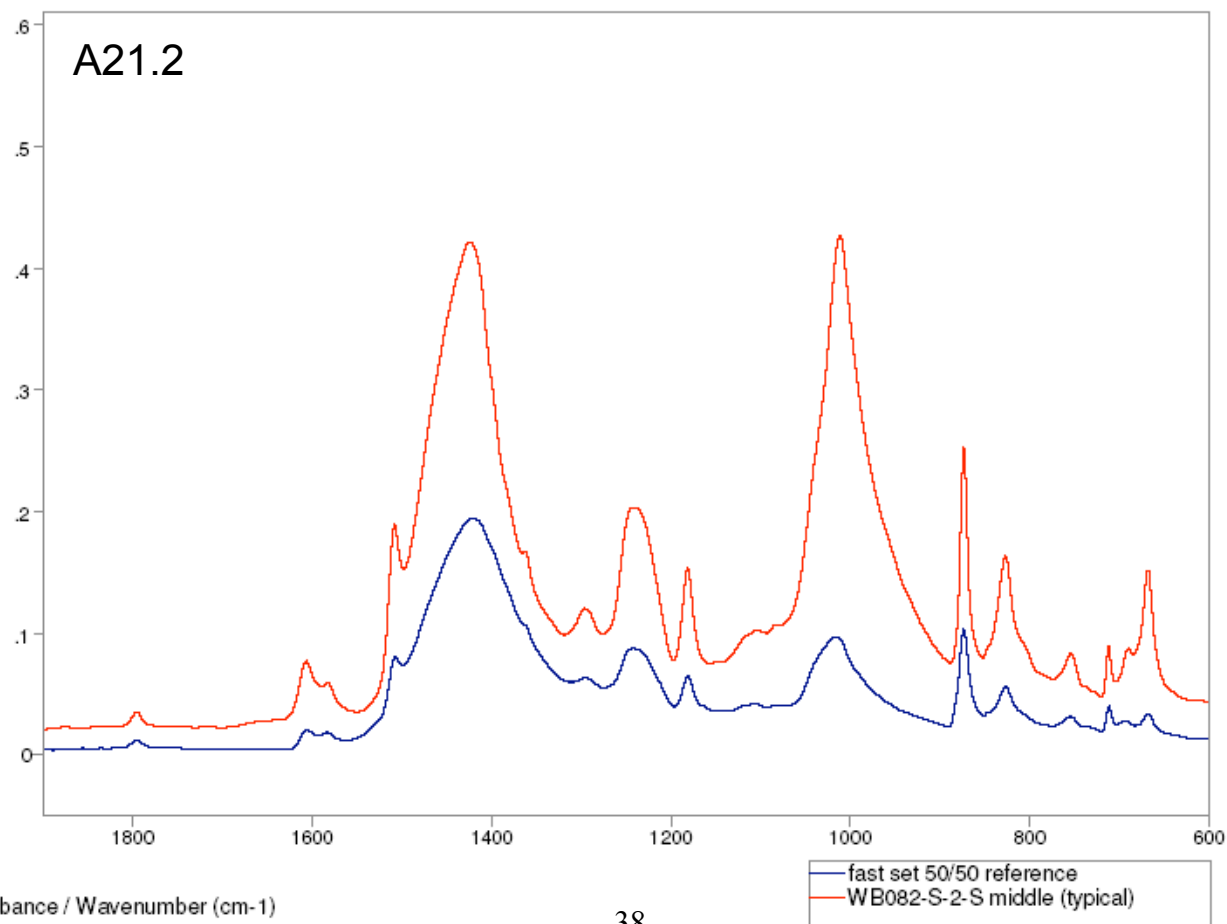
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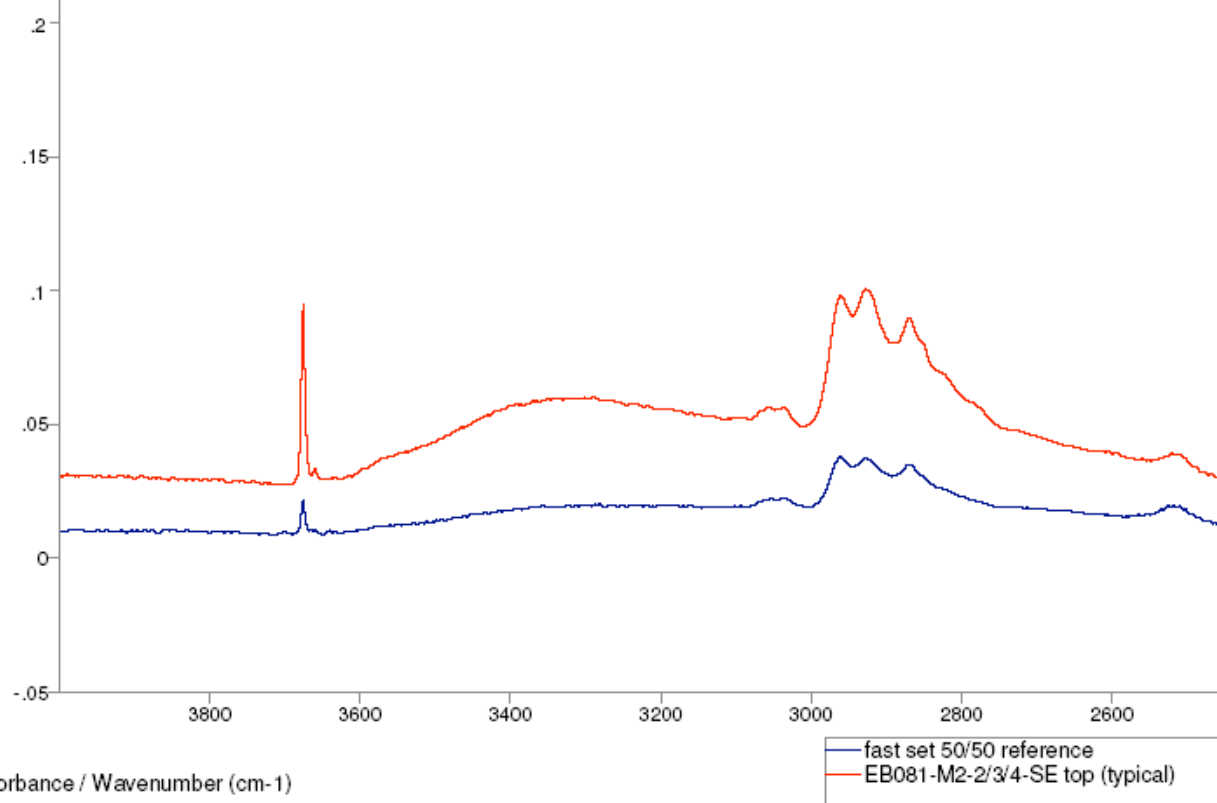
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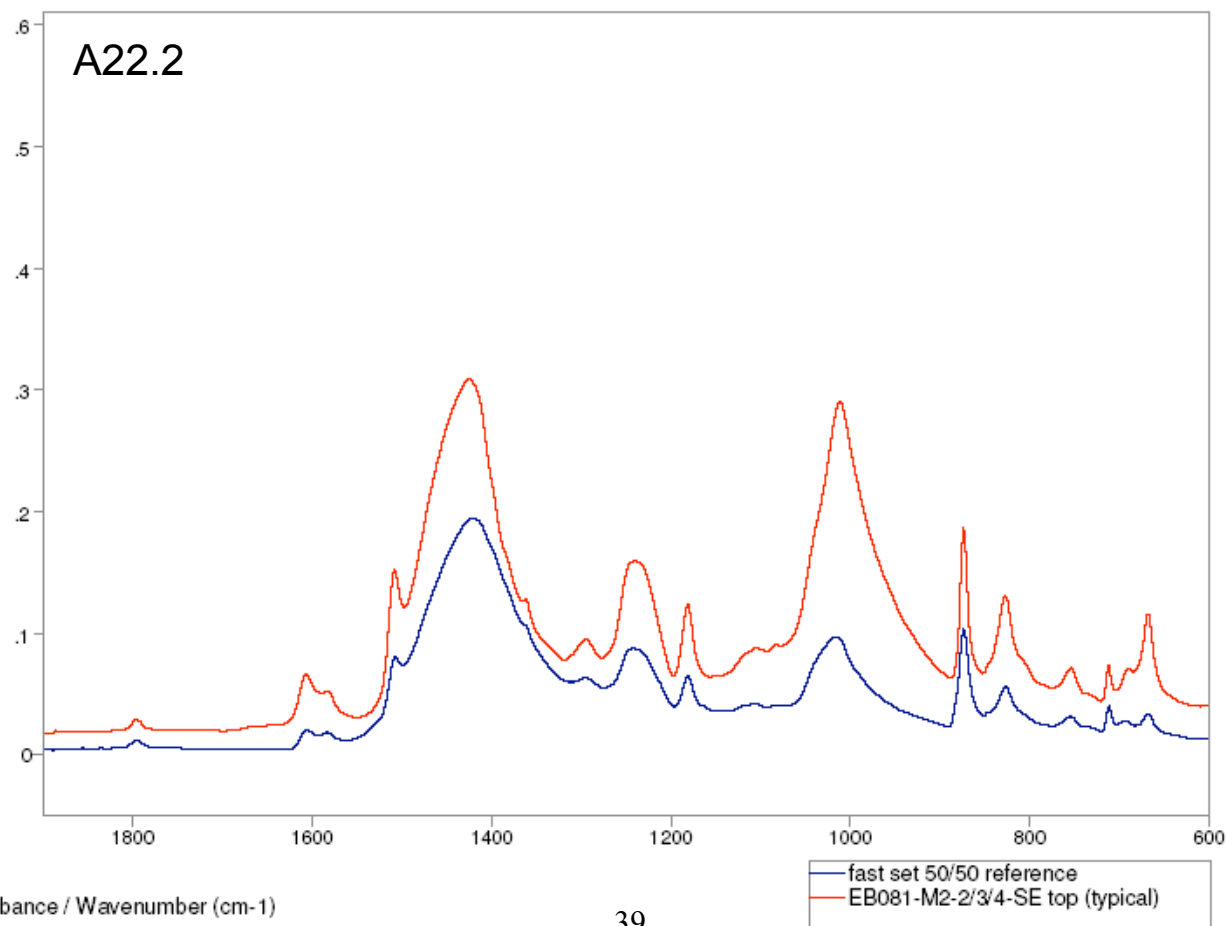
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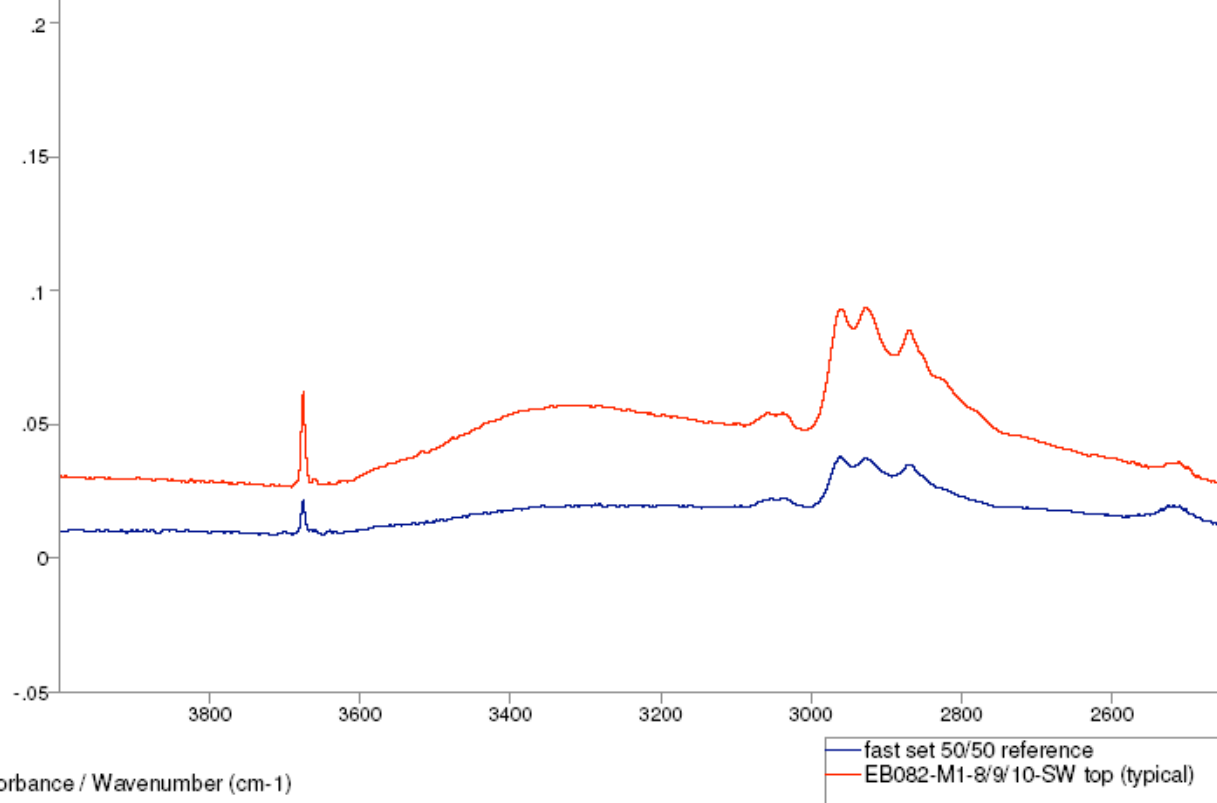
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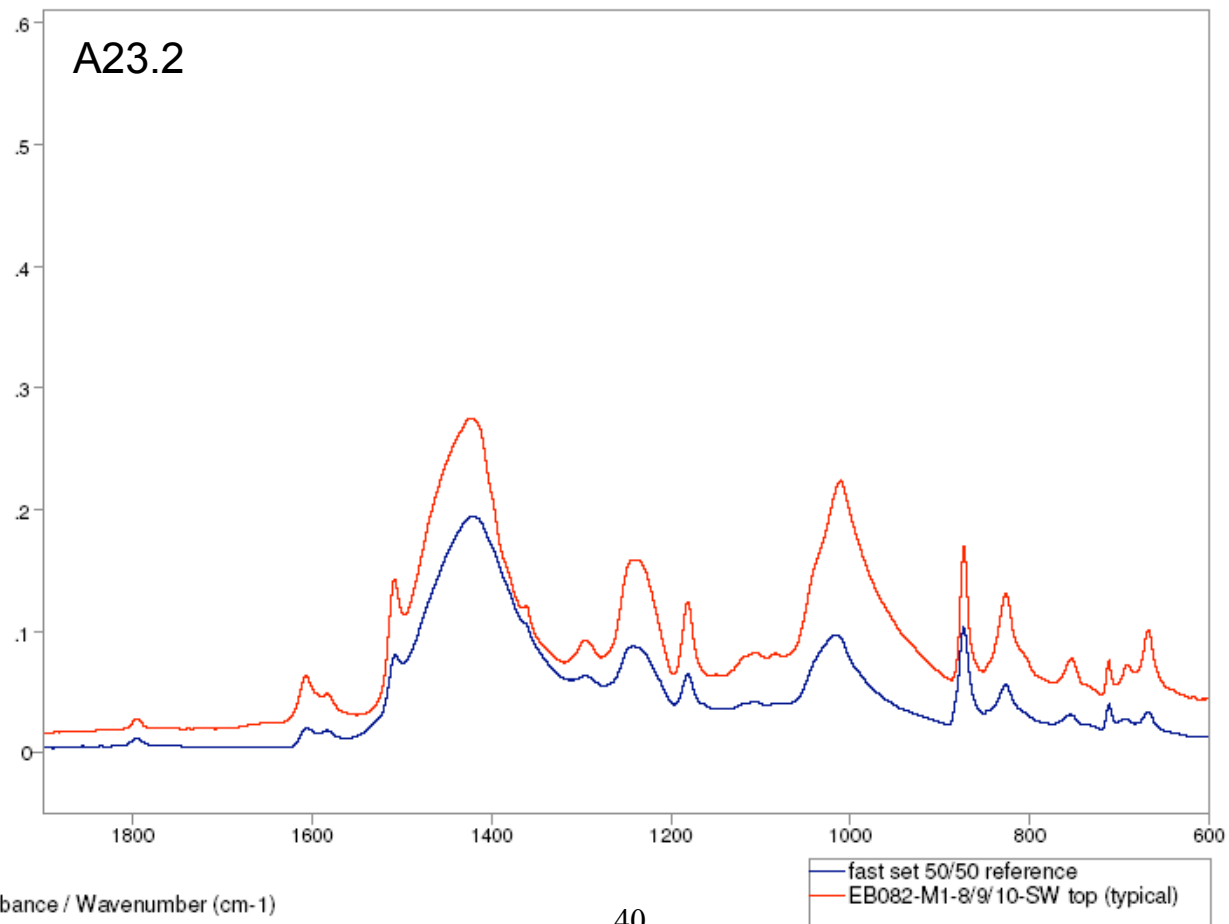
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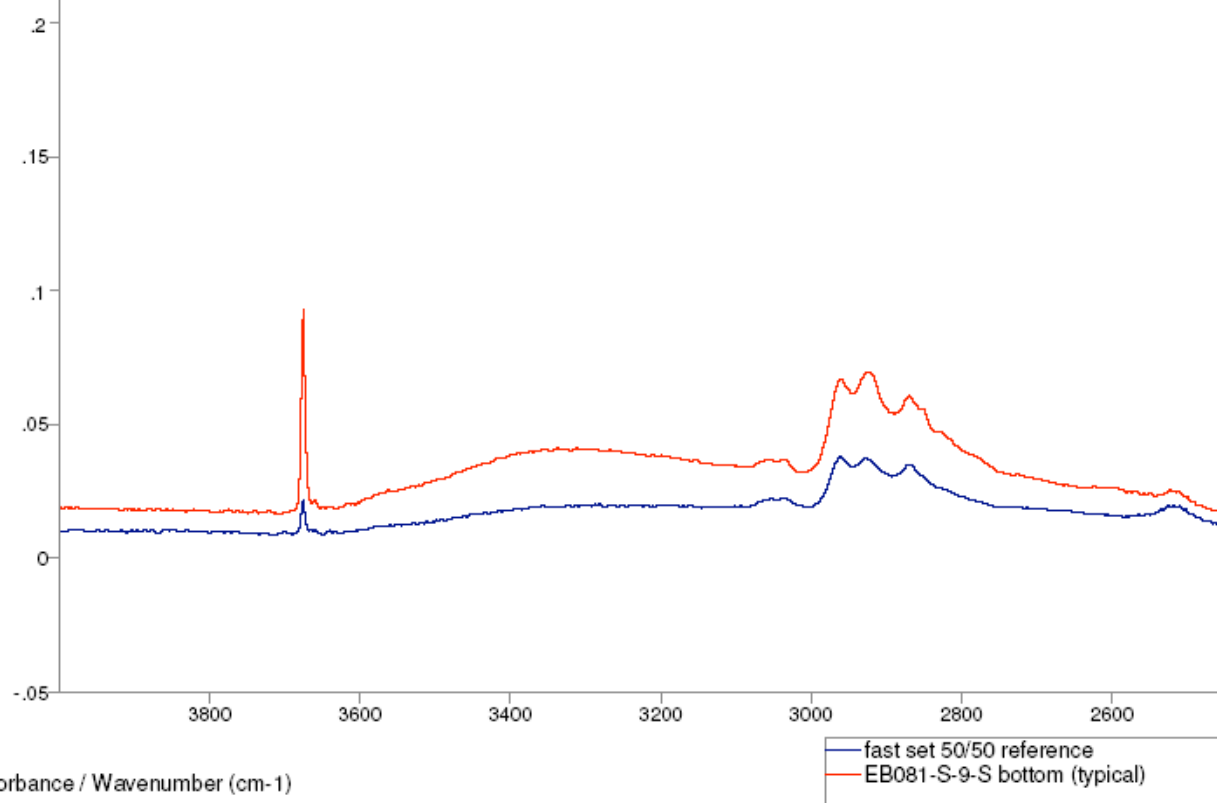
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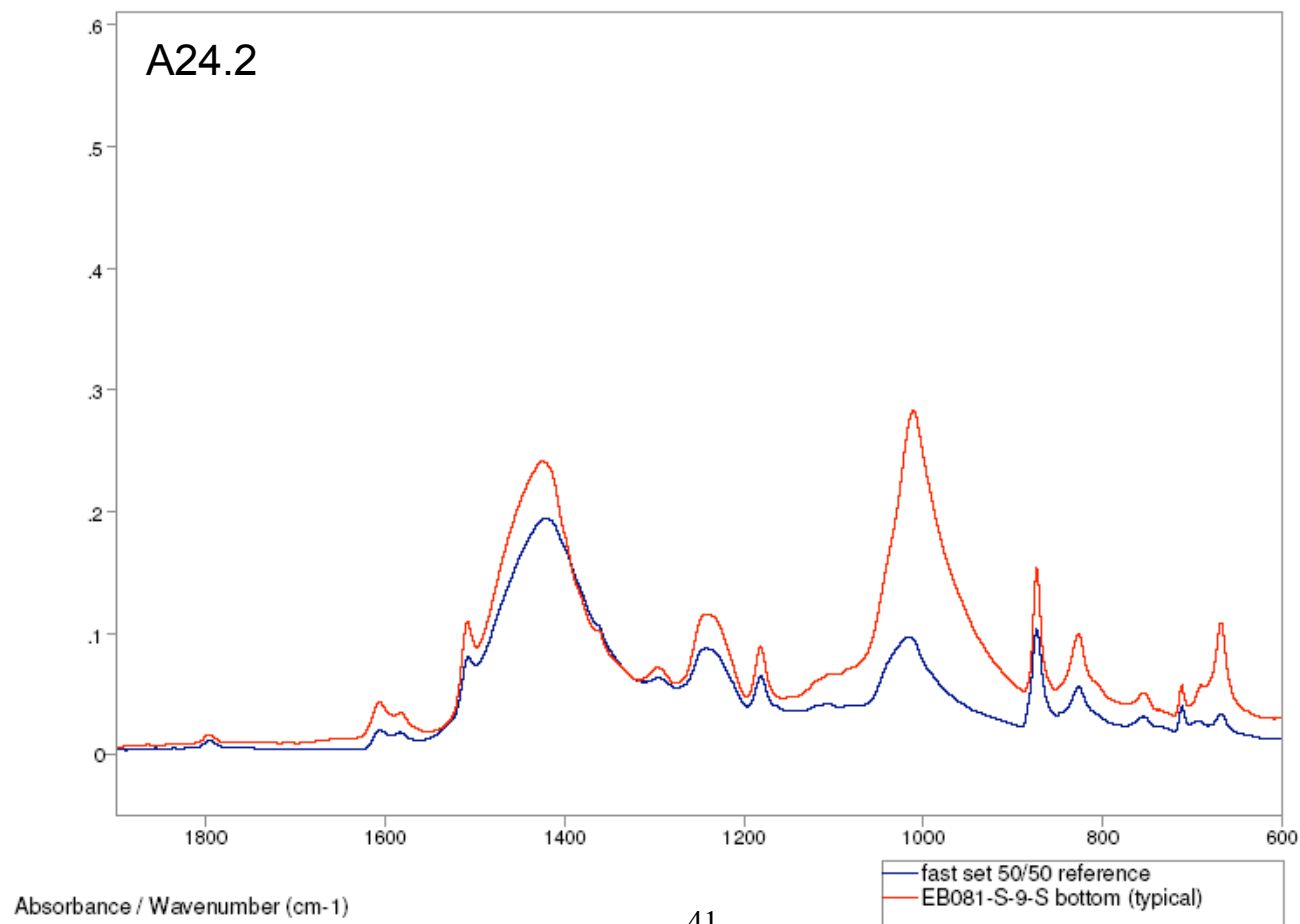
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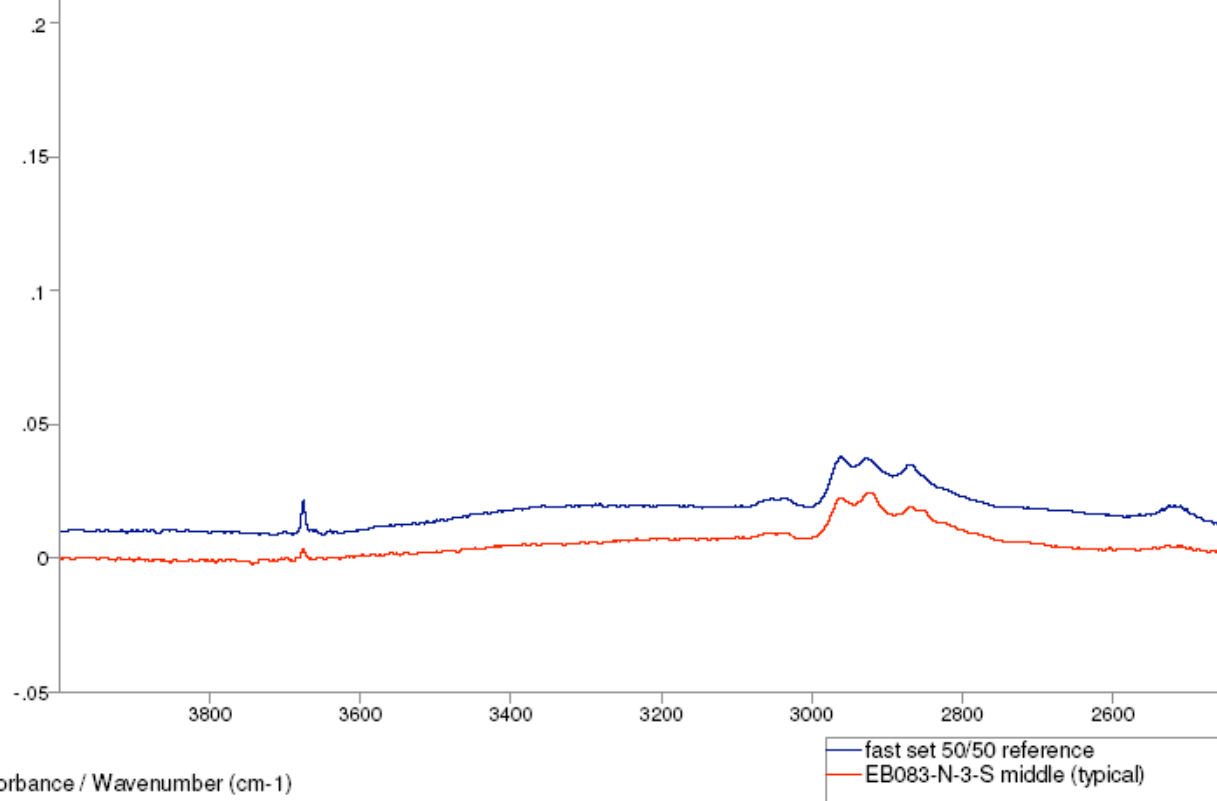
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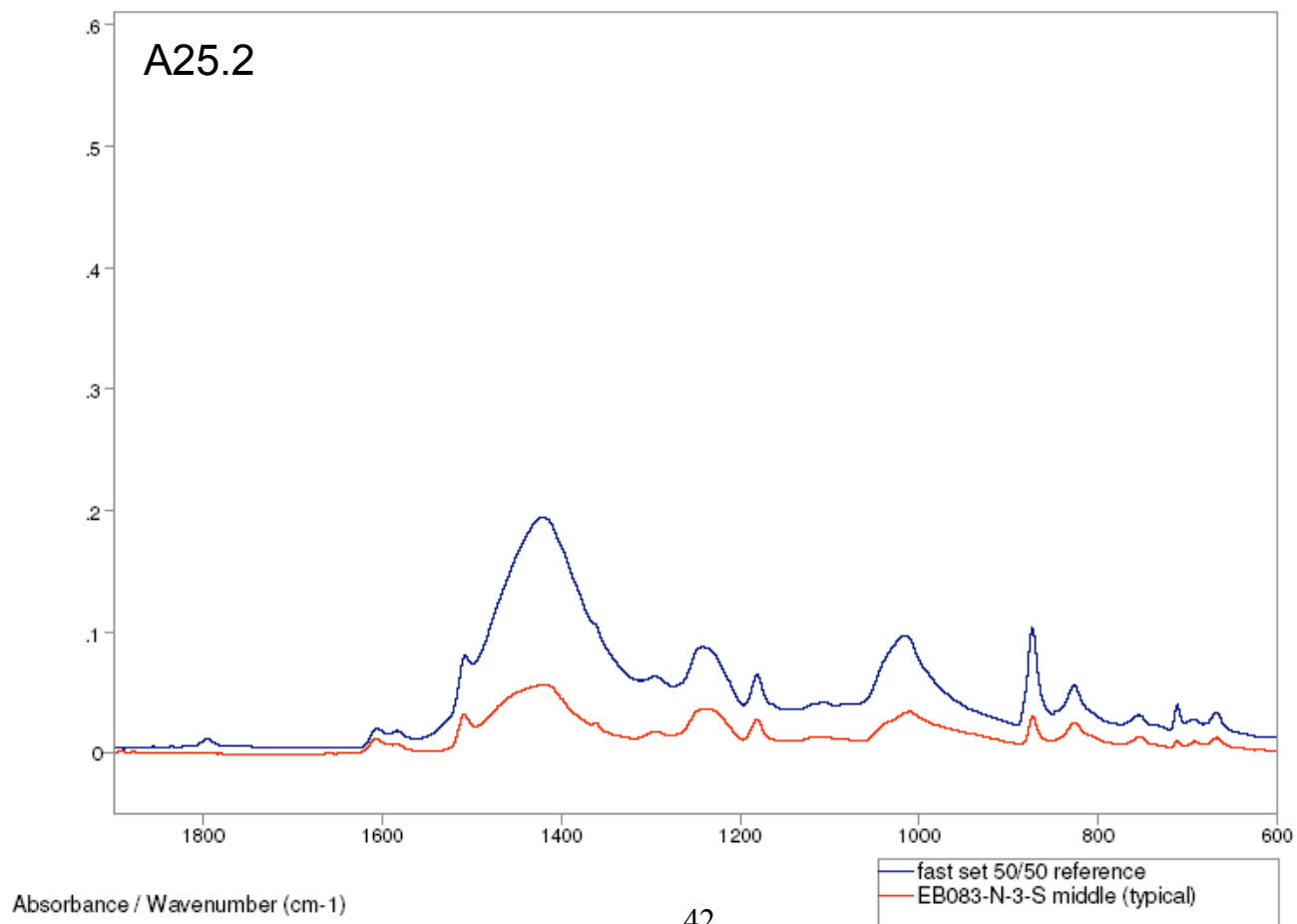
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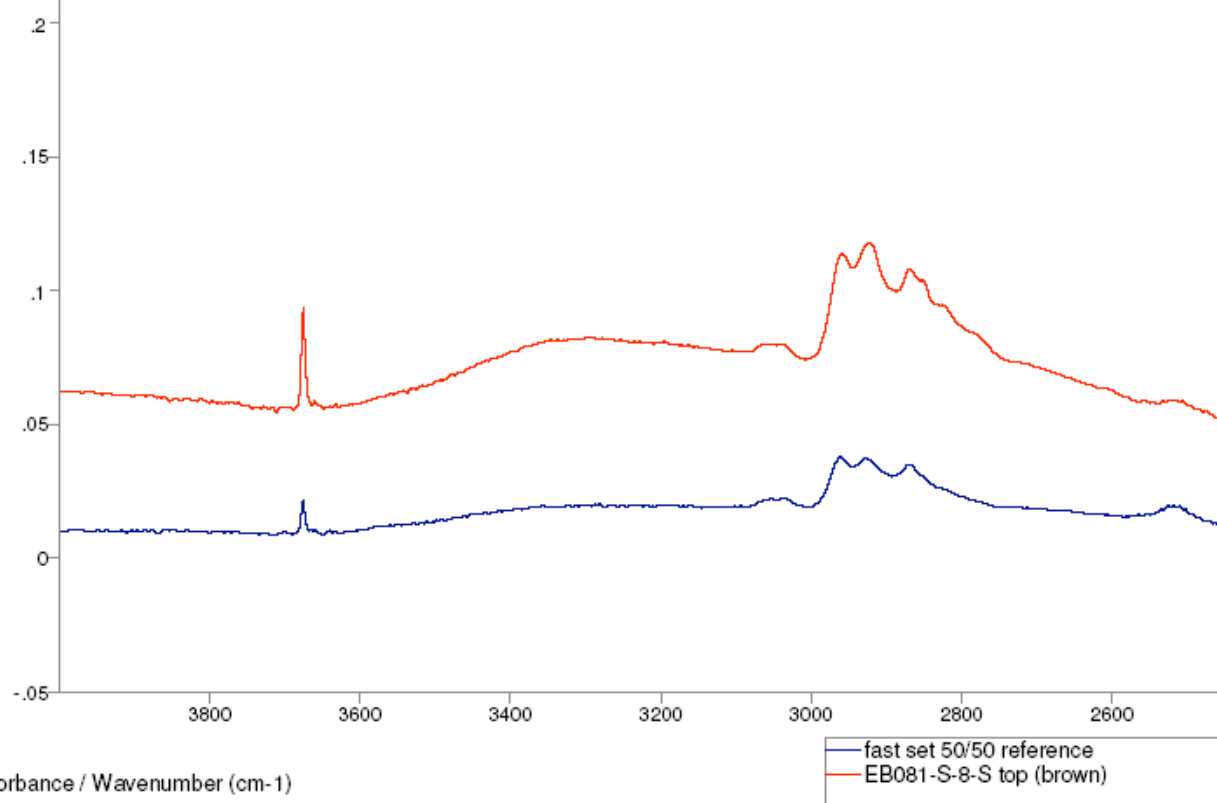
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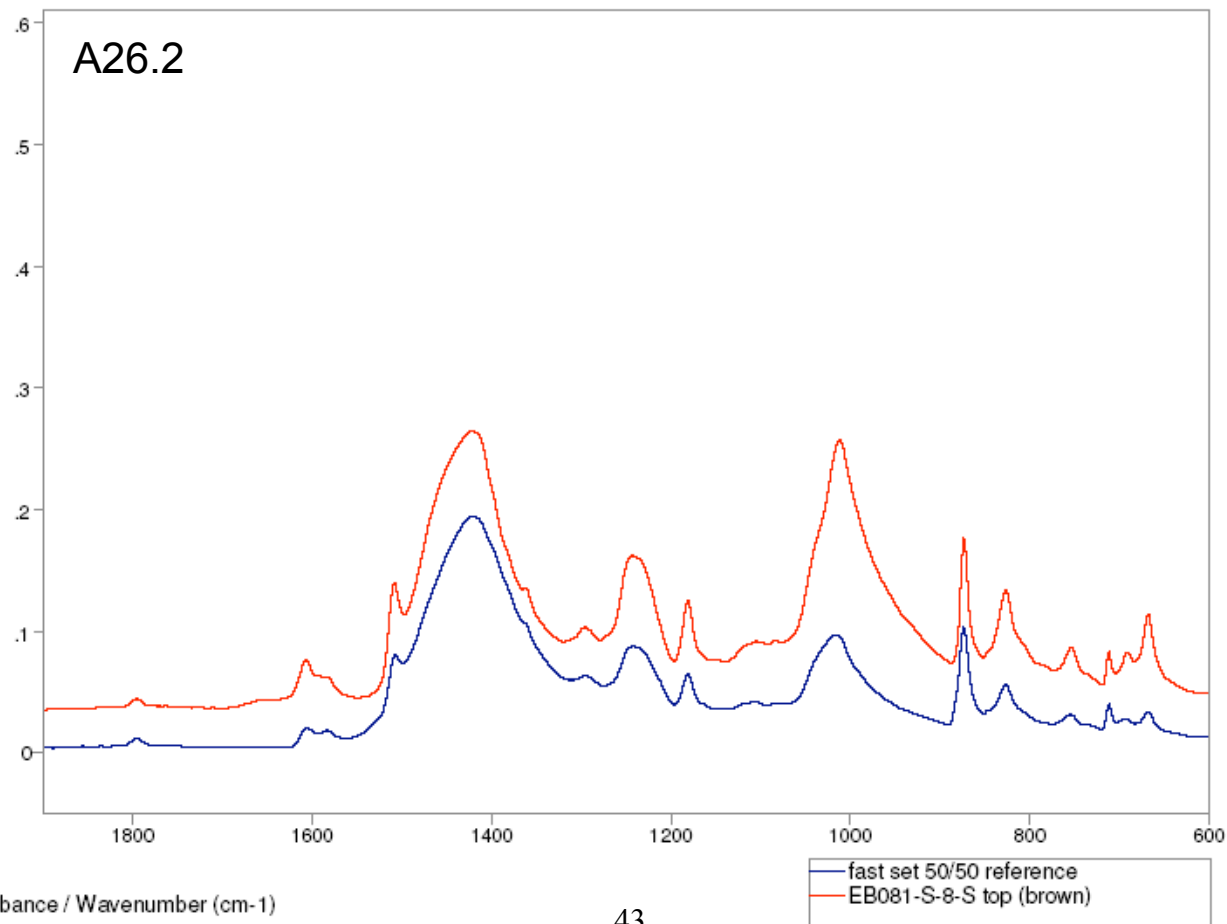
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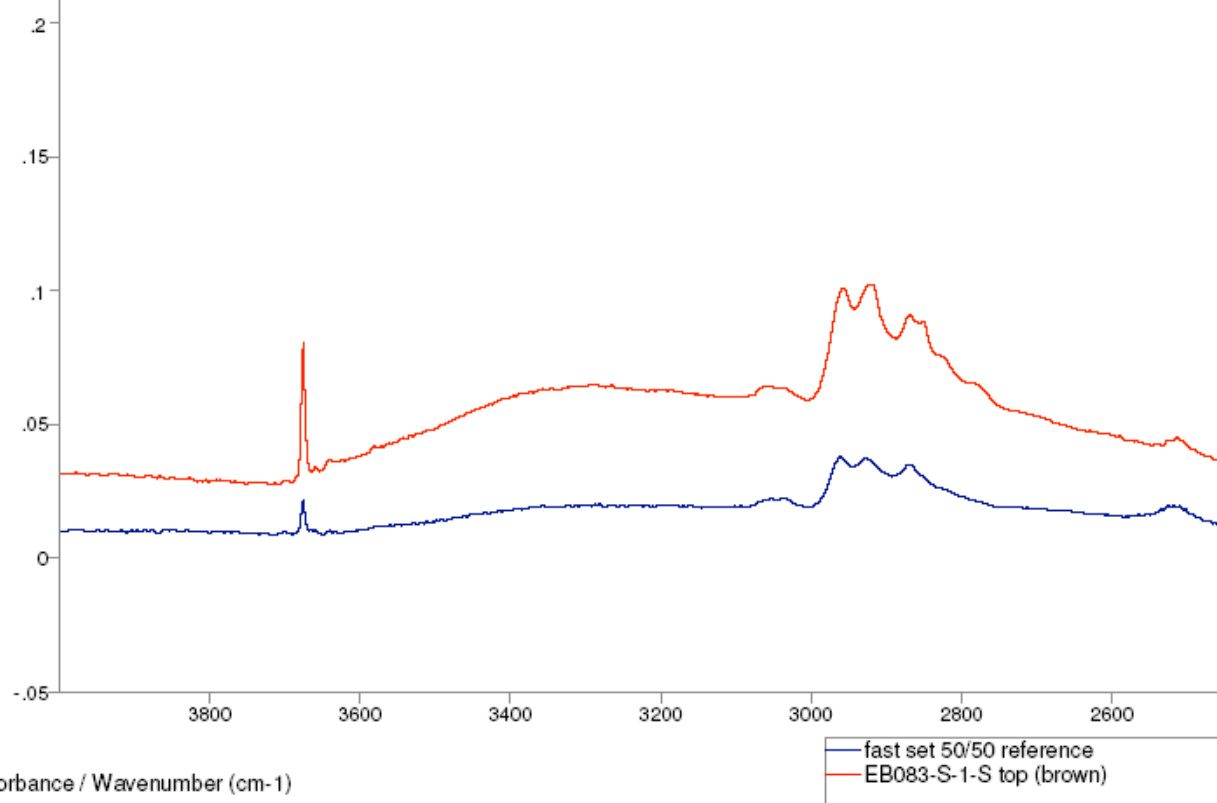
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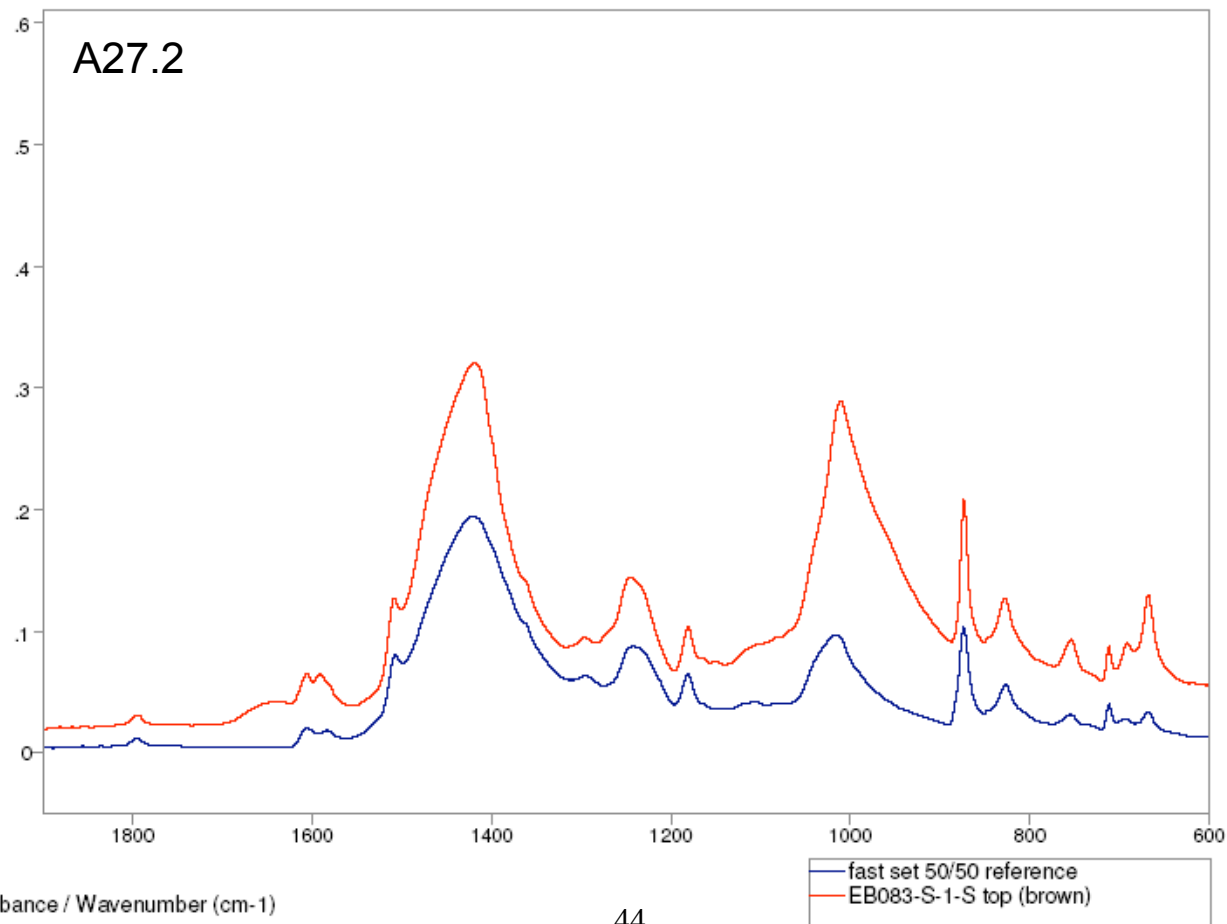
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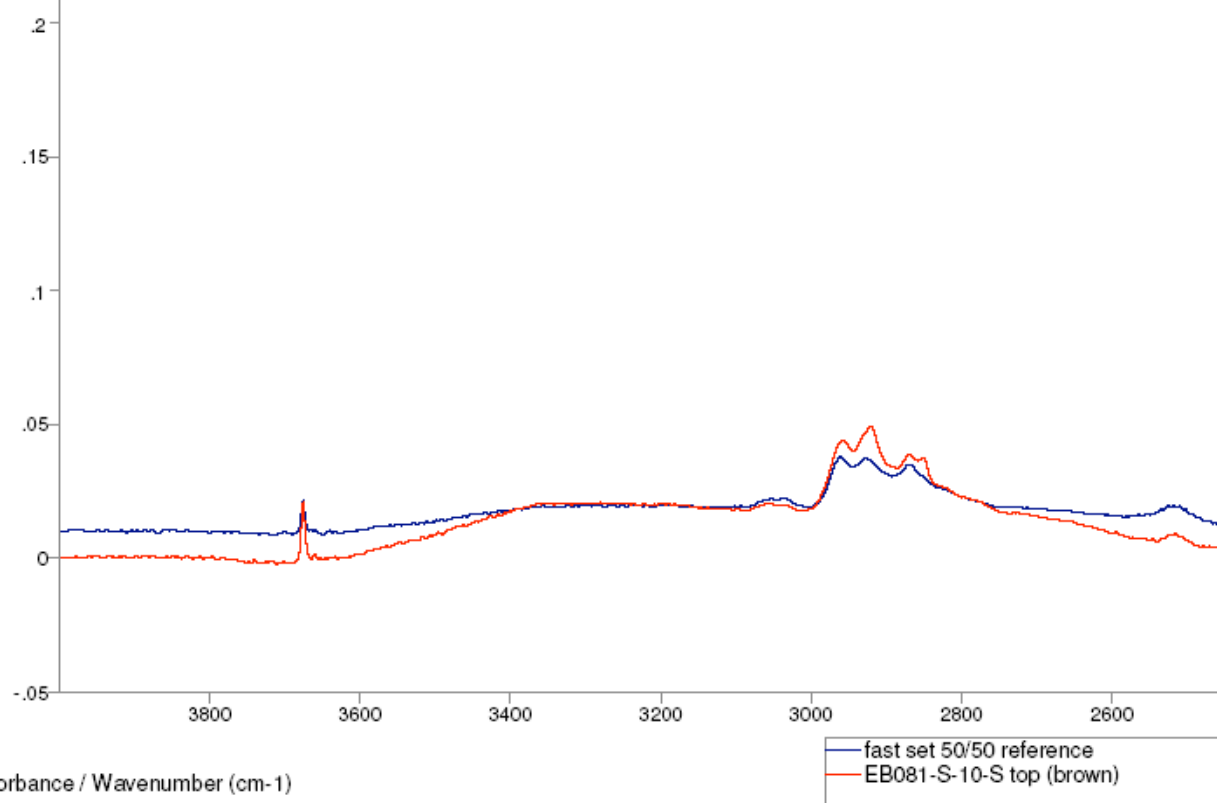
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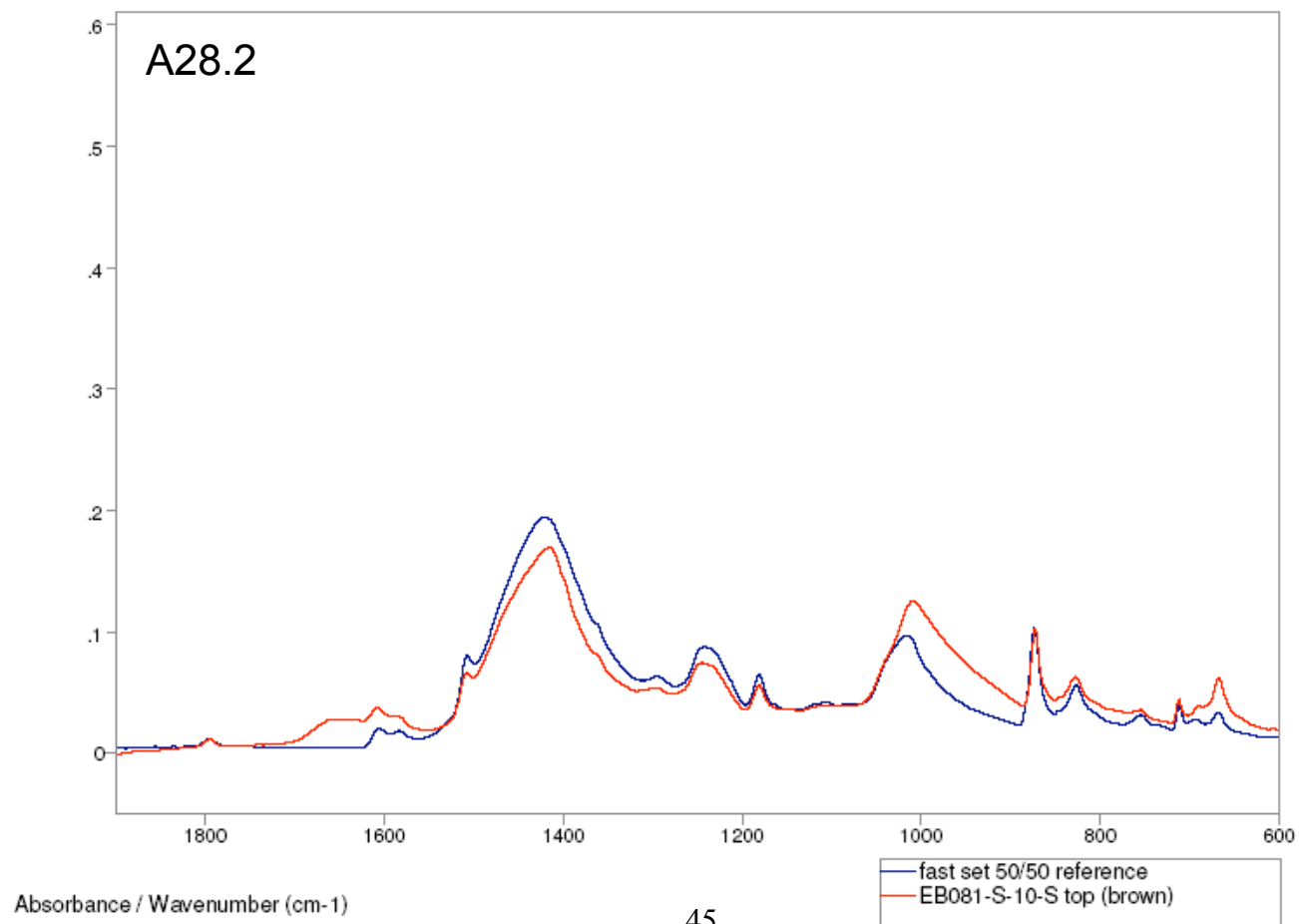
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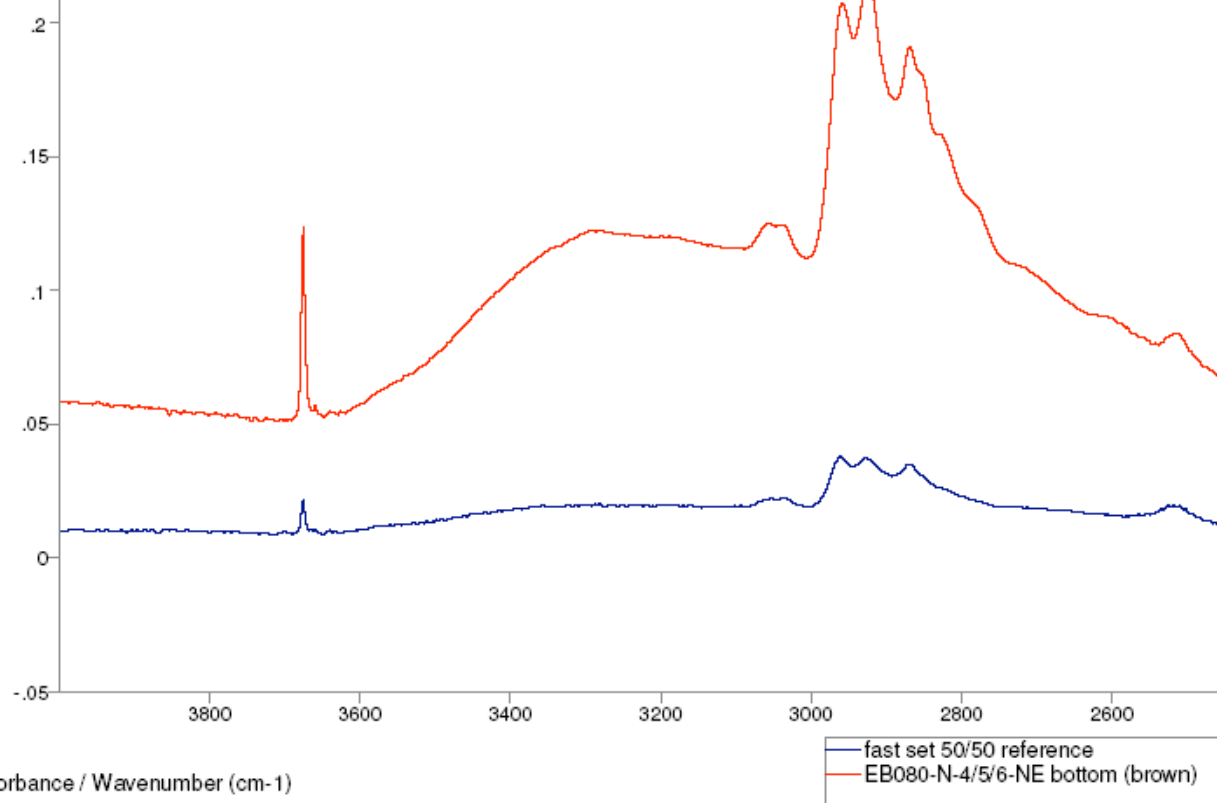
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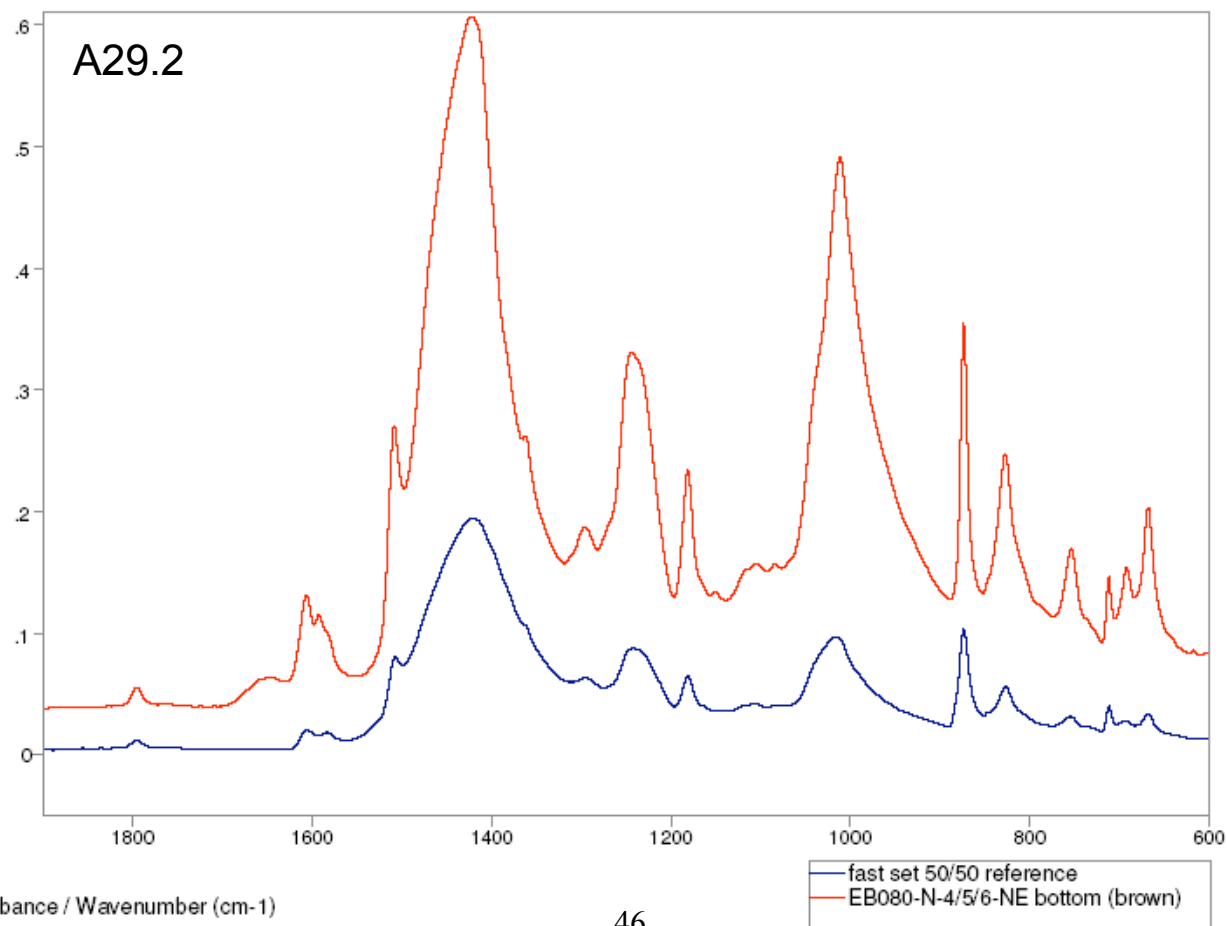
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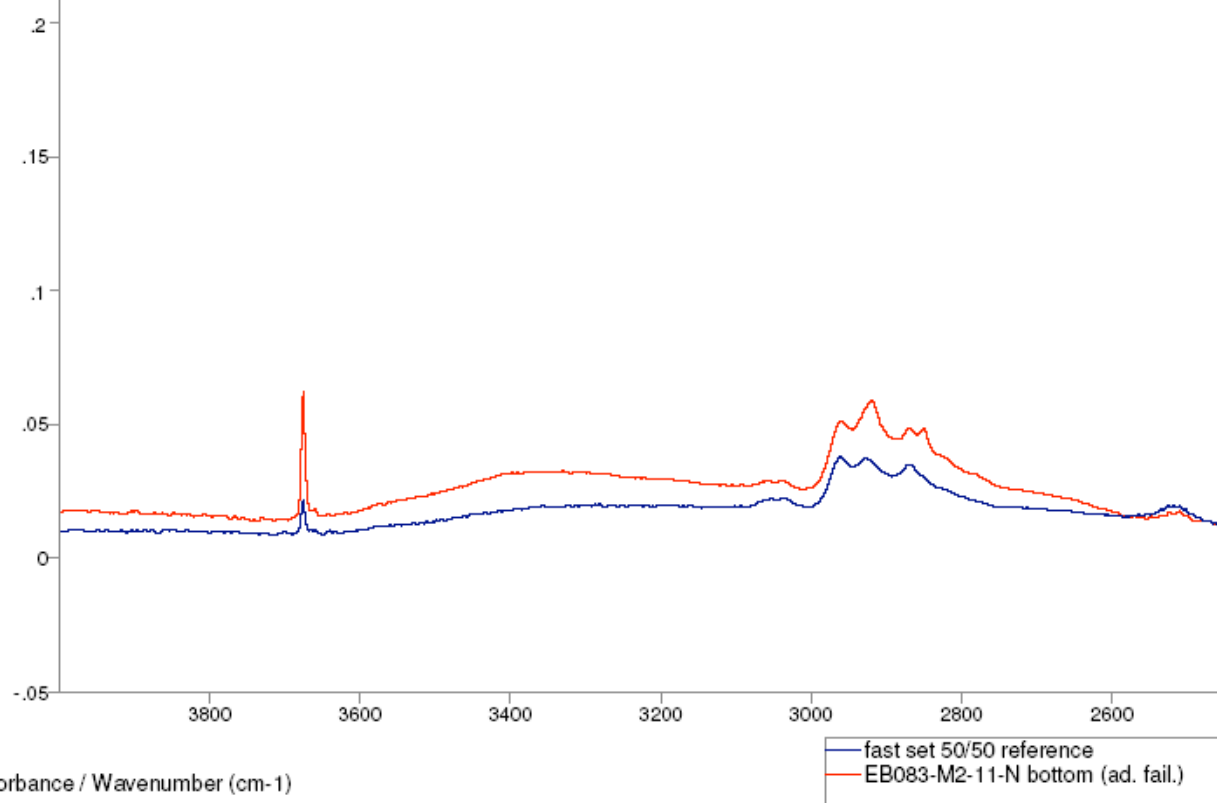
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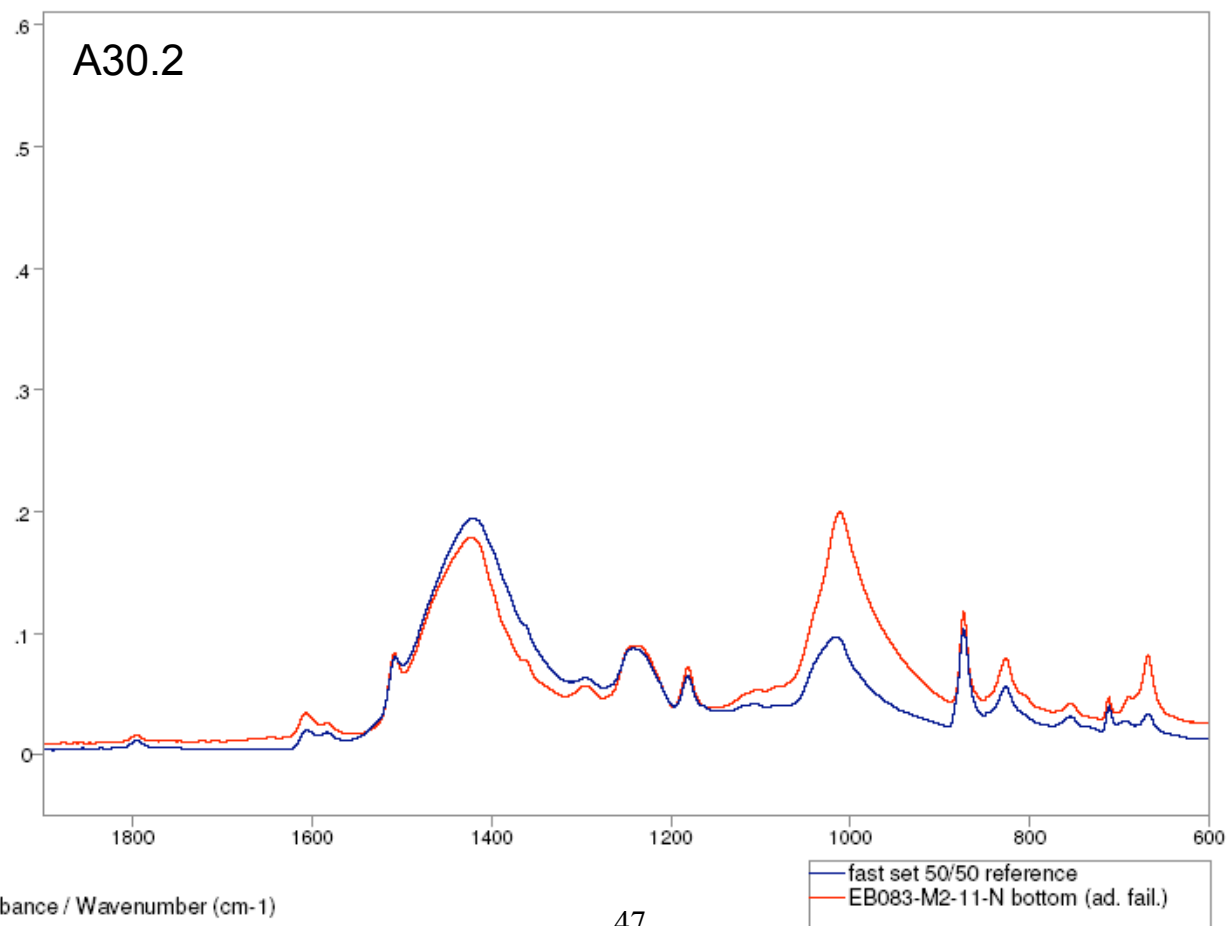
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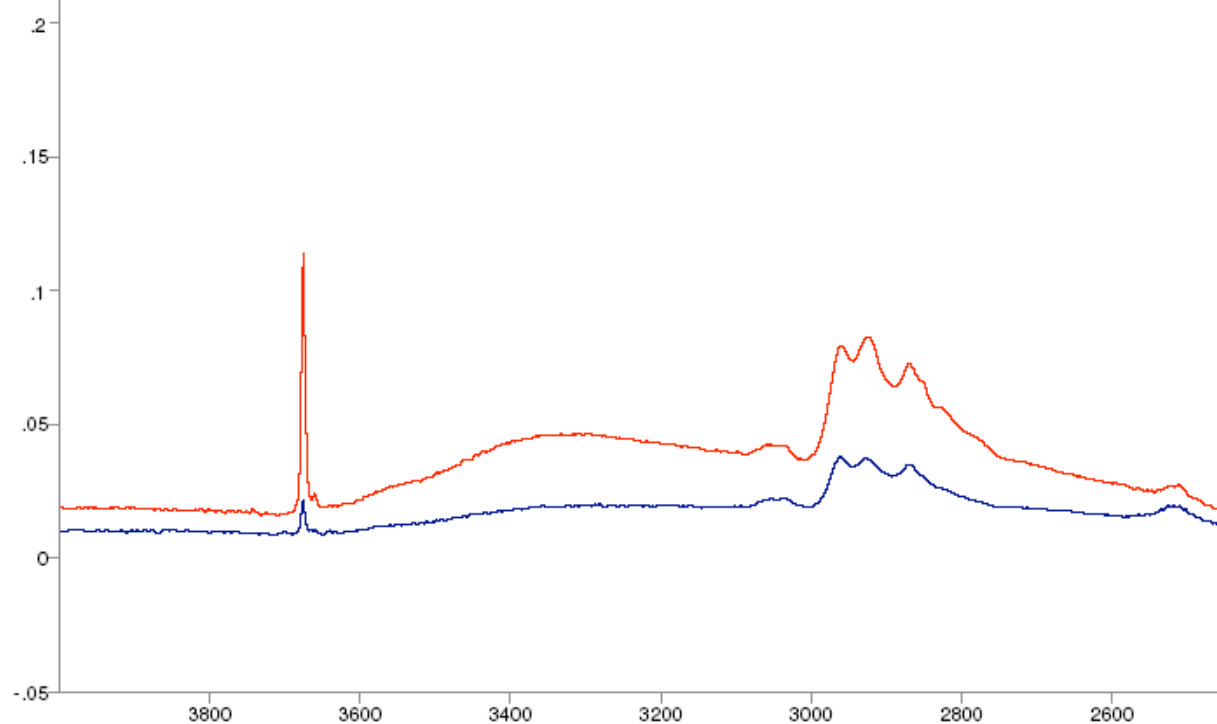
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A30.2



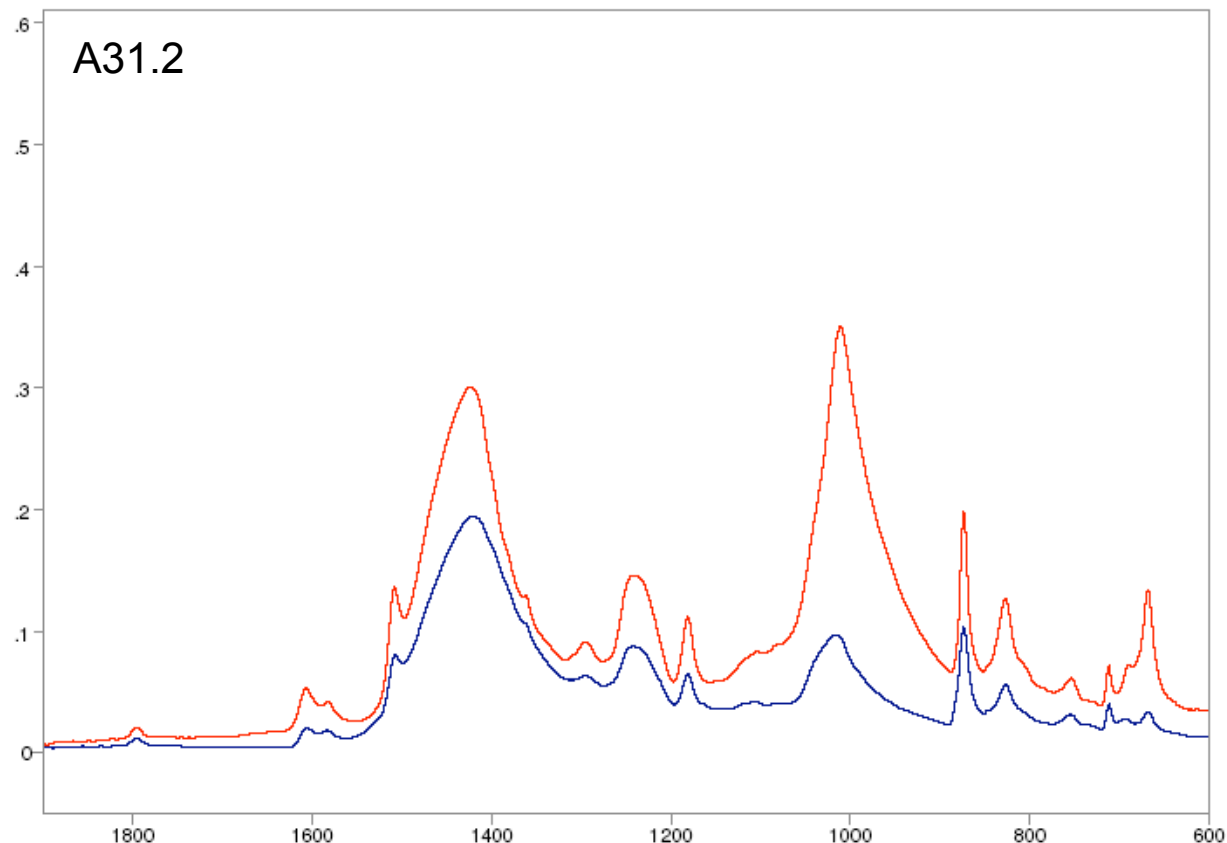
A31.1



Absorbance / Wavenumber (cm-1)

fast set 50/50 reference
EB083-M2-7/8/9-SW bottom (ad. fail.)

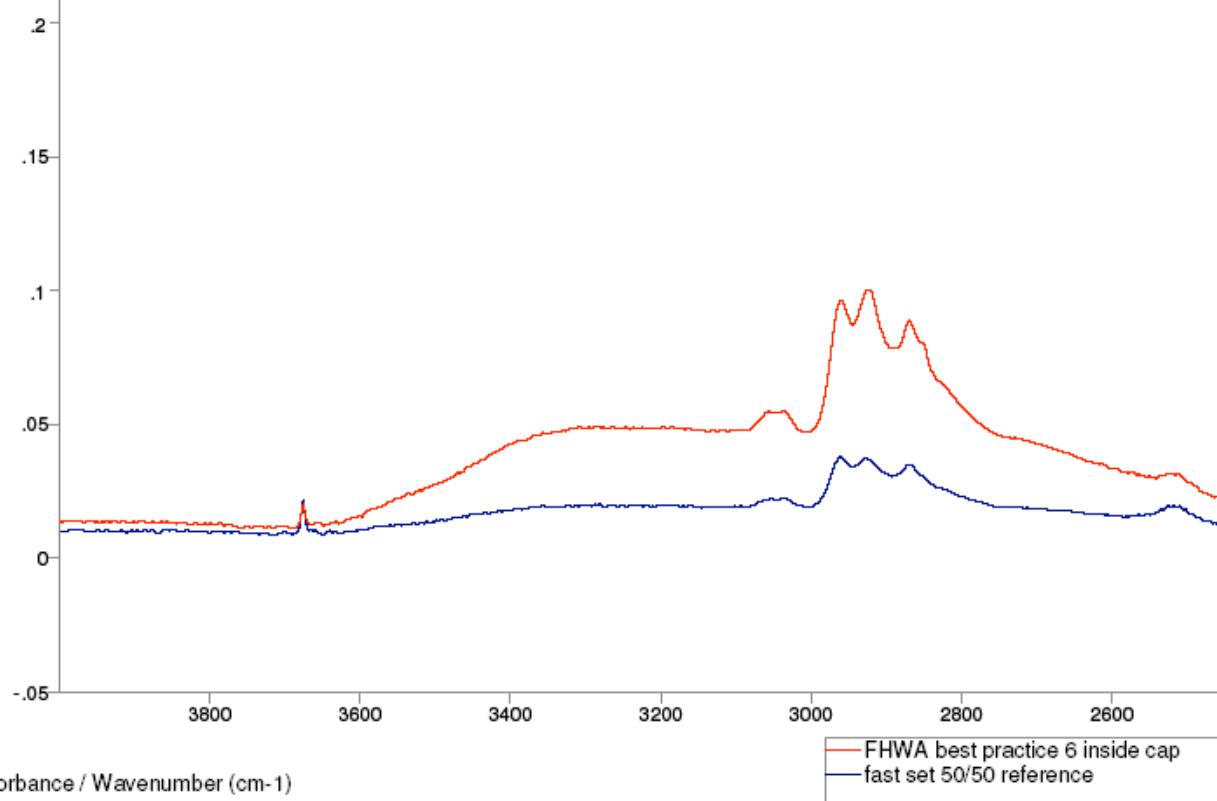
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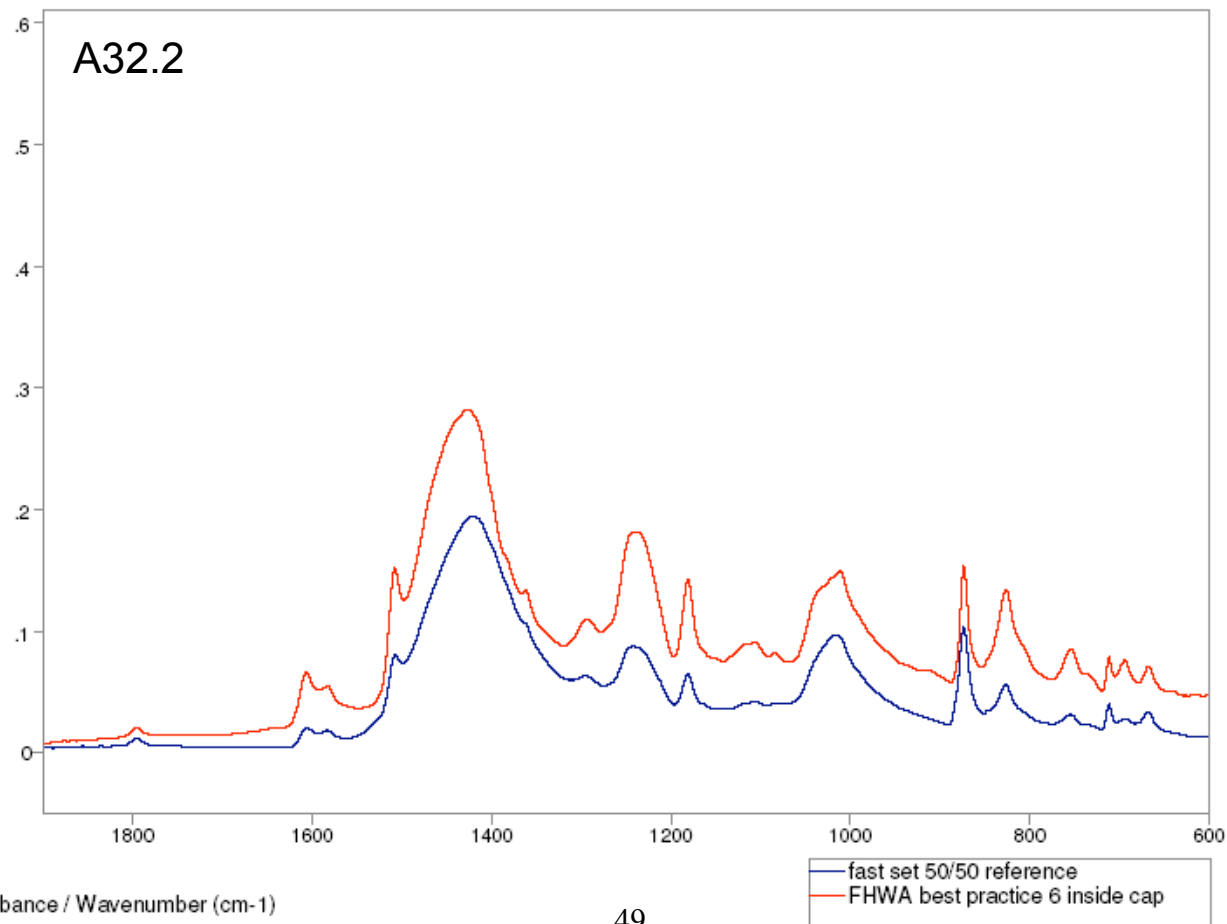
Absorbance / Wavenumber (cm-1)

fast set 50/50 reference
EB083-M2-7/8/9-SW bottom (ad. fail.)

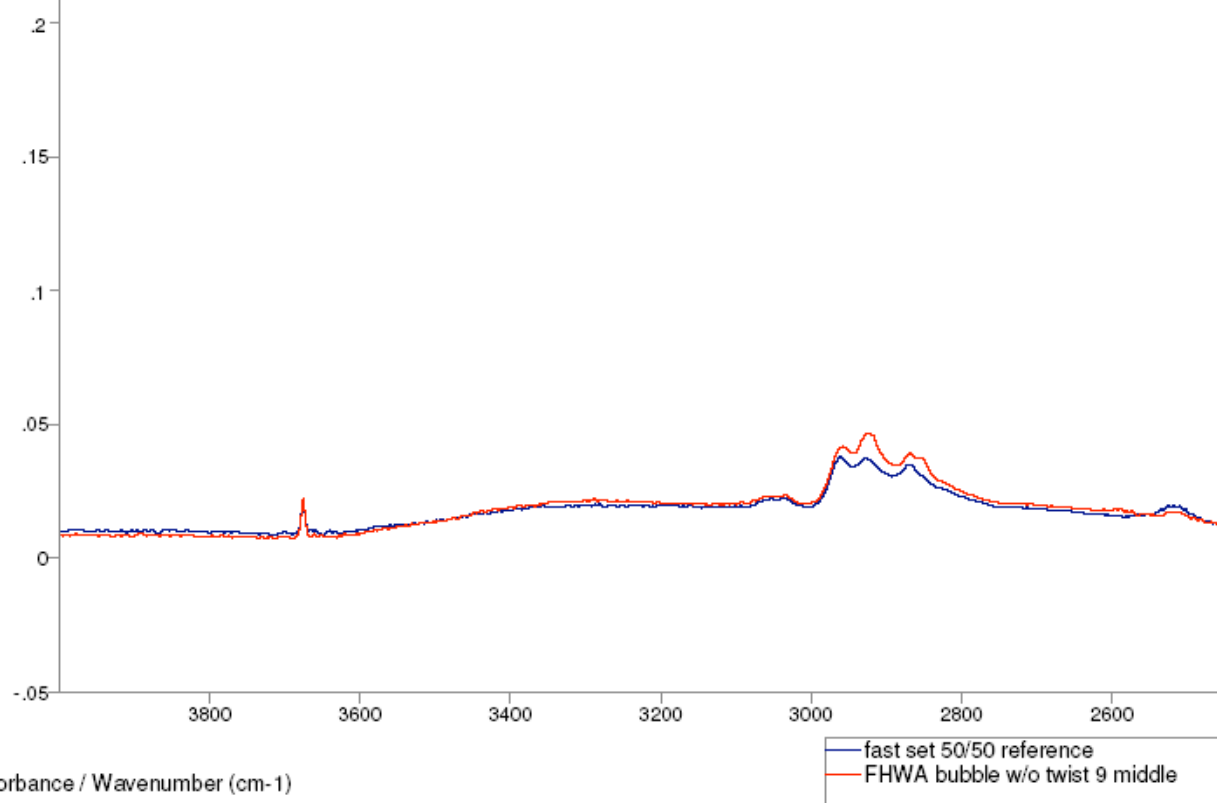
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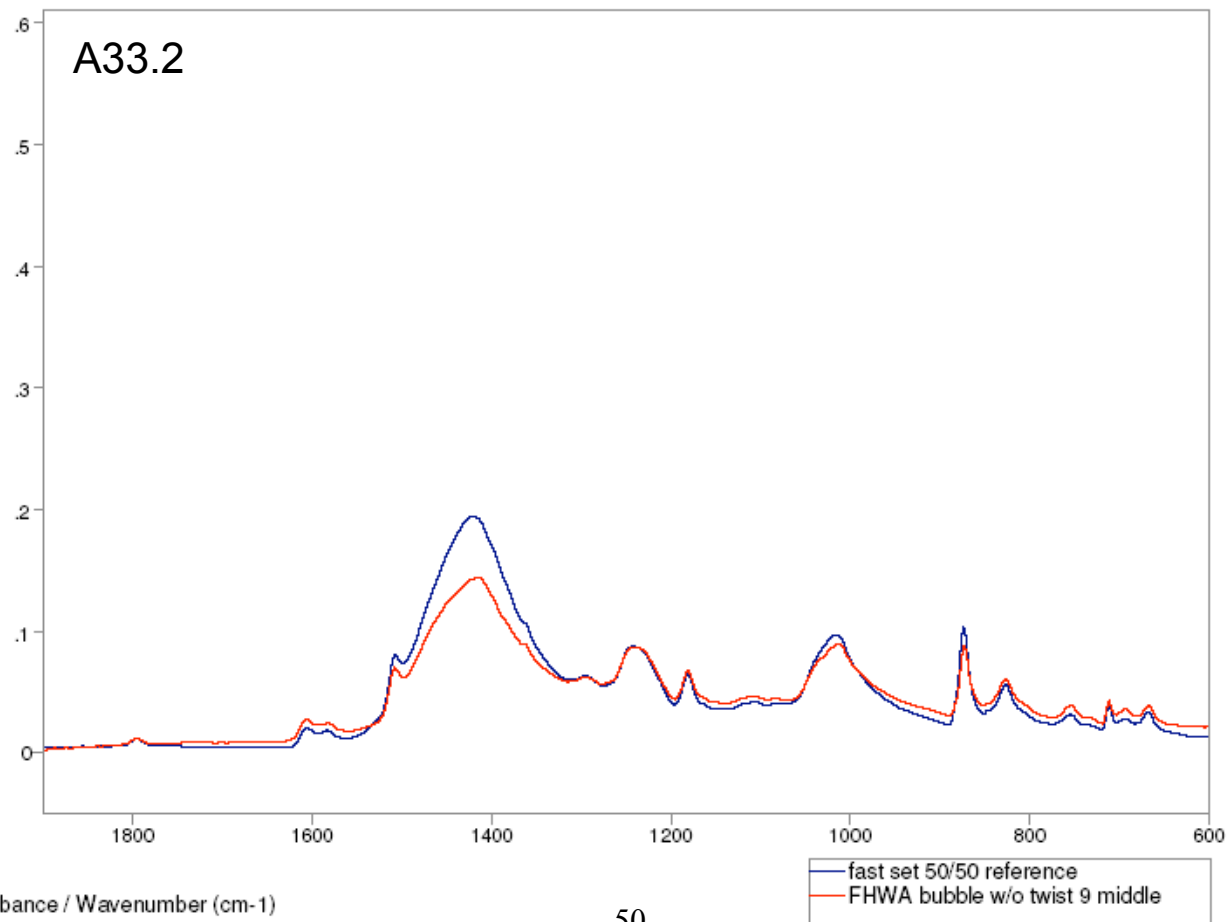
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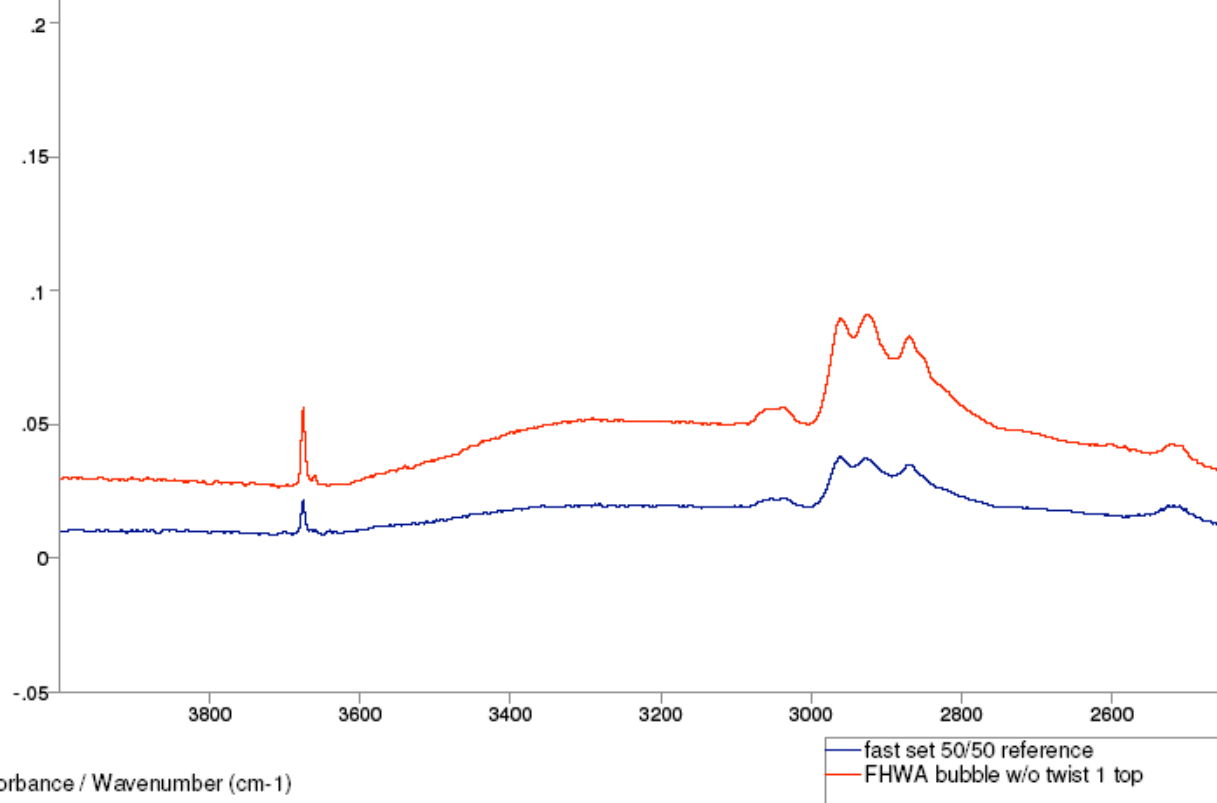
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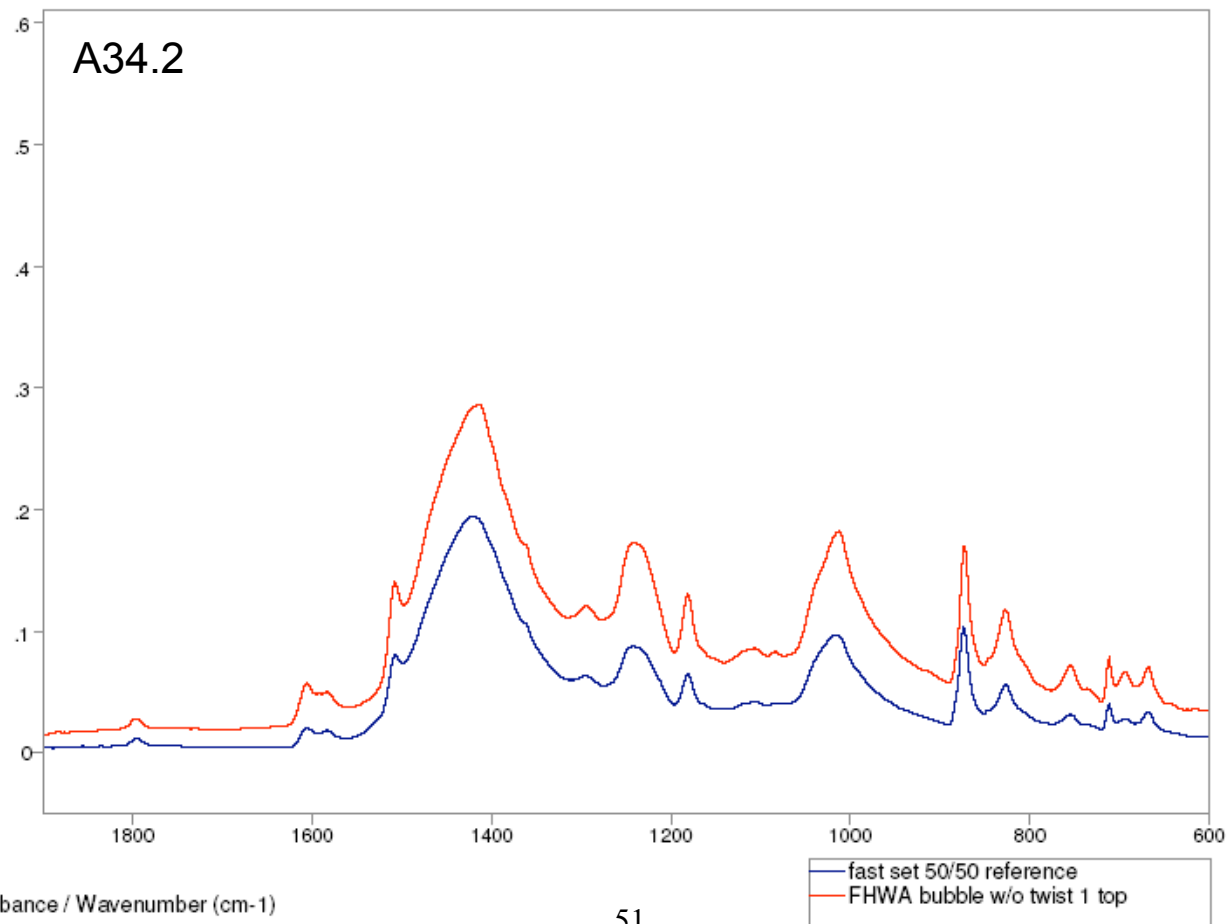
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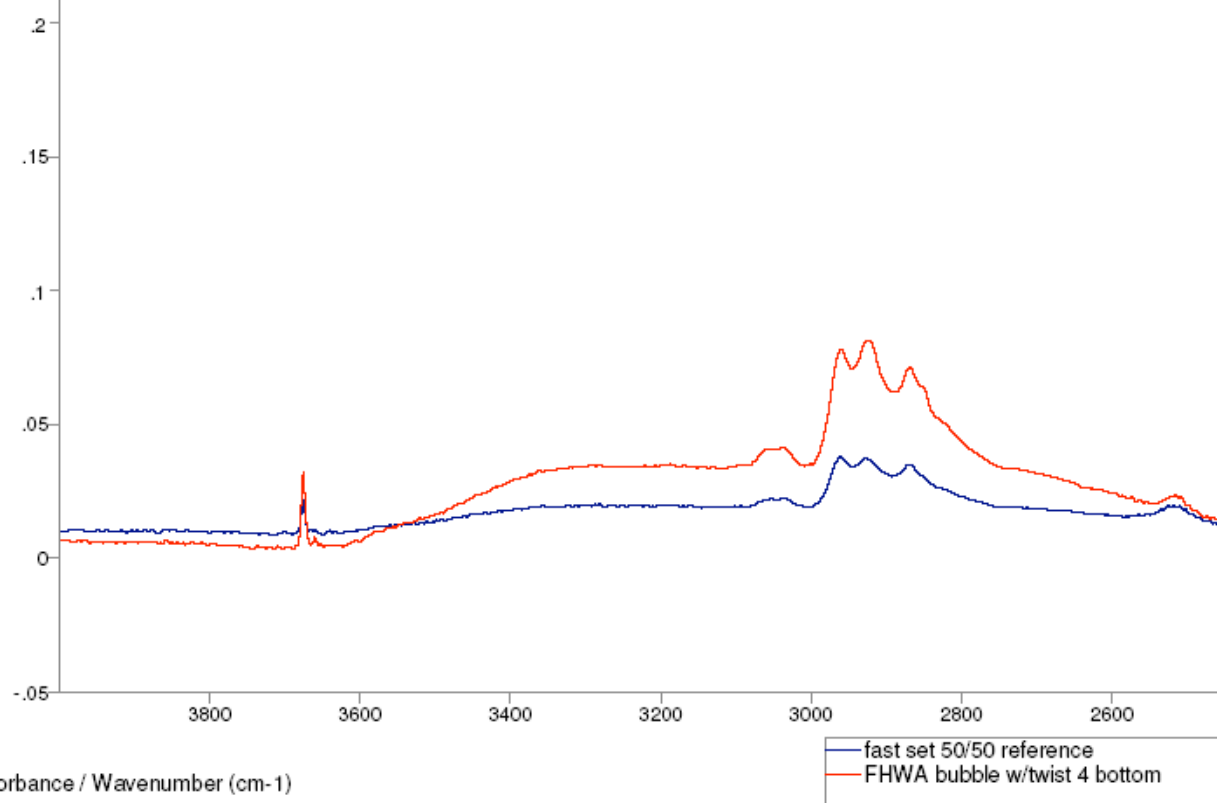
A34.1



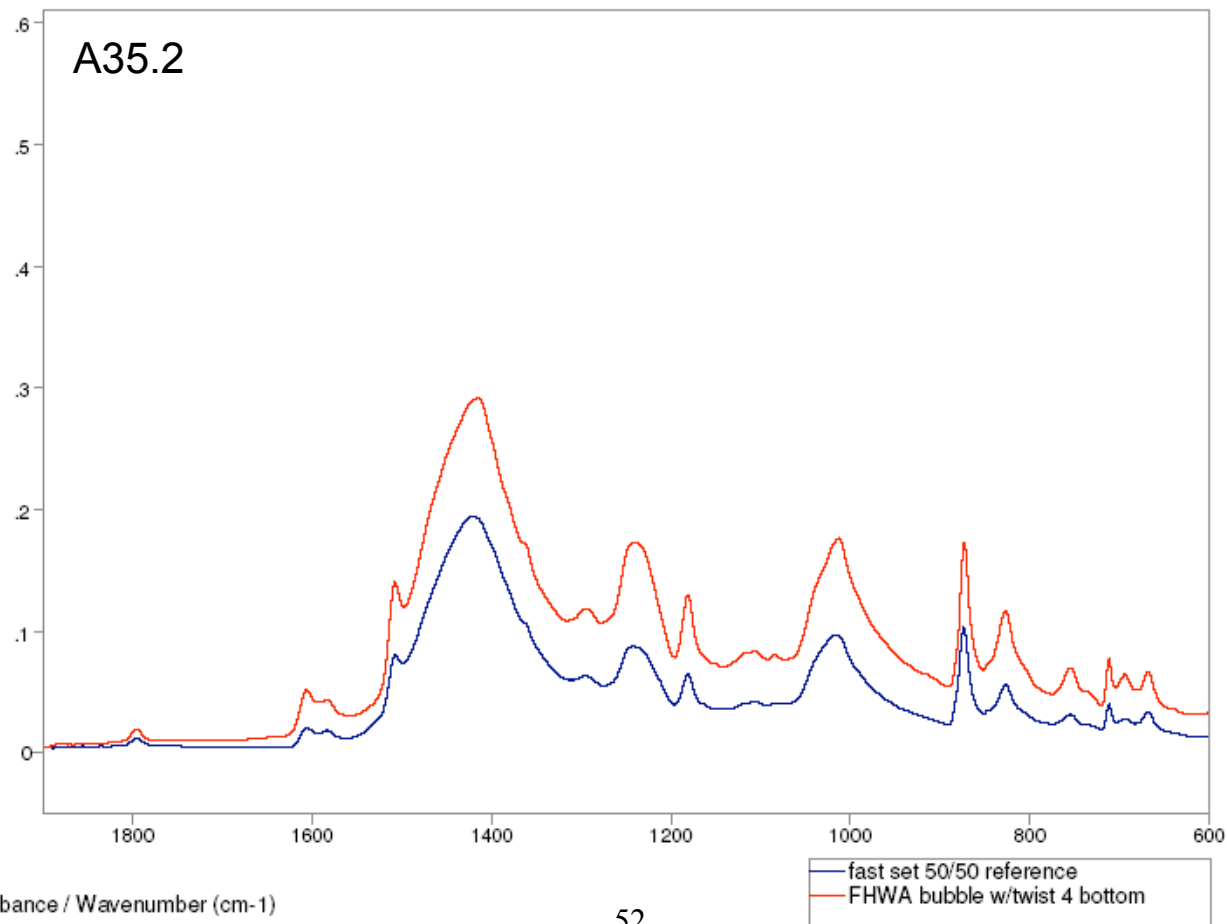
A34.2



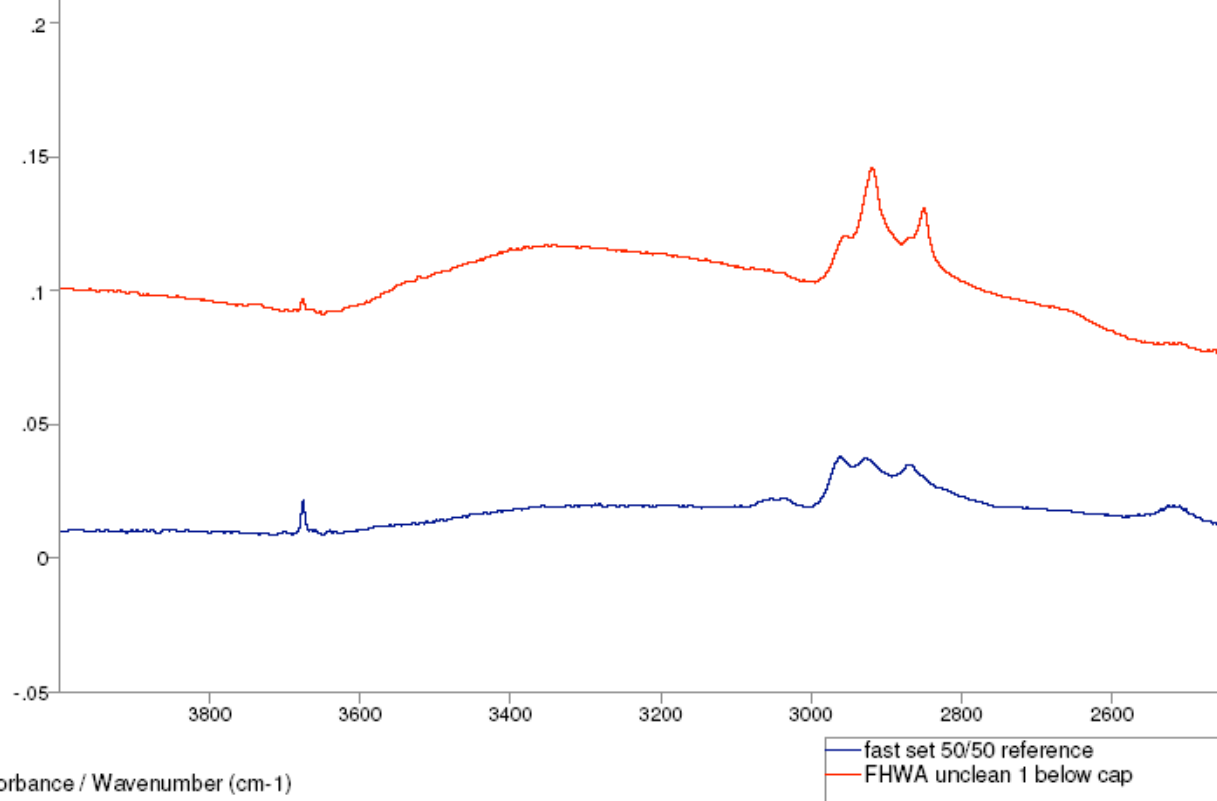
A35.1



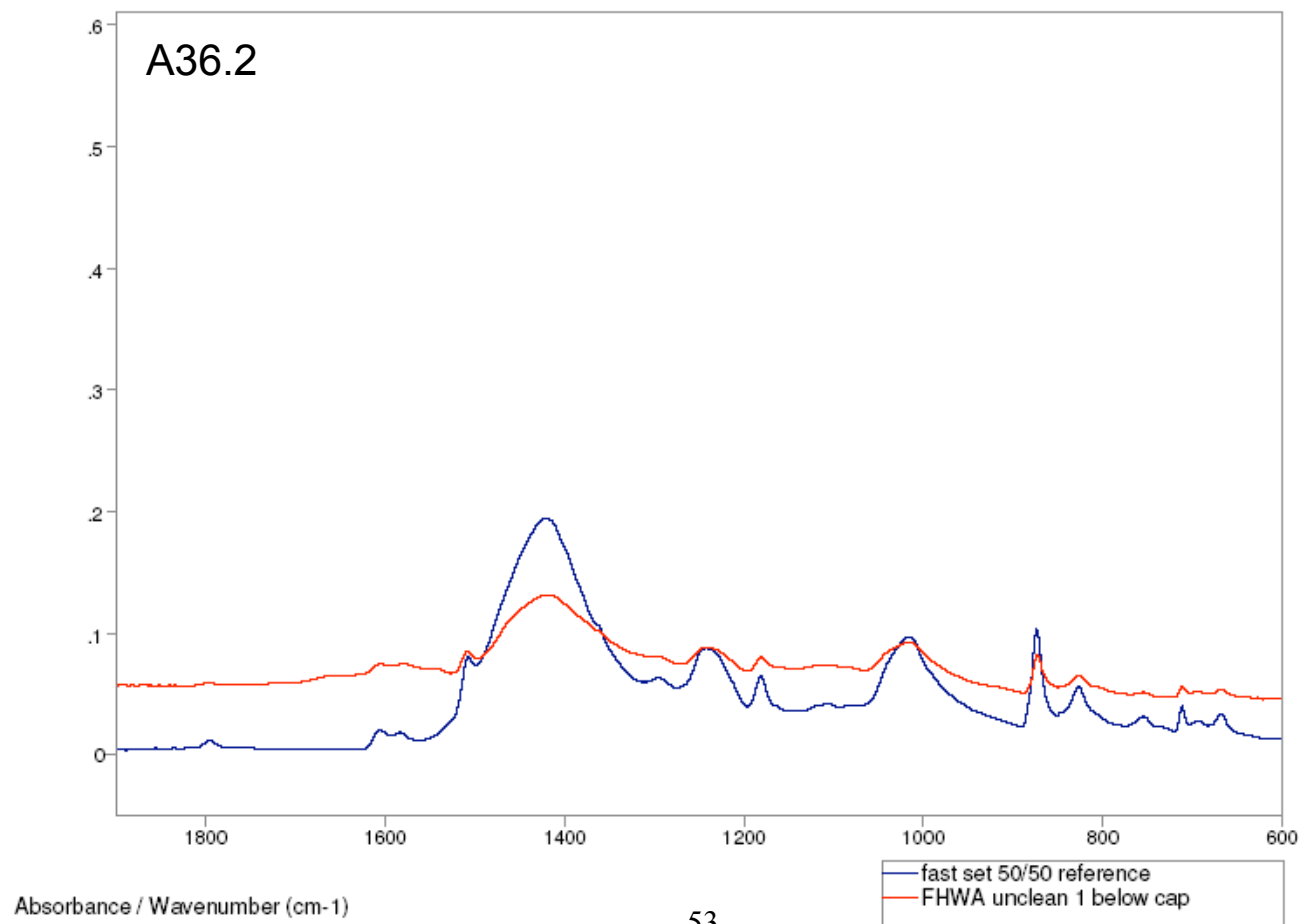
A35.2



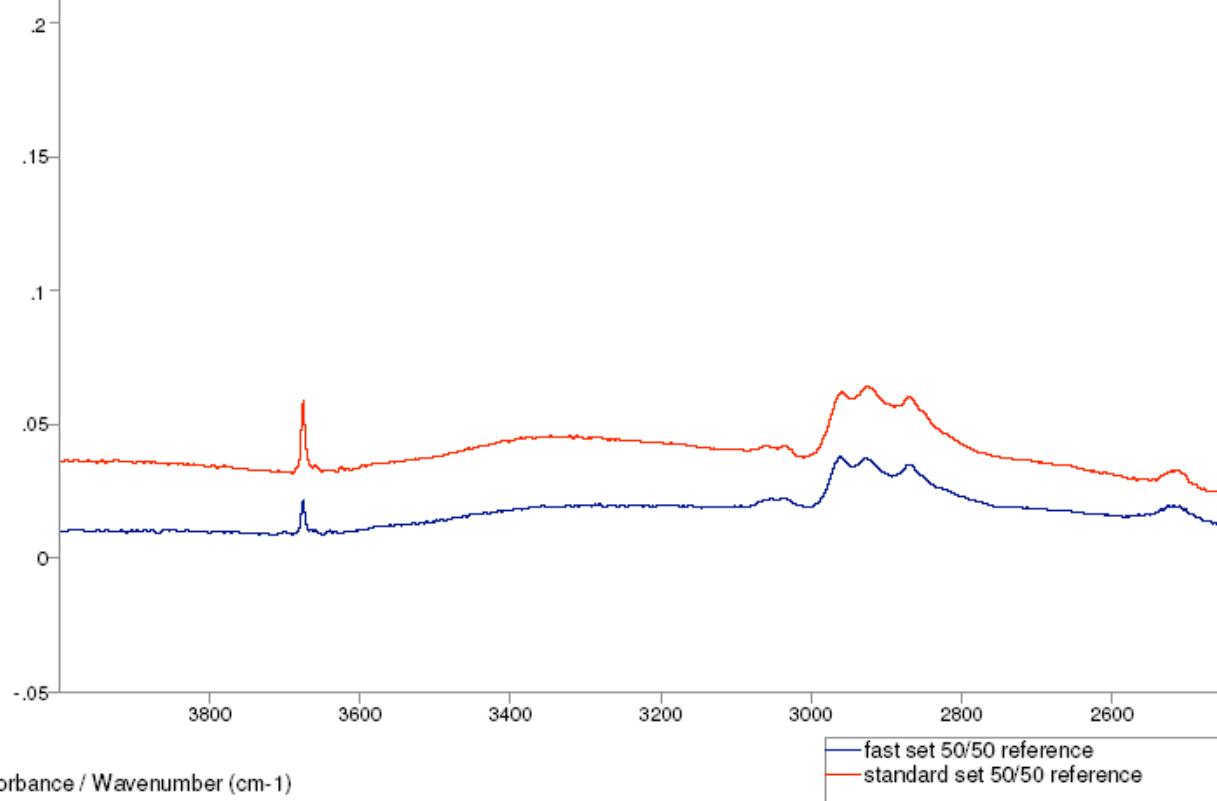
A36.1



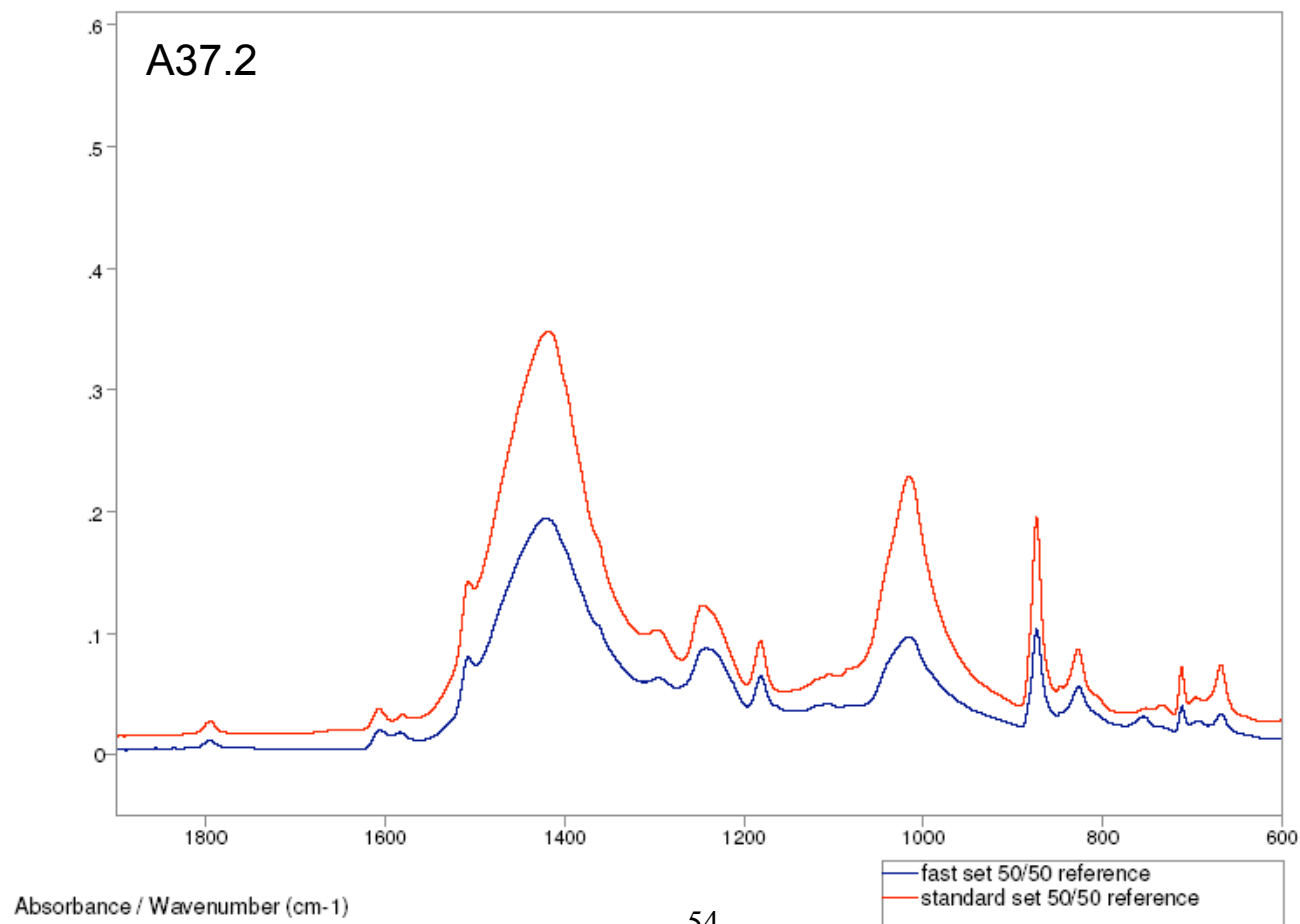
A36.2



A37.1



A37.2



Appendix B: DSC Data

DSC data for all samples examined are presented in this appendix. Each plot includes a first ramp to 150 °C (blue) and a second ramp to 150 °C (red). Glass transition can be seen as a step change in the slope of the heat flow vs. temperature curve. T_g values are labeled as the middle of the transition and are shown next to its temperature ramp in corresponding color. Because glass transitions on the first temperature ramp often merge with an endothermic peak and thus are more complex, an appropriate estimation is made for the T_g value. In cases where this estimation was made, the T_g value is followed by the letter H in parentheses: (H). Labels also indicate endothermic peaks and their normalized size in Joules/gram, as well as any exothermic peaks.

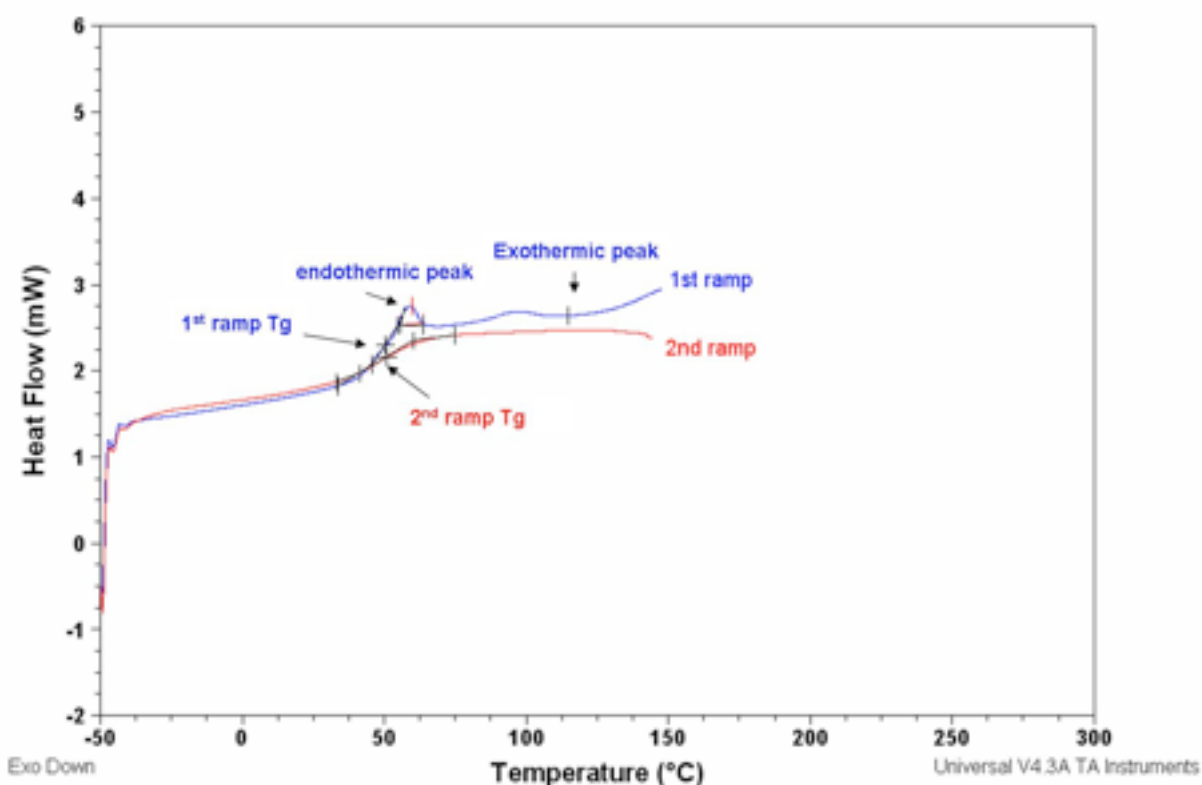
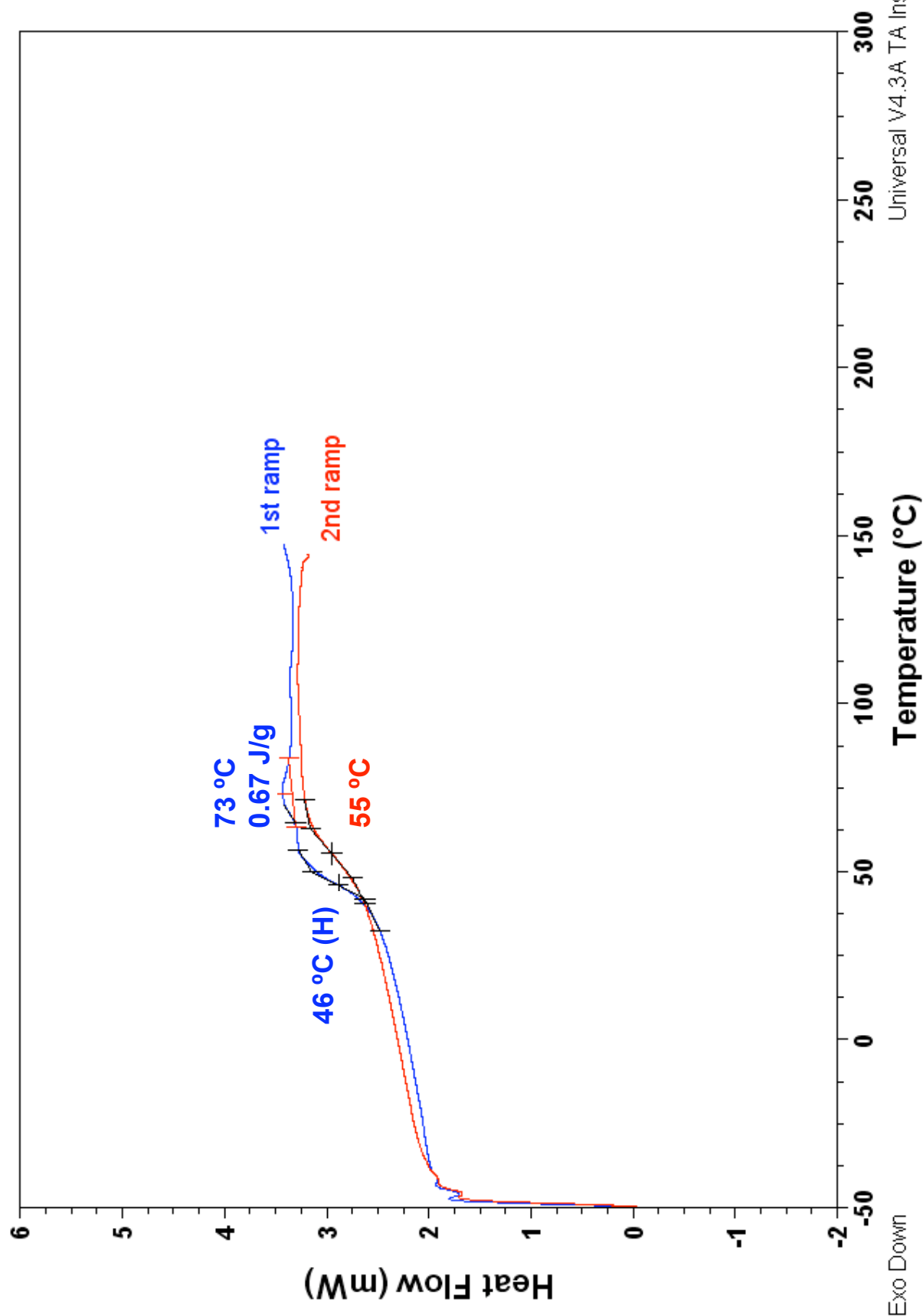


Figure B.0 Schematic of DSC plot showing first temperature ramp (blue), second temperature ramp (red), endothermic peaks, and exothermic peaks.

Sample: anchor 1-2 middle
Size: 5.3000 mg
Method: helen

DSC

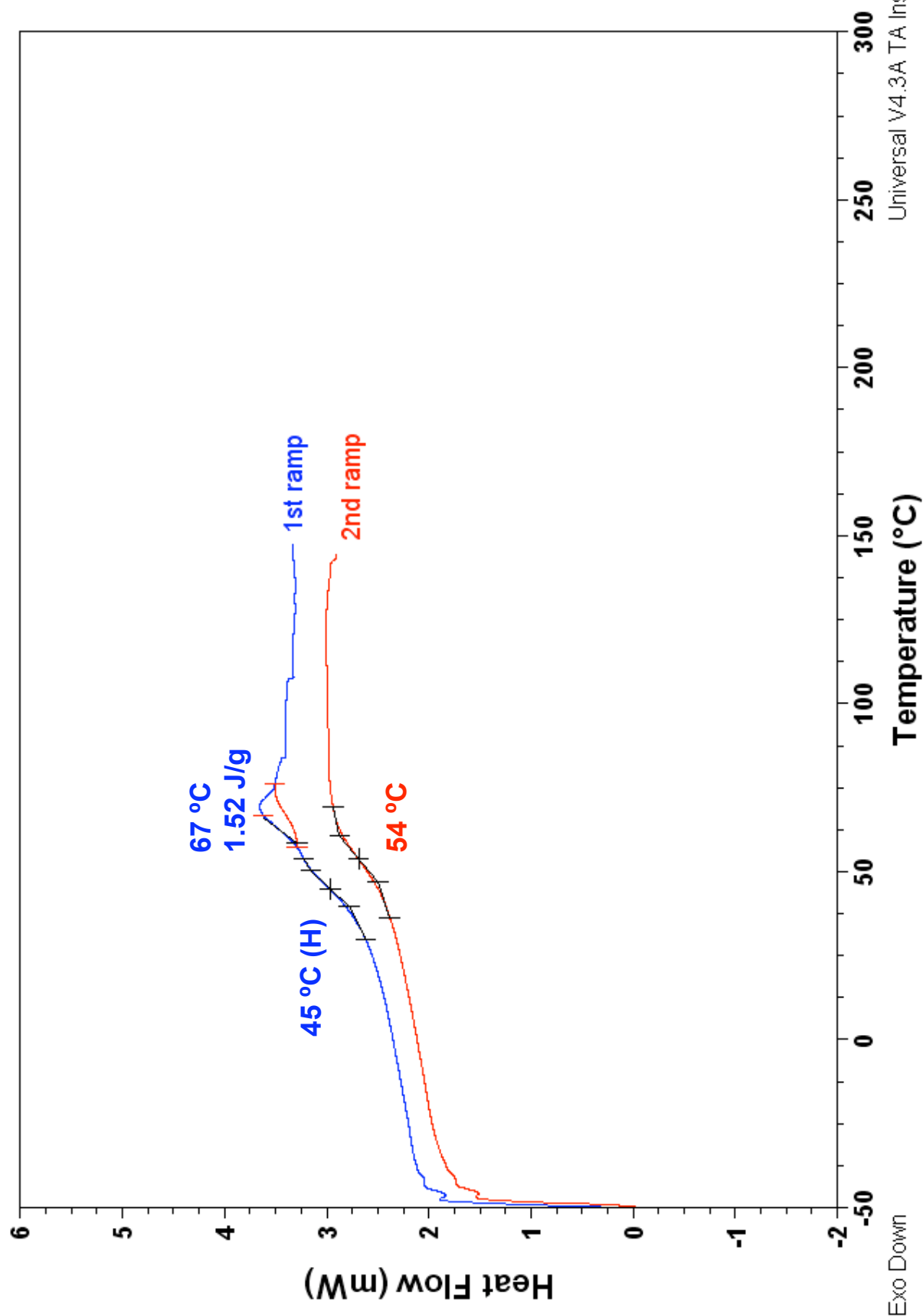
File: C:\NTSB-HWY06MH024-bolt1-2middle.001
Operator: helen
Run Date: 31-Jan-2007 15:53
Instrument: DSC Q1000 V9.8 Build 296



Sample: anchor 1-3 top
Size: 4.6000 mg
Method: helen

DSC

File: C:\NTSB-HWY06MH024-bolt1-3top.001
Operator: helen
Run Date: 02-Feb-2007 10:47
Instrument: DSC Q1000 V9.8 Build 296



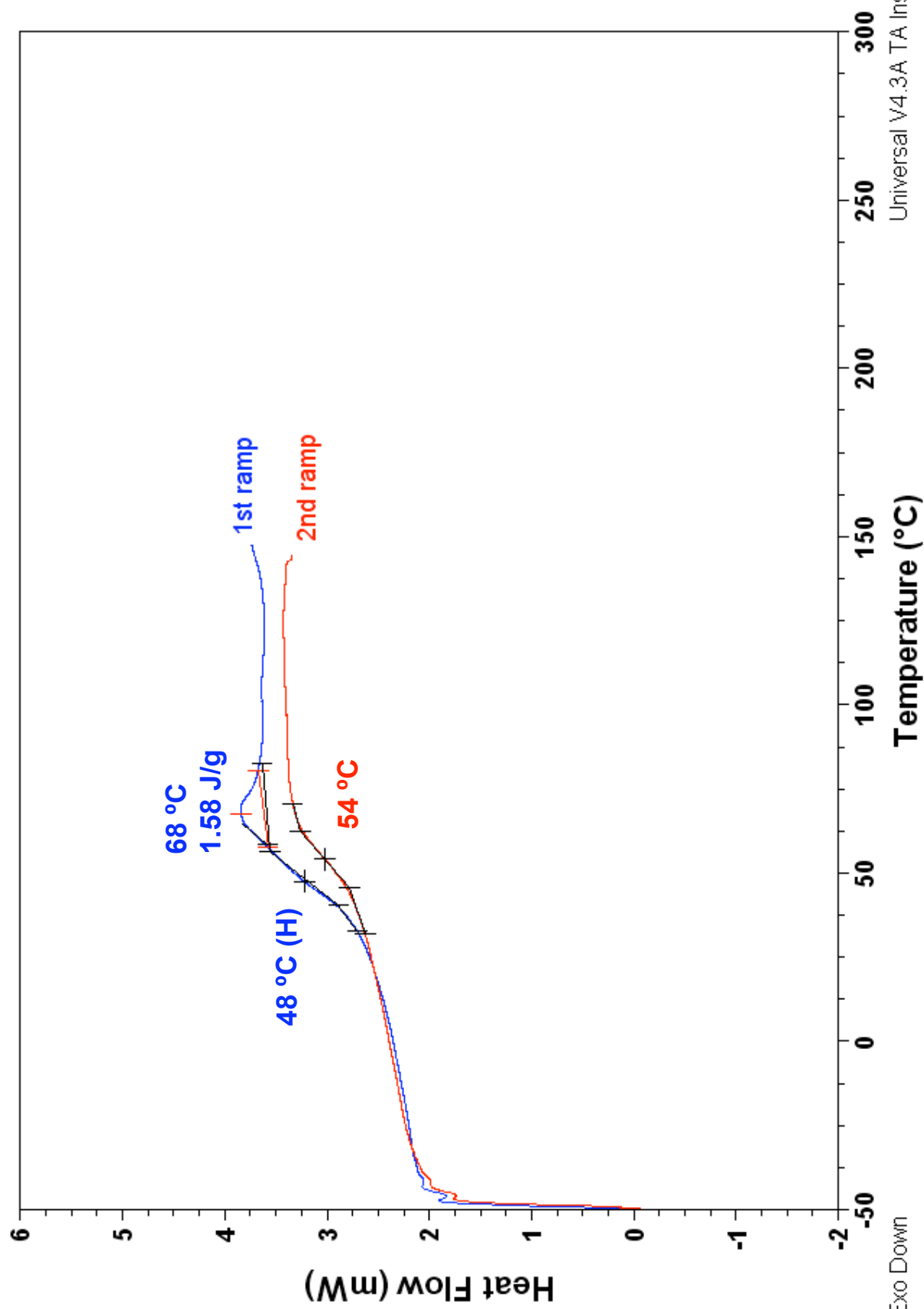
Exo Down

Universal V4.3A TA Instruments

Sample: anchor 1-4 middle
Size: 5.4000 mg
Method: helen

DSC

File: C:\...WTSB-HWY06MH024-bolt1-4middle.001
Operator: helen
Run Date: 02-Feb-2007 11:38
Instrument: DSC Q1000 V9.8 Build 296



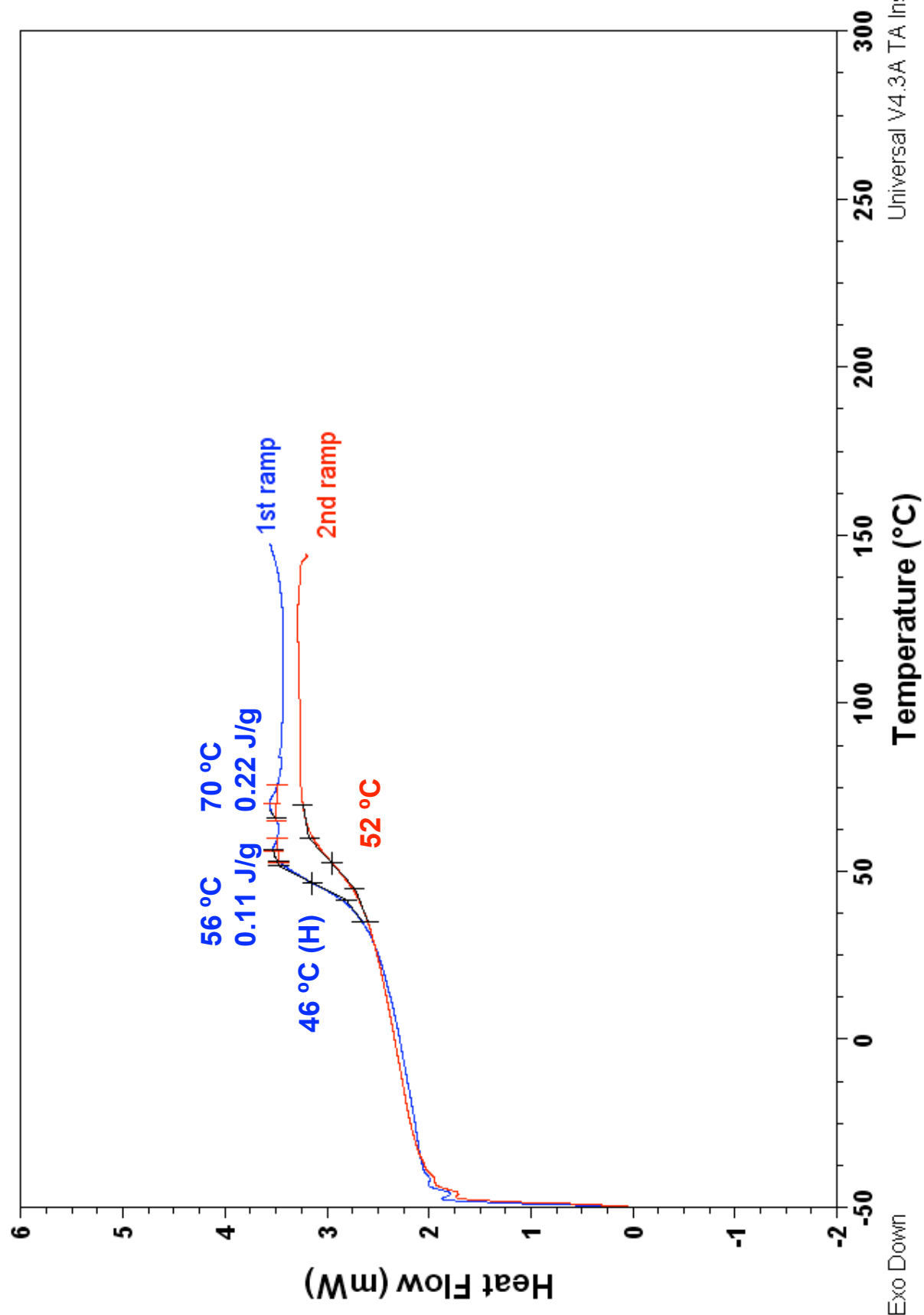
Exo Down

Universal V4.3A TA Instruments

Sample: anchor 2-1 top
Size: 5.4000 mg
Method: helen

DSC

File: C:\...\NTSB-HWY06MH024-bolt2-1top.001
Operator: helen
Run Date: 02-Feb-2007 12:30
Instrument: DSC Q1000 V9.8 Build 296



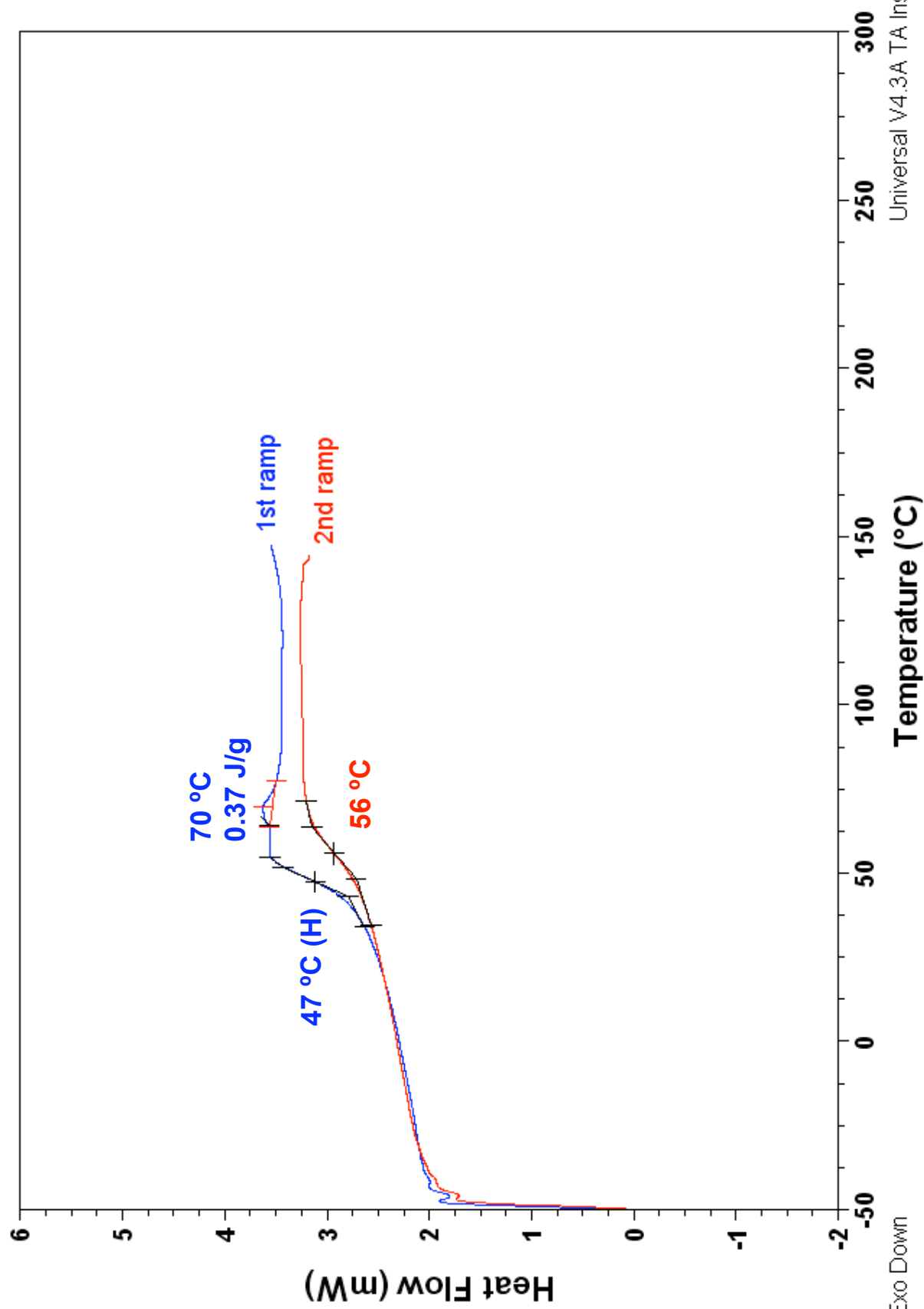
Exo Down

Universal V4.3A TA Instruments

Sample: anchor 2-3 top
Size: 5.2000 mg
Method: helen

DSC

File: C:\NTSB-HWY06MH024-bolt2-3top.001
Operator: helen
Run Date: 02-Feb-2007 14:14
Instrument: DSC Q1000 V9.8 Build 296



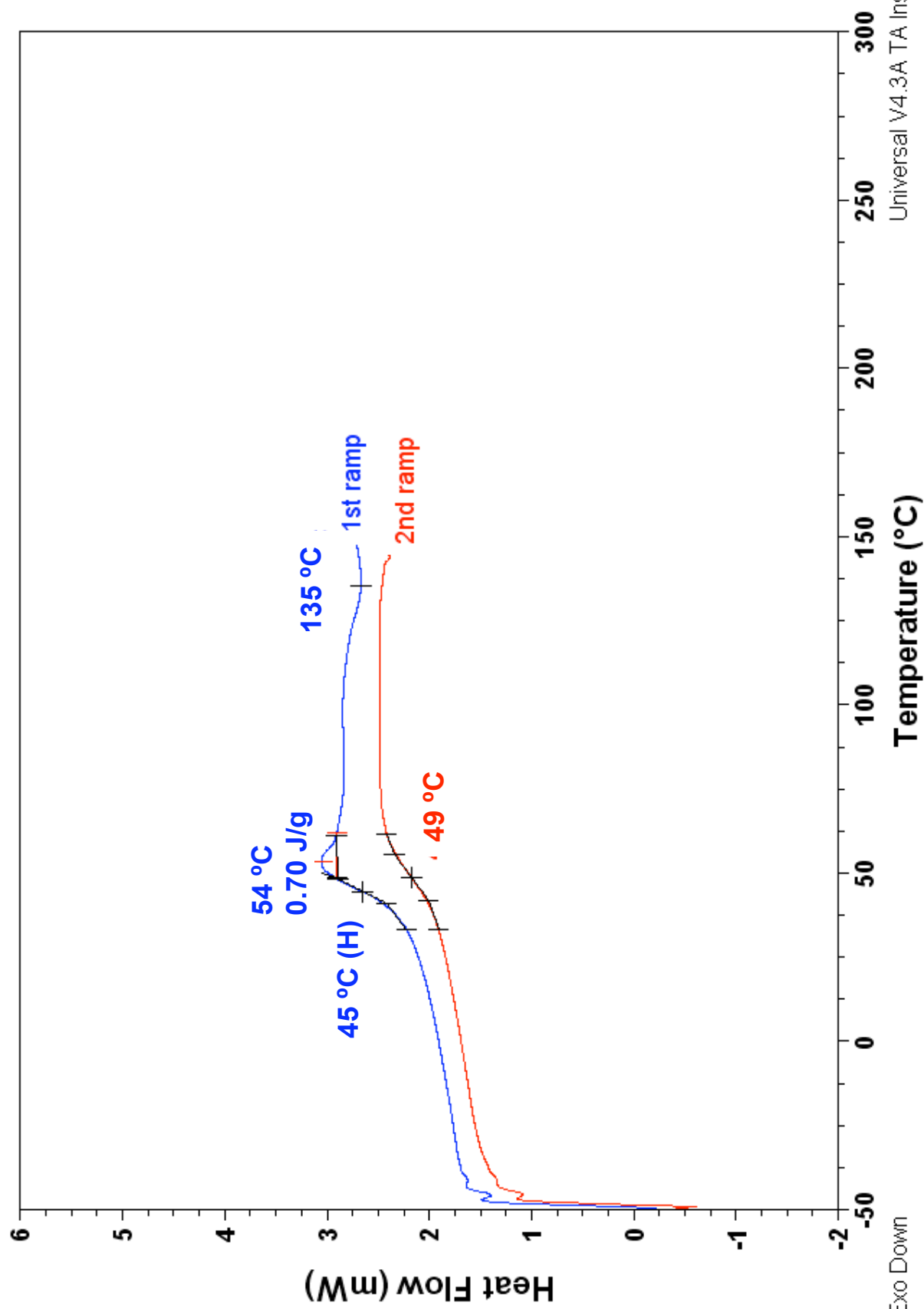
Exo Down

Universal V4.3A TA Instruments

Sample: anchor 3-1 bottom
Size: 5.0000 mg
Method: helen

DSC

File: C:\NTSB-HWY06MH024-bolt3-1bottom.001
Operator: helen
Run Date: 02-Feb-2007 15:05
Instrument: DSC Q1000 V9.8 Build 296



Exo Down

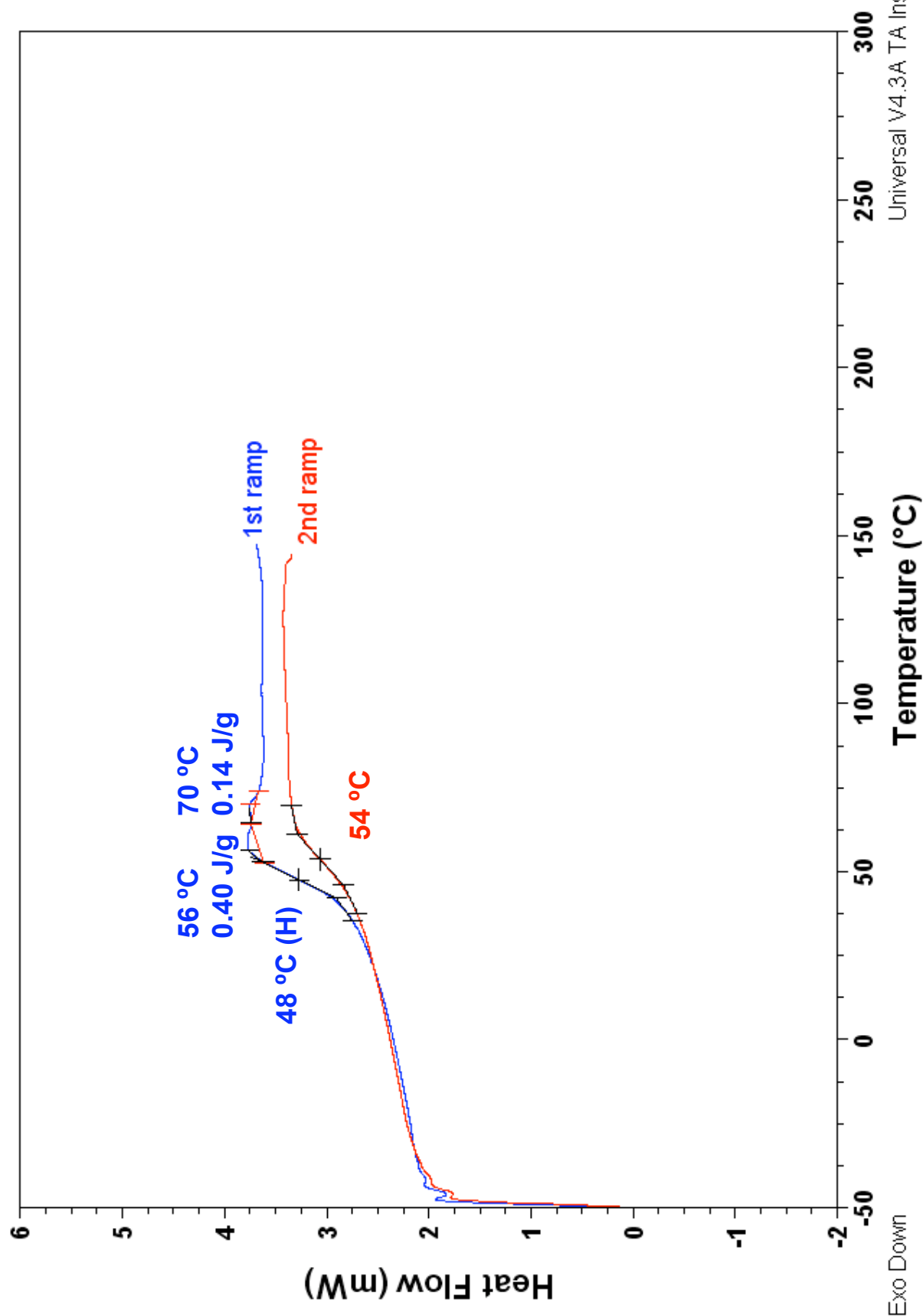
Temperature (°C)

Universal V4.3A TA Instruments

Sample: anchor 3-2 top
Size: 5.7000 mg
Method: helen

DSC

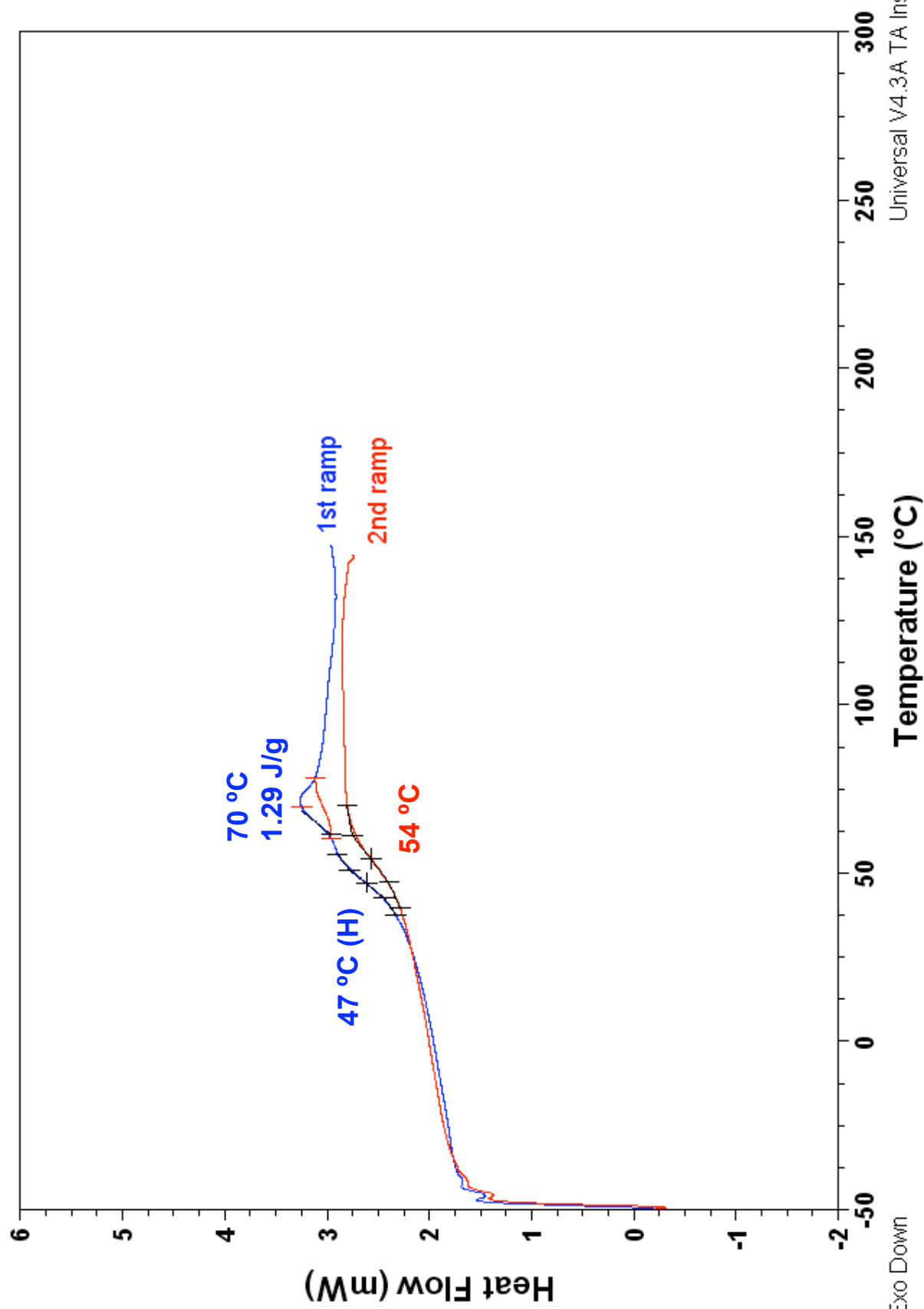
File: C:\NTSB-HWY06MH024-bolt3-2top.001
Operator: helen
Run Date: 02-Feb-2007 15:57
Instrument: DSC Q1000 V9.8 Build 296



Sample: anchor 3-3 middle
Size: 5.0000 mg
Method: helen

DSC

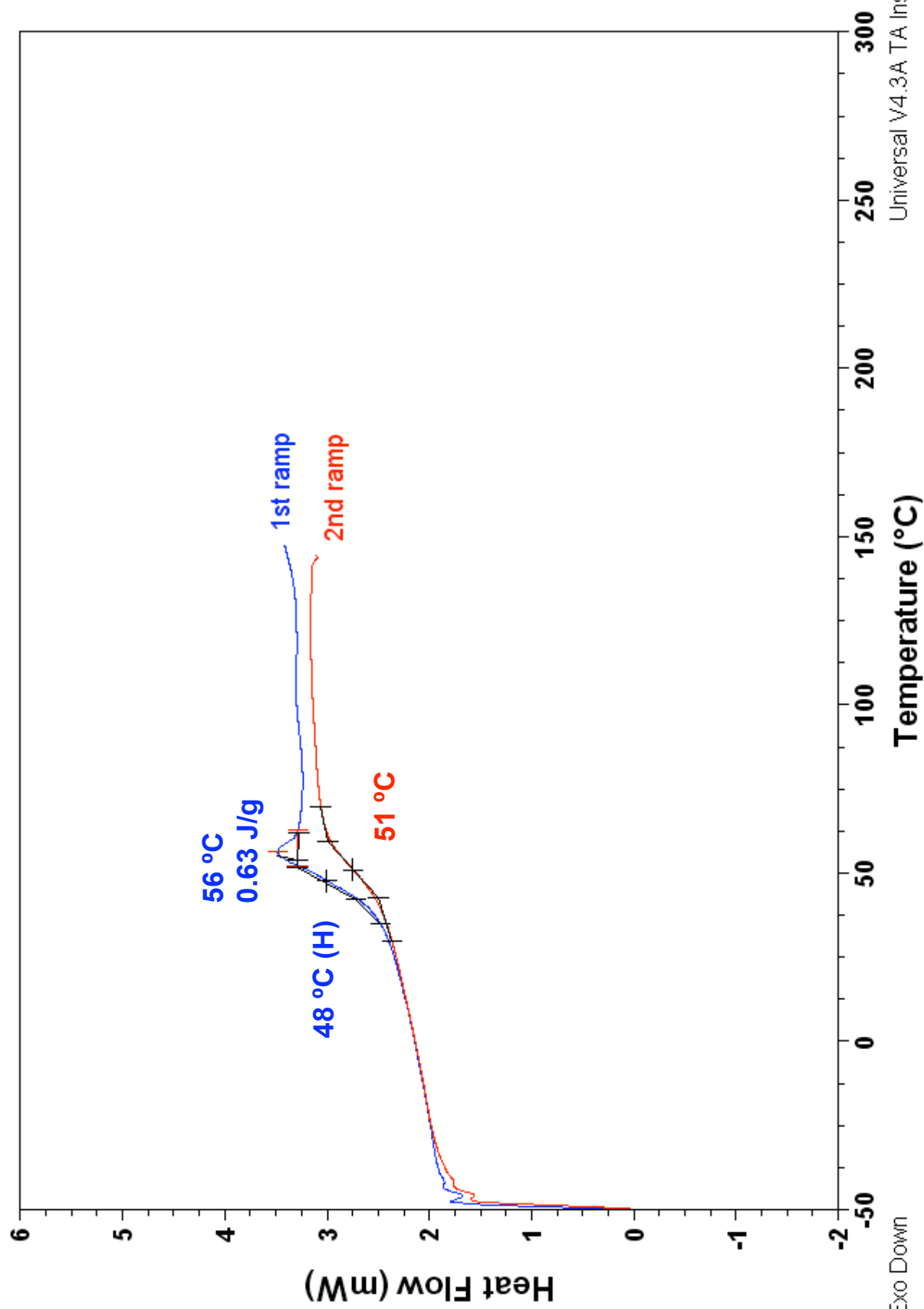
File: C:\NTSB-HWY06MH024-bolt3-3middle.001
Operator: helen
Run Date: 02-Feb-2007 17:41
Instrument: DSC Q1000 V9.8 Build 296



Sample: anchor 3-4 bottom
Size: 5.2000 mg
Method: helen

DSC

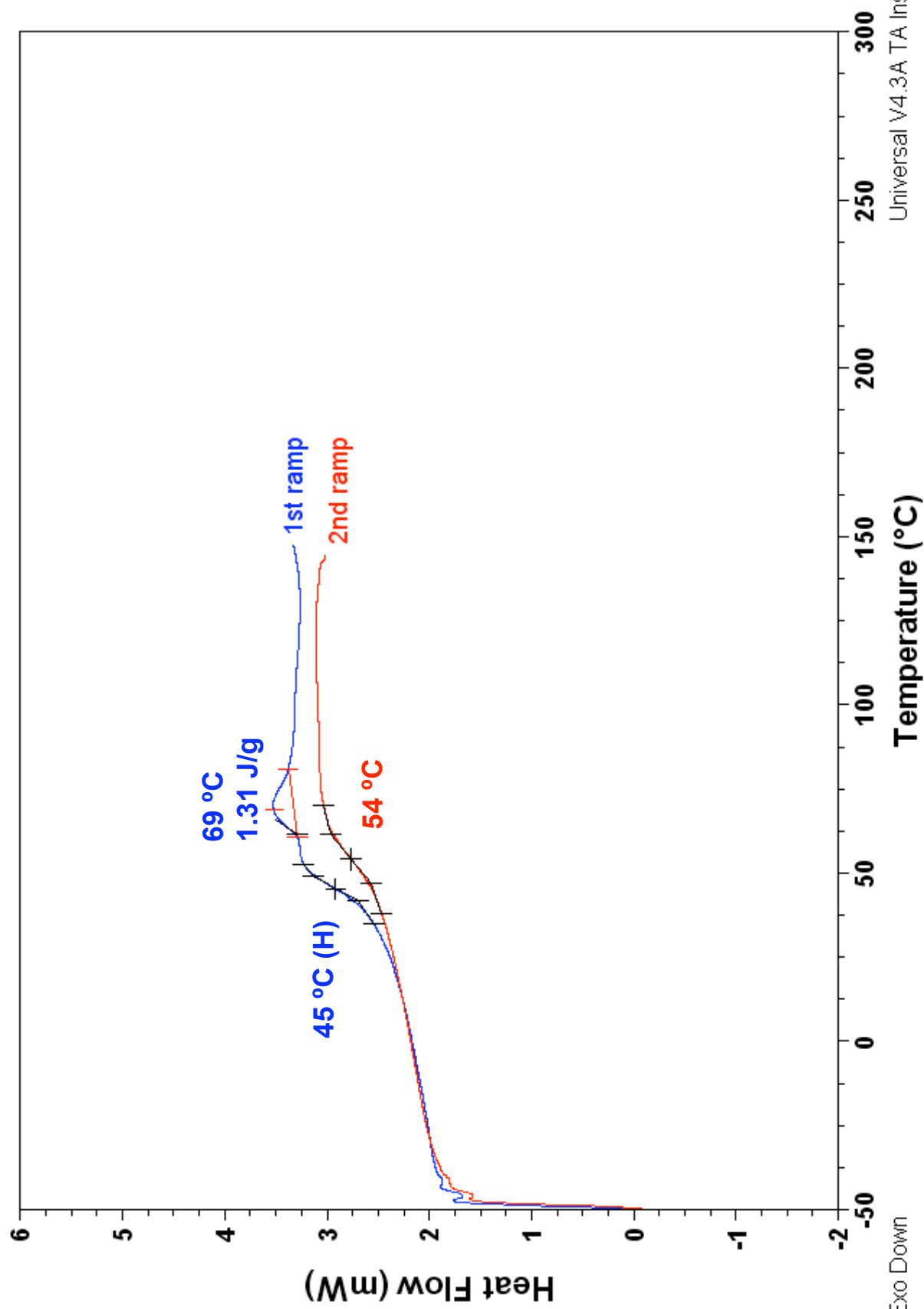
File: C:\...WTSB-HWY06MH024-bolt3-4bottom.001
Operator: helen
Run Date: 30-Jan-2007 15:00
Instrument: DSC Q1000 V9.8 Build 296



Sample: anchor 4-2 middle
Size: 5.2000 mg
Method: helen

DSC

File: ... \NTSB-HWY06MH024-bolt4-2middle-2.001
Operator: helen
Run Date: 02-Feb-2007 18:32
Instrument: DSC Q1000 V9.8 Build 296



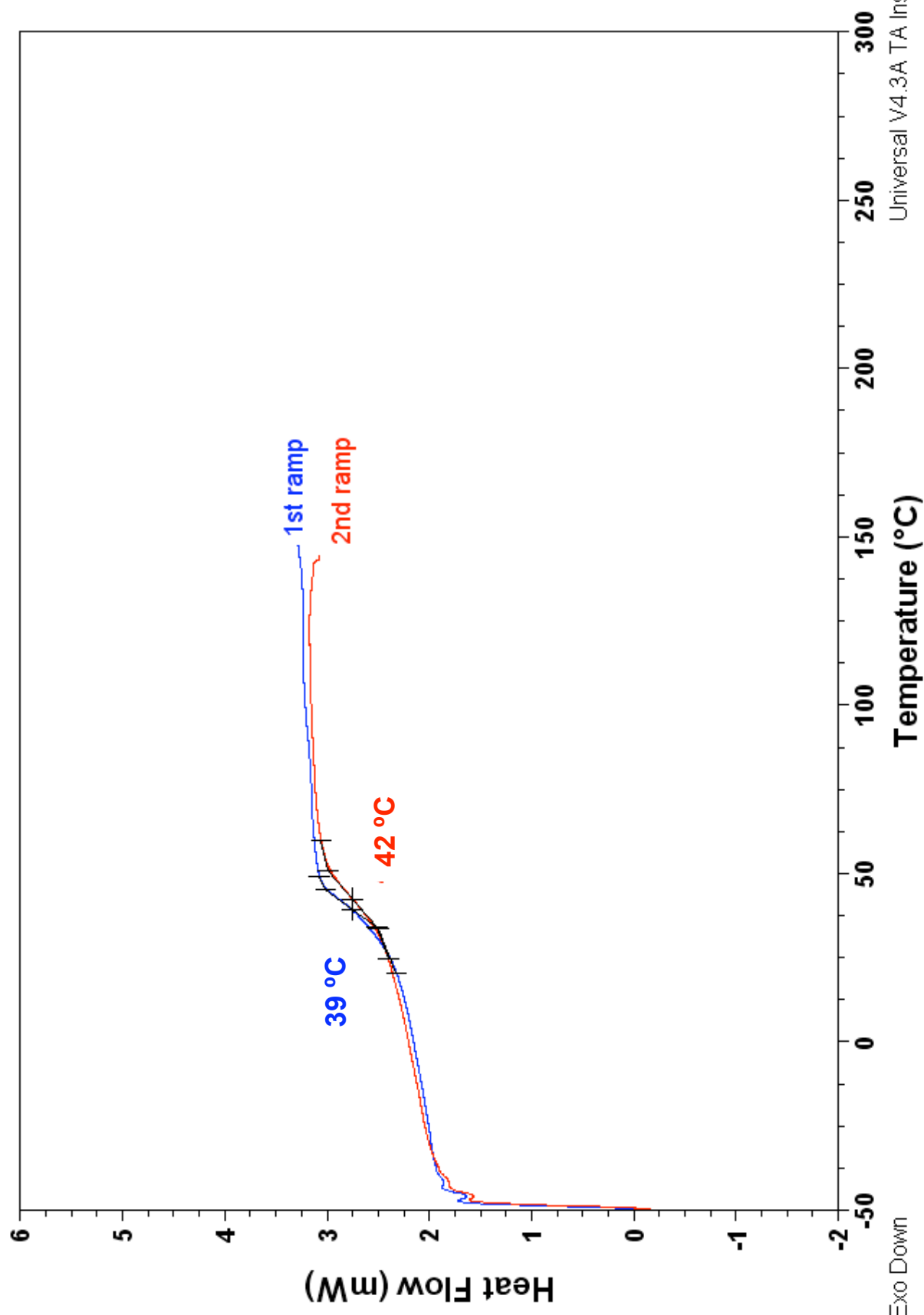
Exo Down

Universal V4.3A TA Instruments

Sample: anchor 5-1 middle
Size: 5.5000 mg
Method: helen

DSC

File: C:\NTSB-HWY06MH024-bolt5-1middle.001
Operator: helen
Run Date: 31-Jan-2007 12:26
Instrument: DSC Q1000 V9.8 Build 296



Exo Down

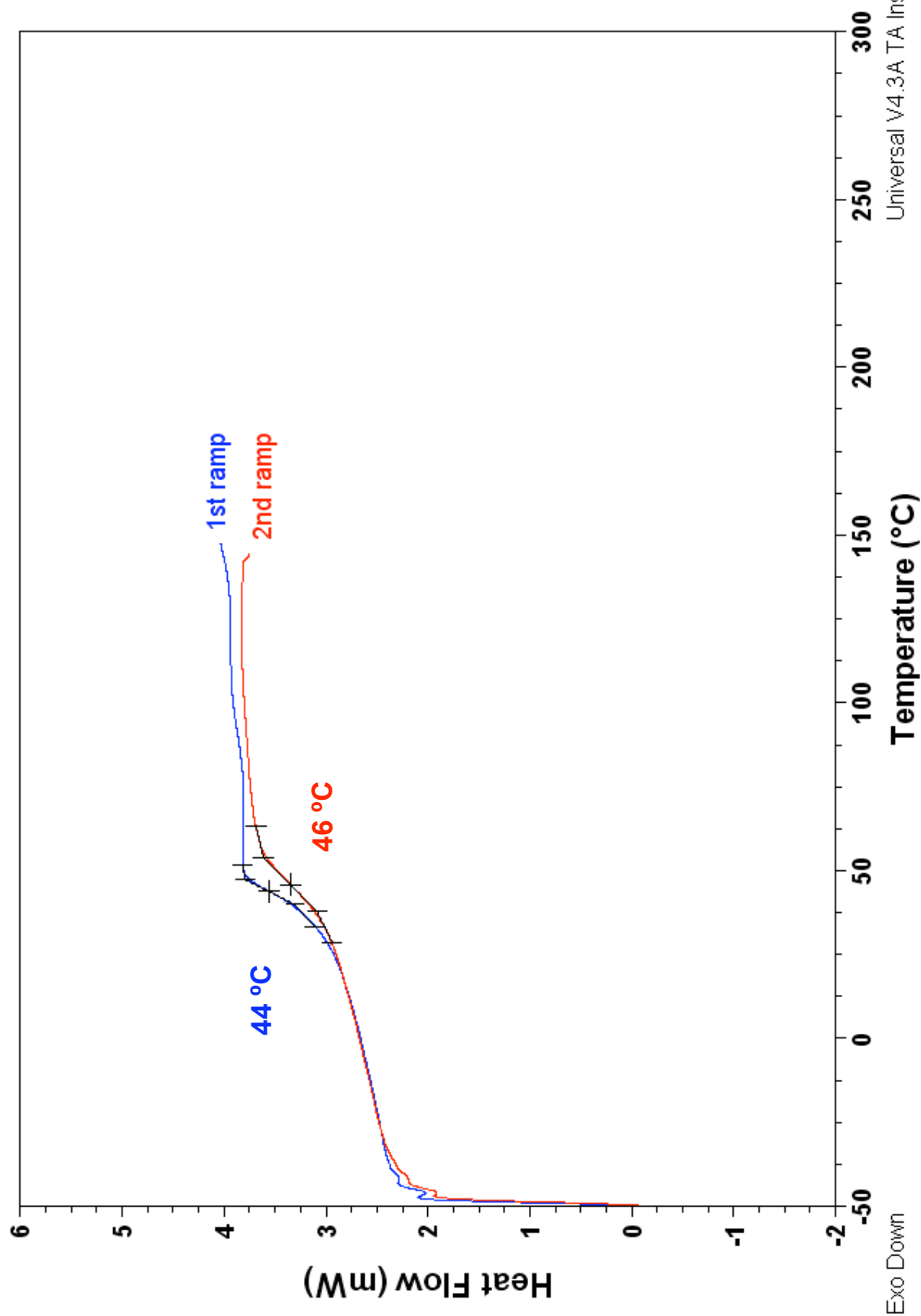
Temperature (°C)

Universal V4.3A TA Instruments

Sample: anchor 5-2 bottom
Size: 5.7000 mg
Method: helen

DSC

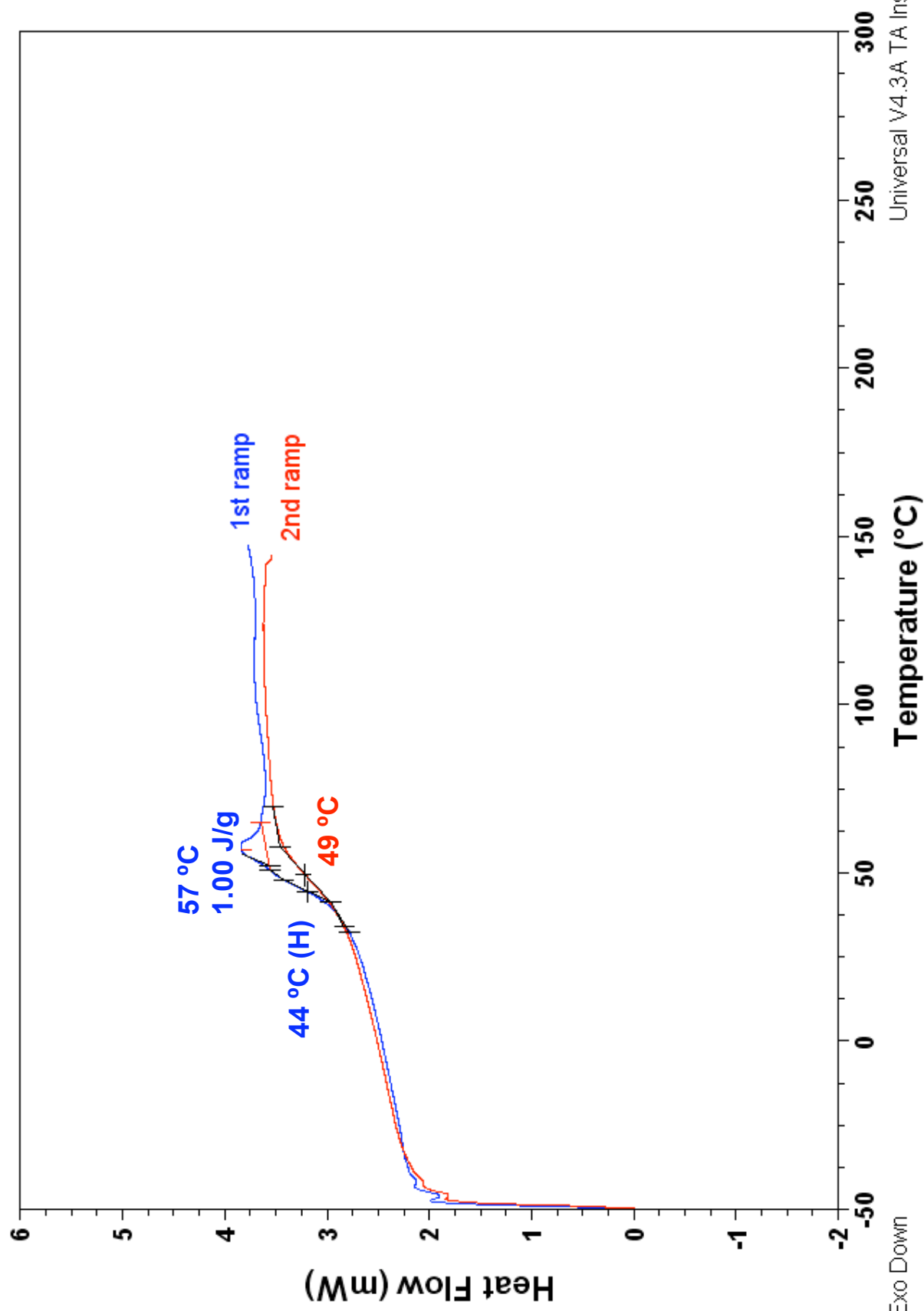
File: C:\NTSB-HWY06MH024-bolt5-2bottom.001
Operator: helen
Run Date: 31-Jan-2007 14:10
Instrument: DSC Q1000 V9.8 Build 296



Sample: anchor 5-3 bottom
Size: 5.4000 mg
Method: helen

DSC

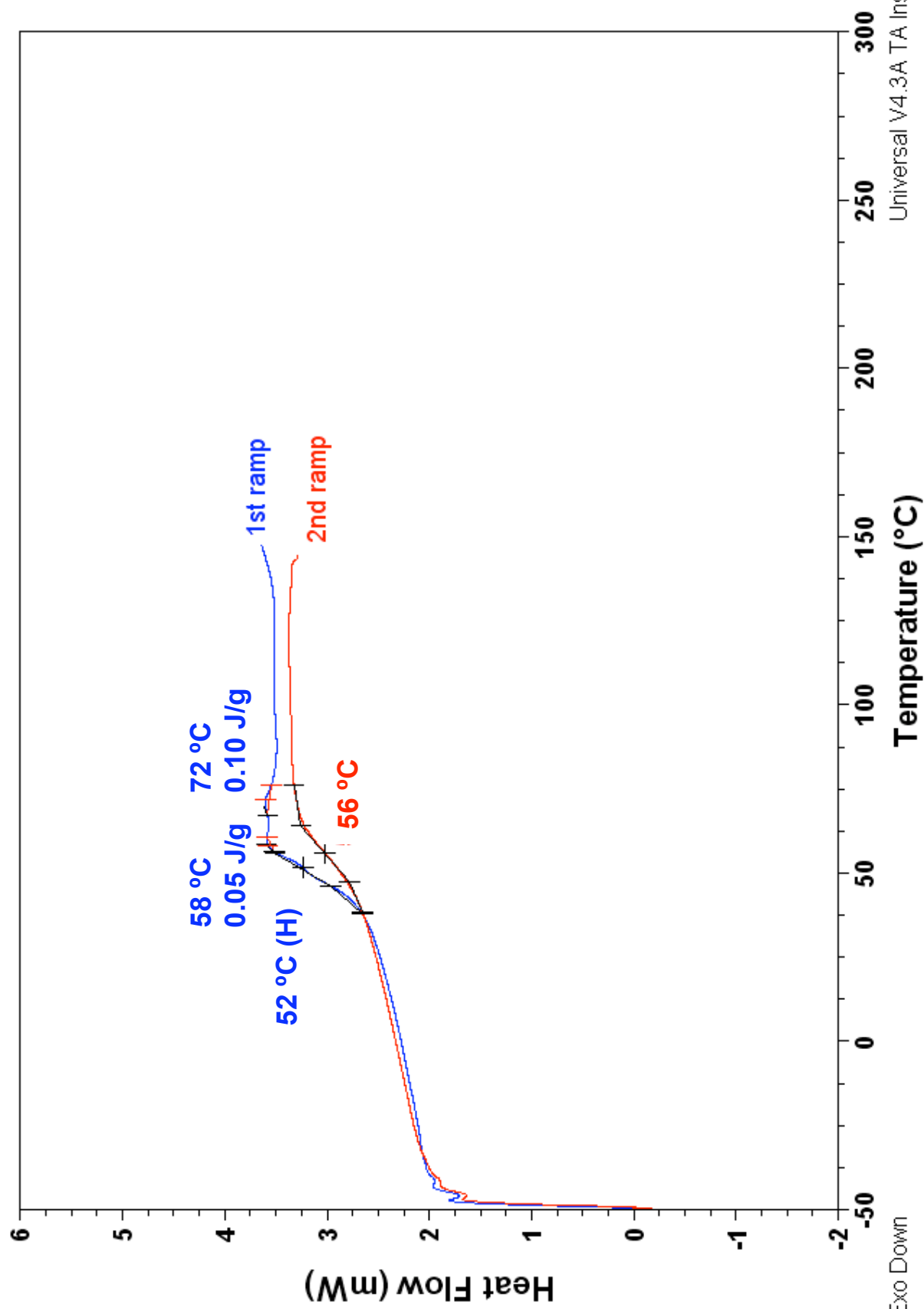
File: C:\...WTSB-HWY06MH024-bolt5-3bottom.001
Operator: helen
Run Date: 31-Jan-2007 15:01
Instrument: DSC Q1000 V9.8 Build 296



Sample: anchor 6-1 top
Size: 5.8000 mg
Method: helen

DSC

File: C:\NTSB-HWY06MH024-bolt6-1top.001
Operator: helen
Run Date: 31-Jan-2007 19:20
Instrument: DSC Q1000 V9.8 Build 296



Exo Down

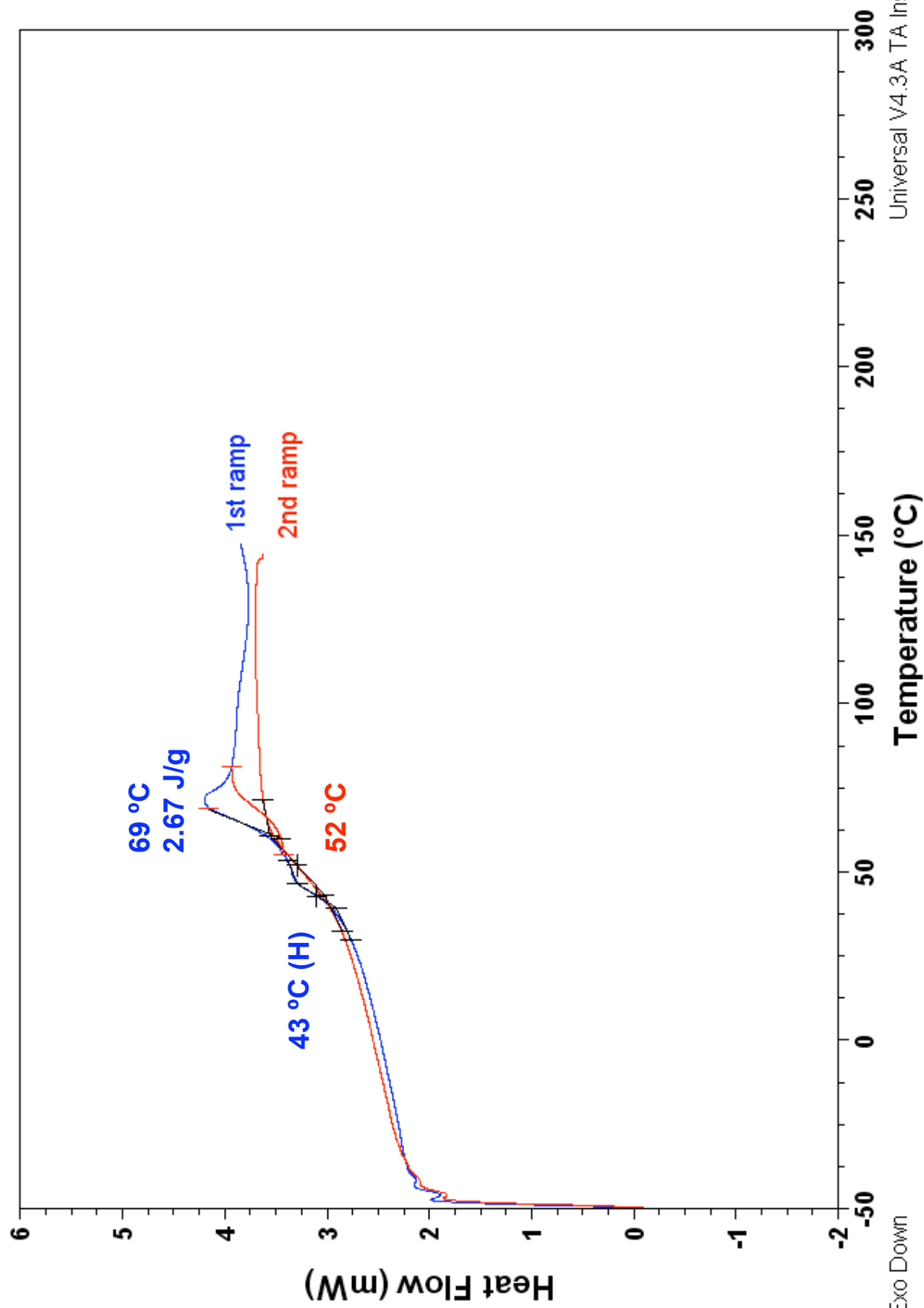
Temperature (°C)

Universal V4.3A TA Instruments

Sample: anchor 6-2 middle
Size: 6.0000 mg
Method: helen

DSC

File: C:\...WTSB-HWY06MH024-bolt6-2middle.001
Operator: helen
Run Date: 31-Jan-2007 18:28
Instrument: DSC Q1000 V9.8 Build 296



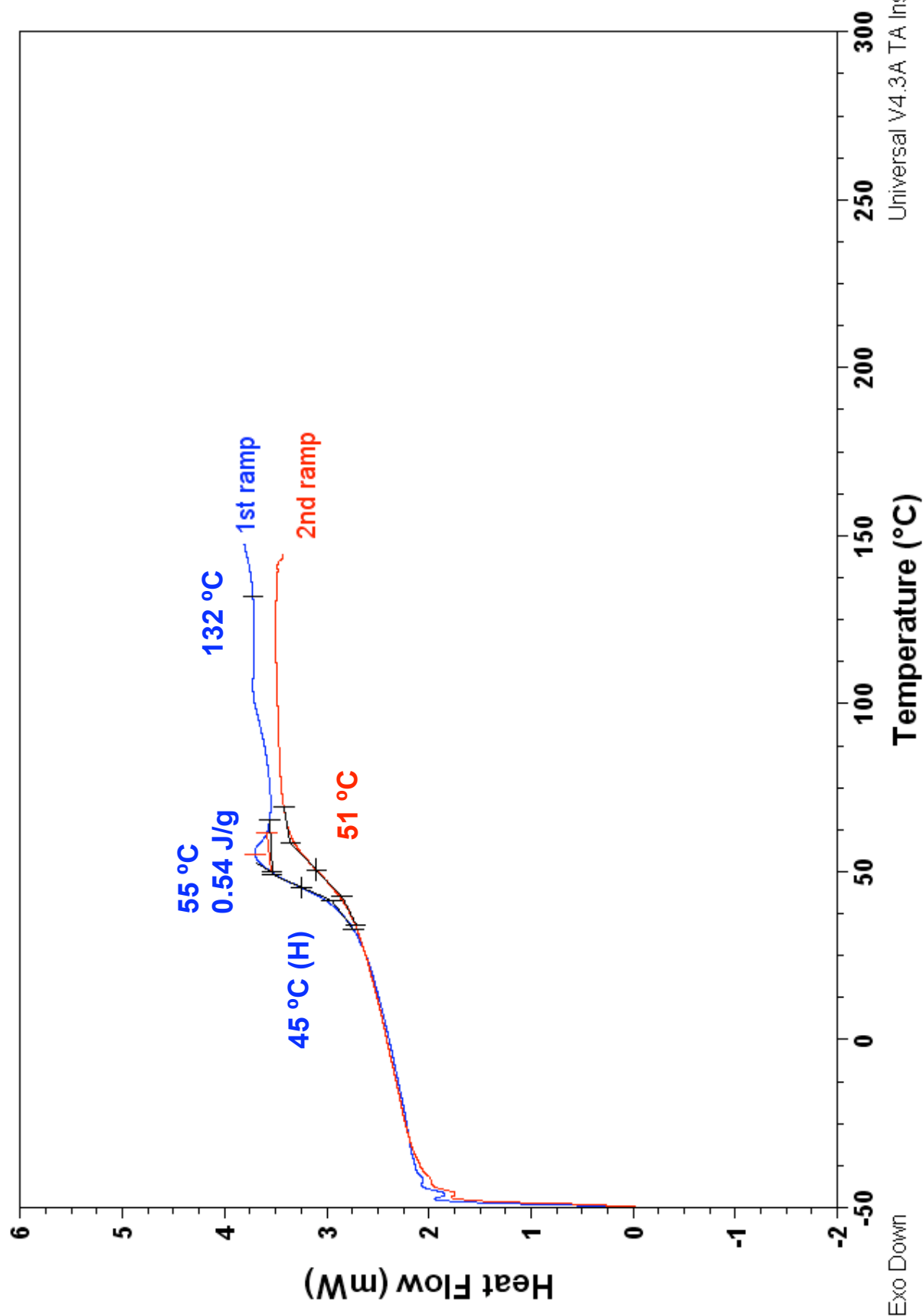
Exo Down

Universal V4.3A TA Instruments

Sample: anchor 6-3 bottom
Size: 5.6000 mg
Method: helen

DSC

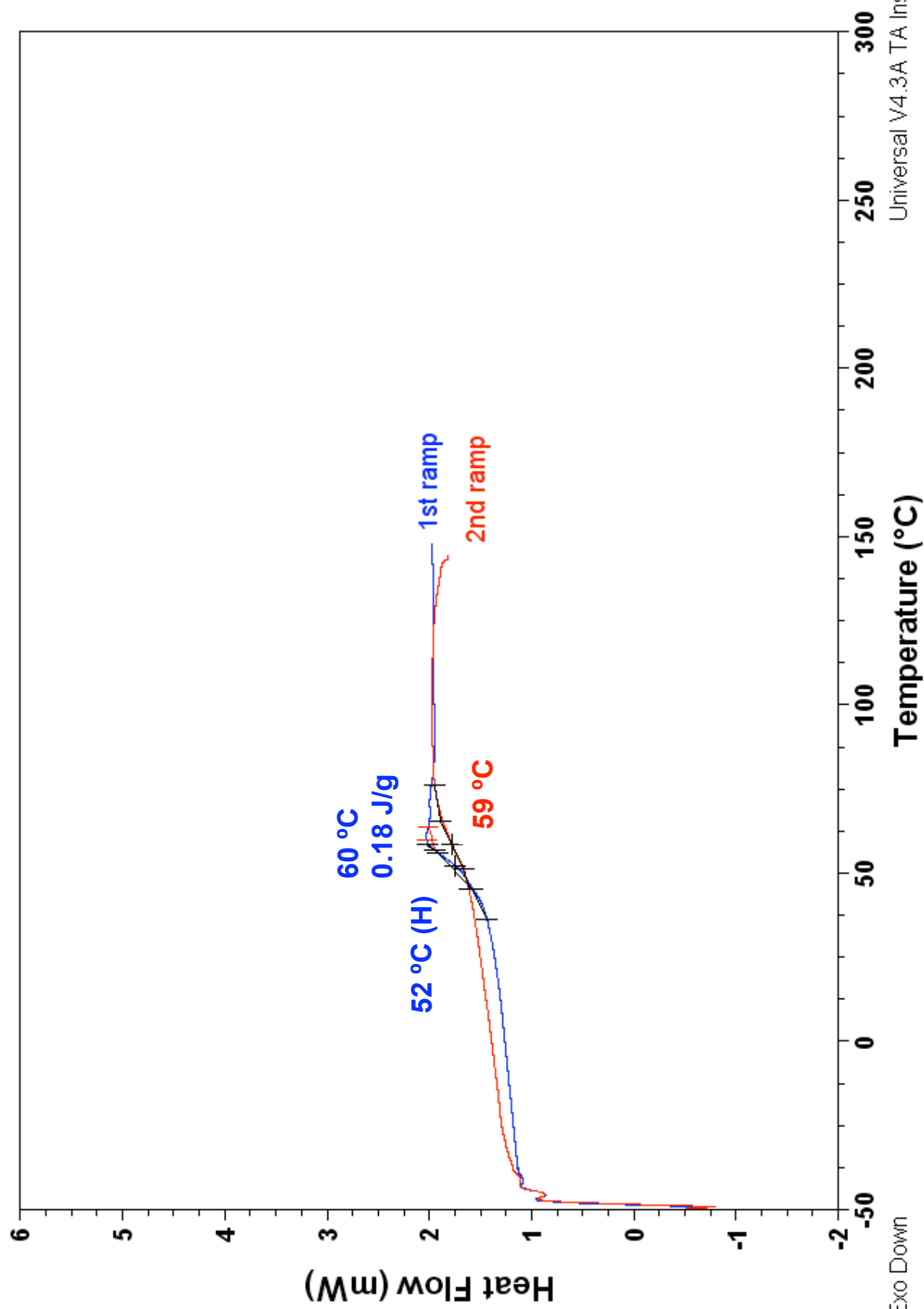
File: C:\NTSB-HWY06MH024-bolt6-3bottom.001
Operator: helen
Run Date: 31-Jan-2007 17:37
Instrument: DSC Q1000 V9.8 Build 296



Sample: anchor E1-1 top
Size: 4.0000 mg
Method: helen

DSC

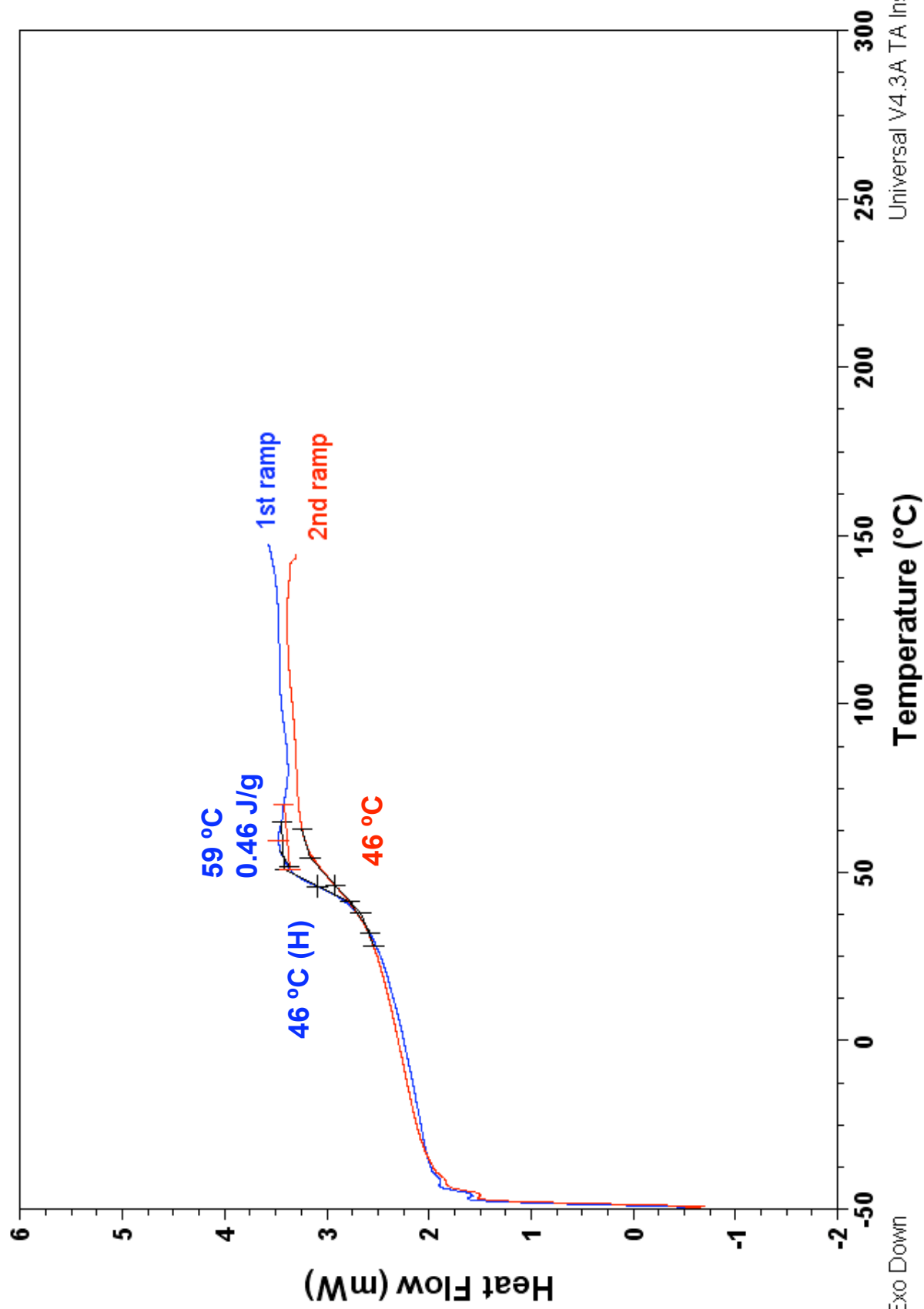
File: C:\NTSB-HWY06MH024-boltE1-1top.001
Operator: helen
Run Date: 31-Jan-2007 10:43
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB080-M1-3/4/5-SW mid (typical)
Size: 5.9000 mg
Method: Helen

DSC

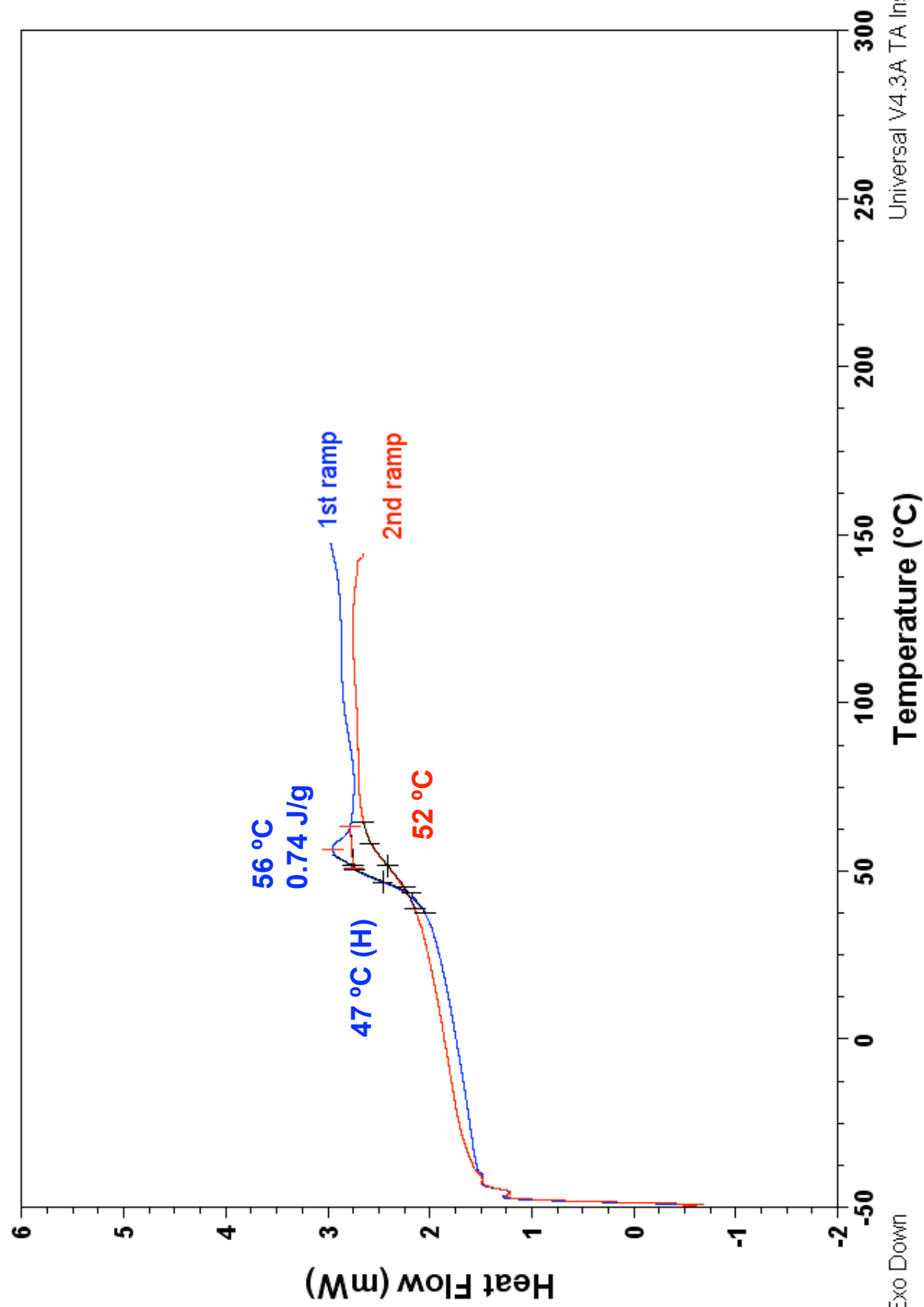
File: NTSB-HWY06MH024-EB080-M1-345-SWmid.C
Operator: Helen
Run Date: 06-Feb-2007 15:37
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB080-N-4/5/6-NW bottom (typica)
Size: 5.1000 mg
Method: Helen

DSC

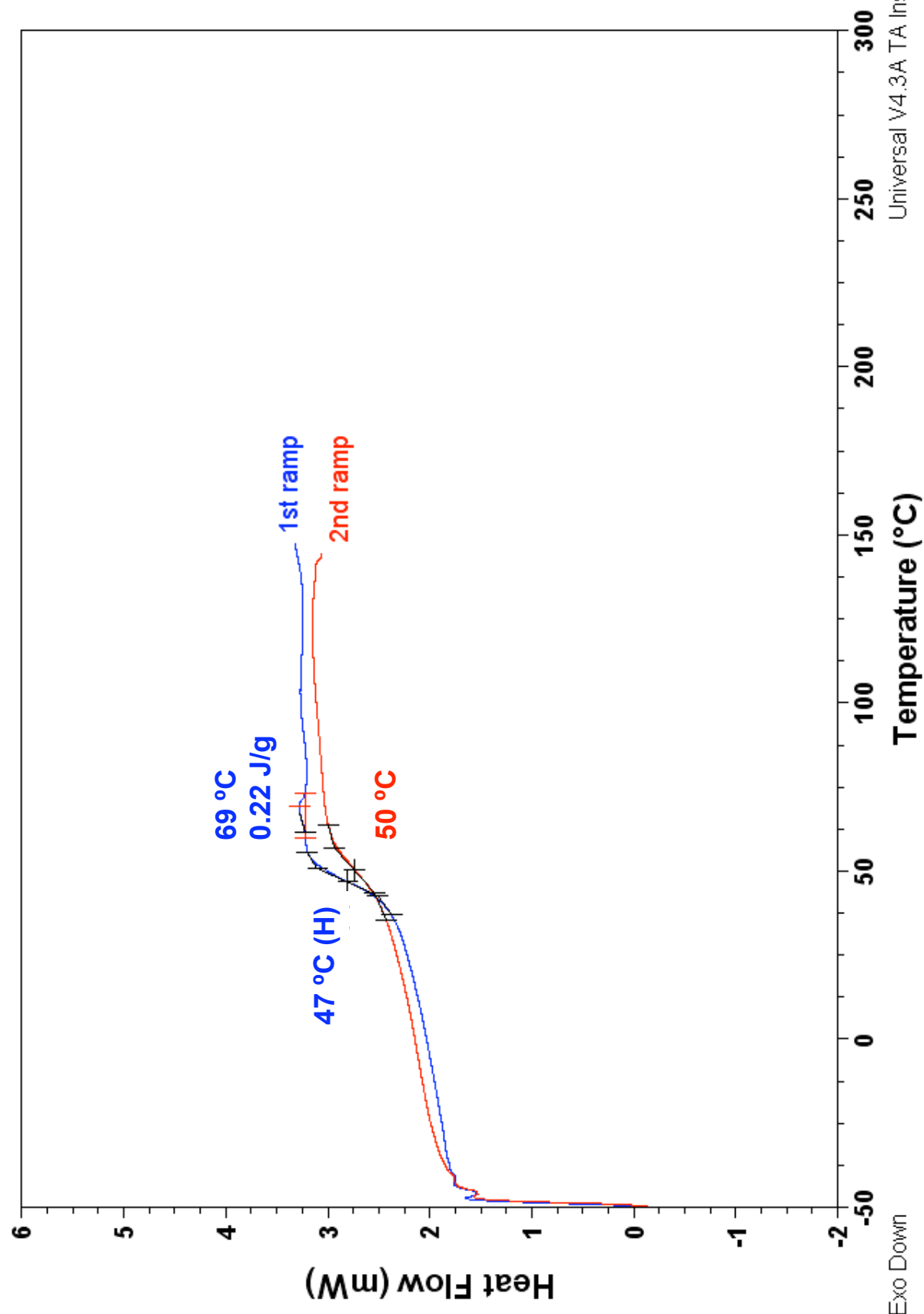
File: NTSB-HWY06MH024-EB080-N-456-NWbottom
Operator: Helen
Run Date: 06-Feb-2007 14:45
Instrument: DSC Q1000 V9.8 Build 296



Sample: WB084-N-4/5/6-NW bottom (typical)
Size: 5.4000 mg
Method: Helen

DSC

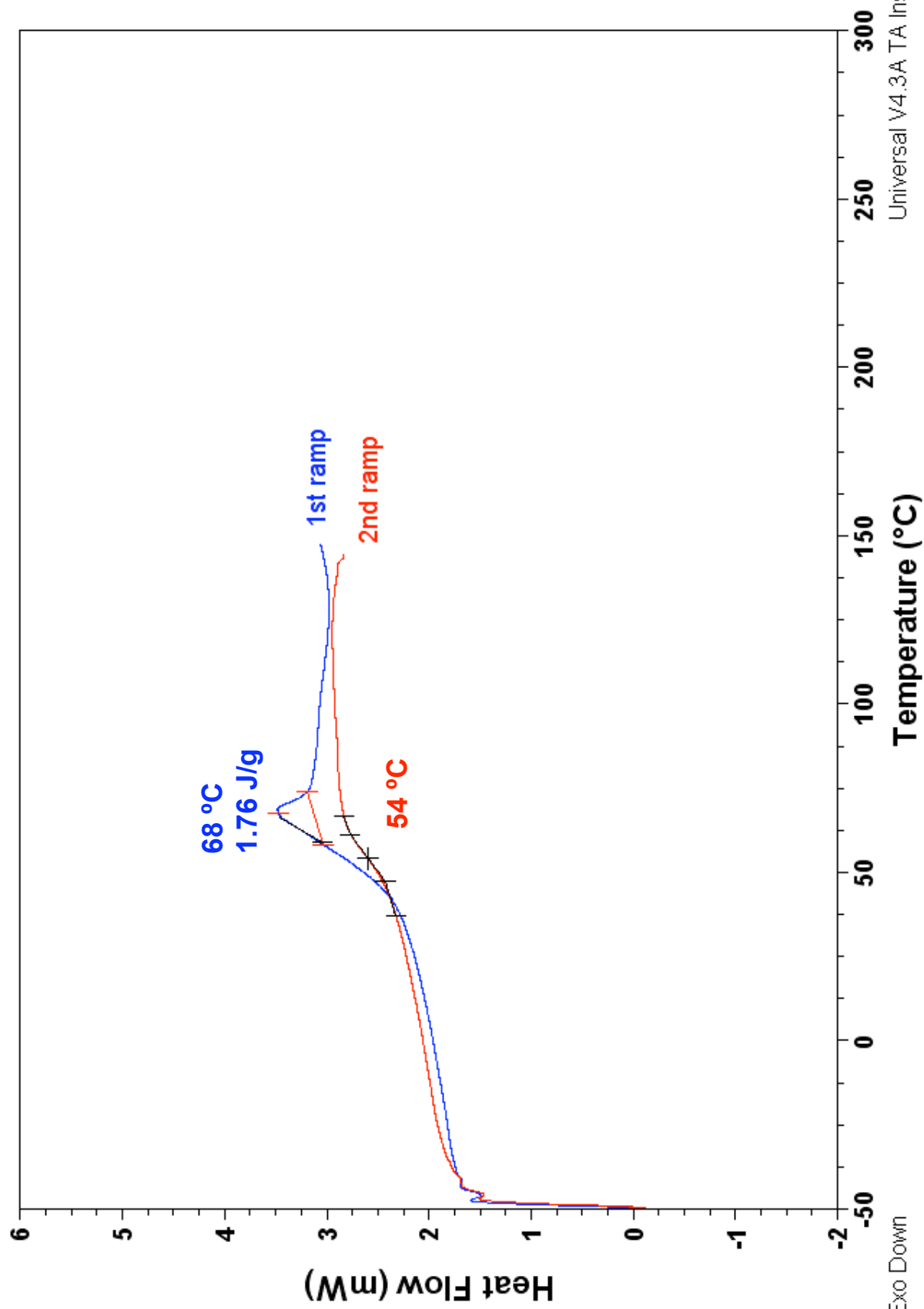
File: NTSB-HWY06MH024-WB084-N-456-NWbottom
Operator: helen
Run Date: 07-Feb-2007 23:51
Instrument: DSC Q1000 V9.8 Build 296



Sample: WB082-S-2-S mid (typical)
Size: 5.2000 mg
Method: Helen

DSC

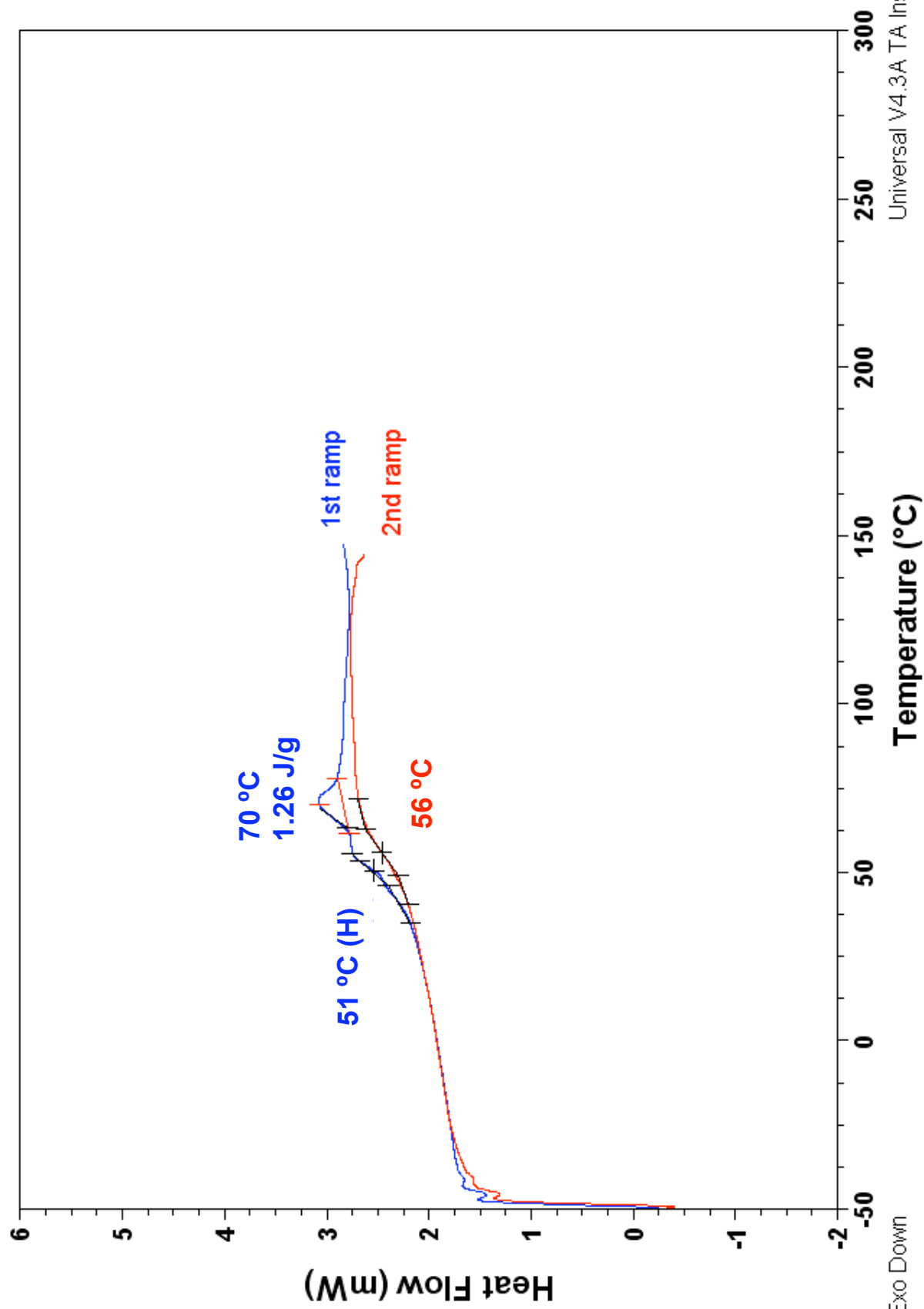
File: C:\...\NTSB-HWY06MH024-WB082-S-2-Smid.0C
Operator: Helen
Run Date: 12-Feb-2007 13:48
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB081-M2-2/3/4-SE top (typical)
Size: 4.8000 mg
Method: Helen

DSC

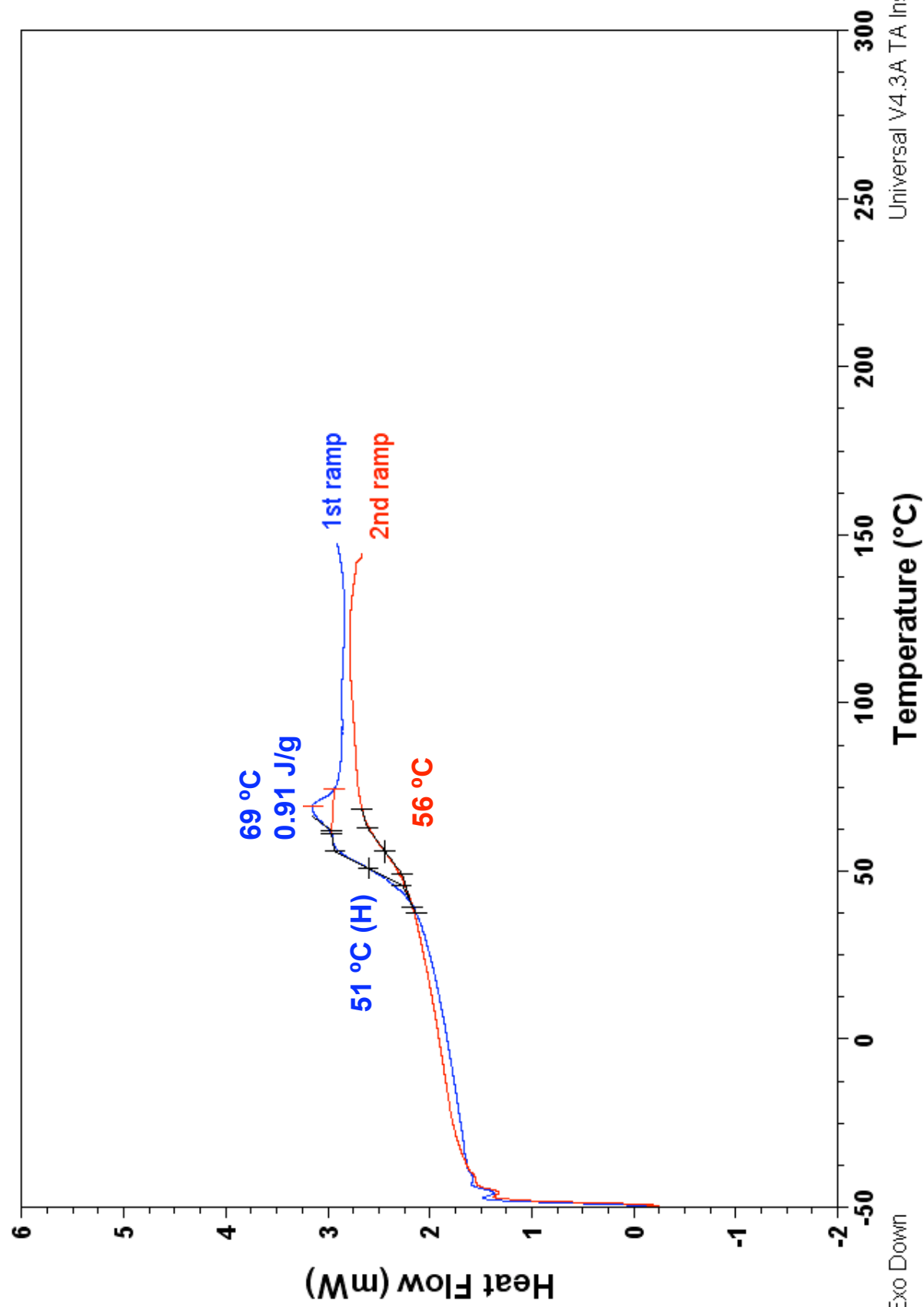
File: NTSB-HWY06MH024-EB081-M2-234-SEtop.00
Operator: Helen
Run Date: 12-Feb-2007 14:40
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB082-M1-8/9/10-SW top (typical)
Size: 4.9000 mg
Method: Helen

DSC

File: NTSB-HWY06MH024-EB082-M1-8910SWtop.C
Operator: Helen
Run Date: 12-Feb-2007 18:07
Instrument: DSC Q1000 V9.8 Build 296



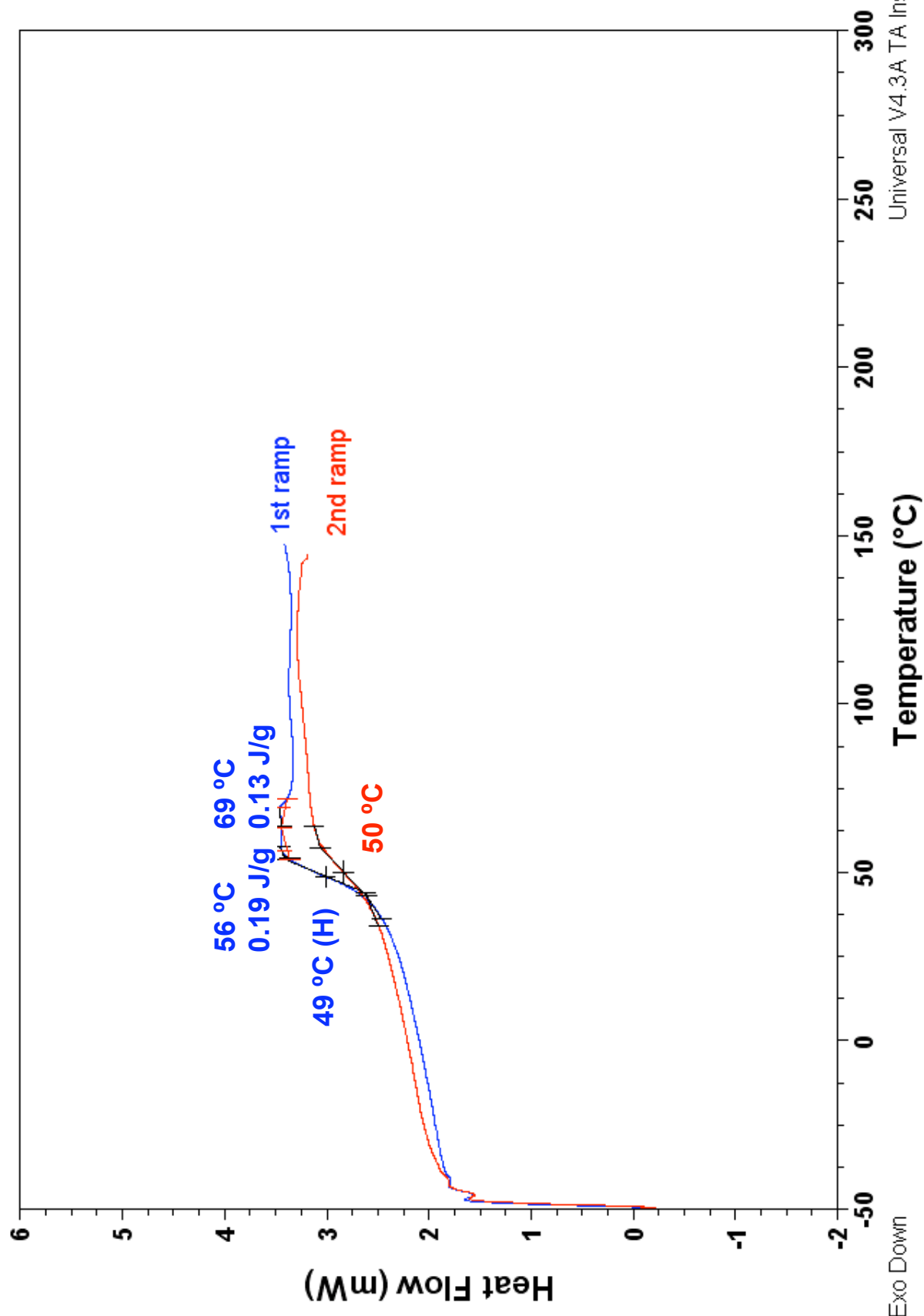
Exo Down

Universal V4.3A TA Instruments

Sample: EB081-S-9-S bottom (typical)
Size: 5.2000 mg
Method: Helen

DSC

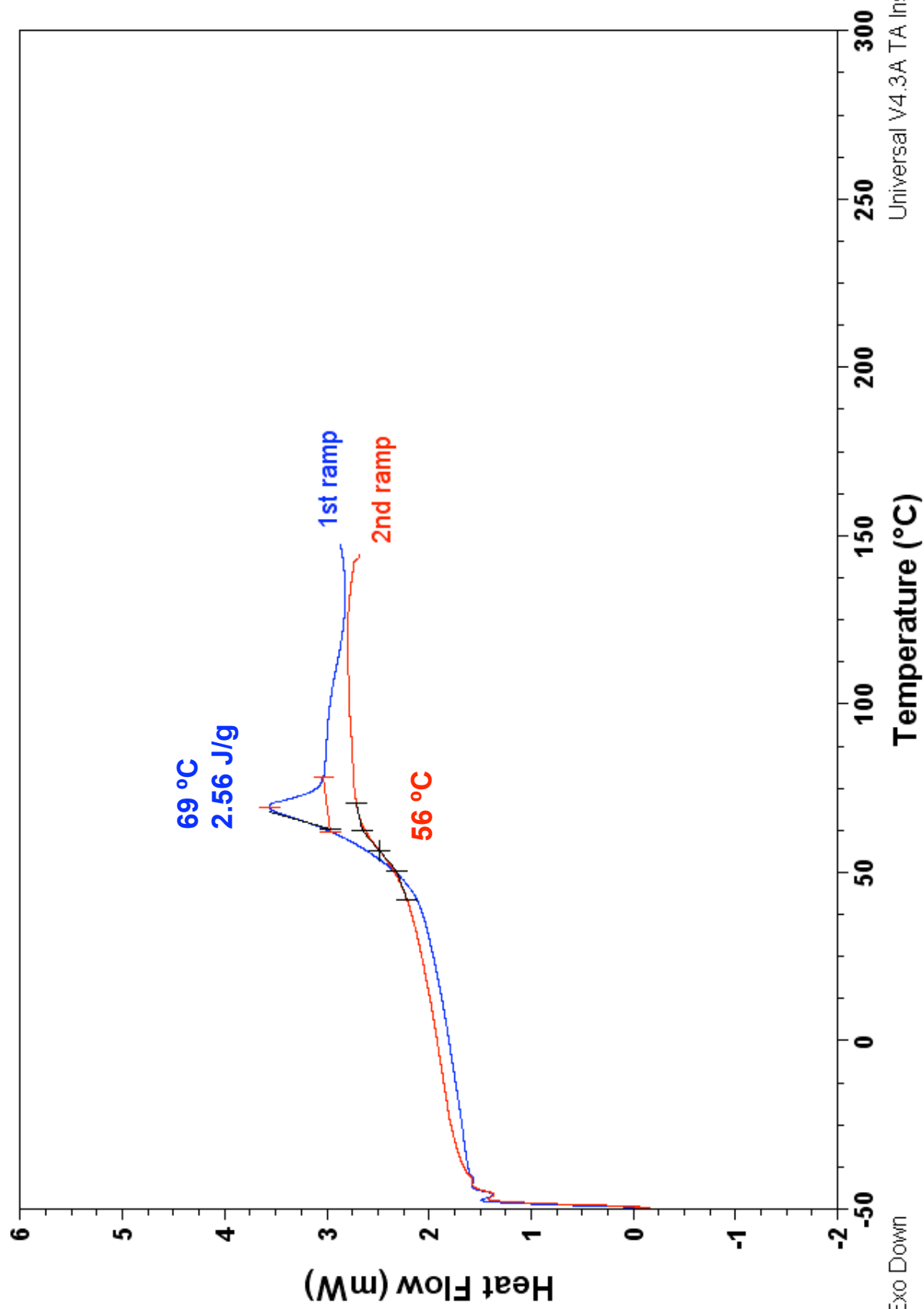
File: NTSB-HWY06MH024-EB081-S-9-S bottom.001
Operator: Helen
Run Date: 12-Feb-2007 17:15
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB083-N-3-S mid (typical)
Size: 4.9000 mg
Method: Helen

DSC

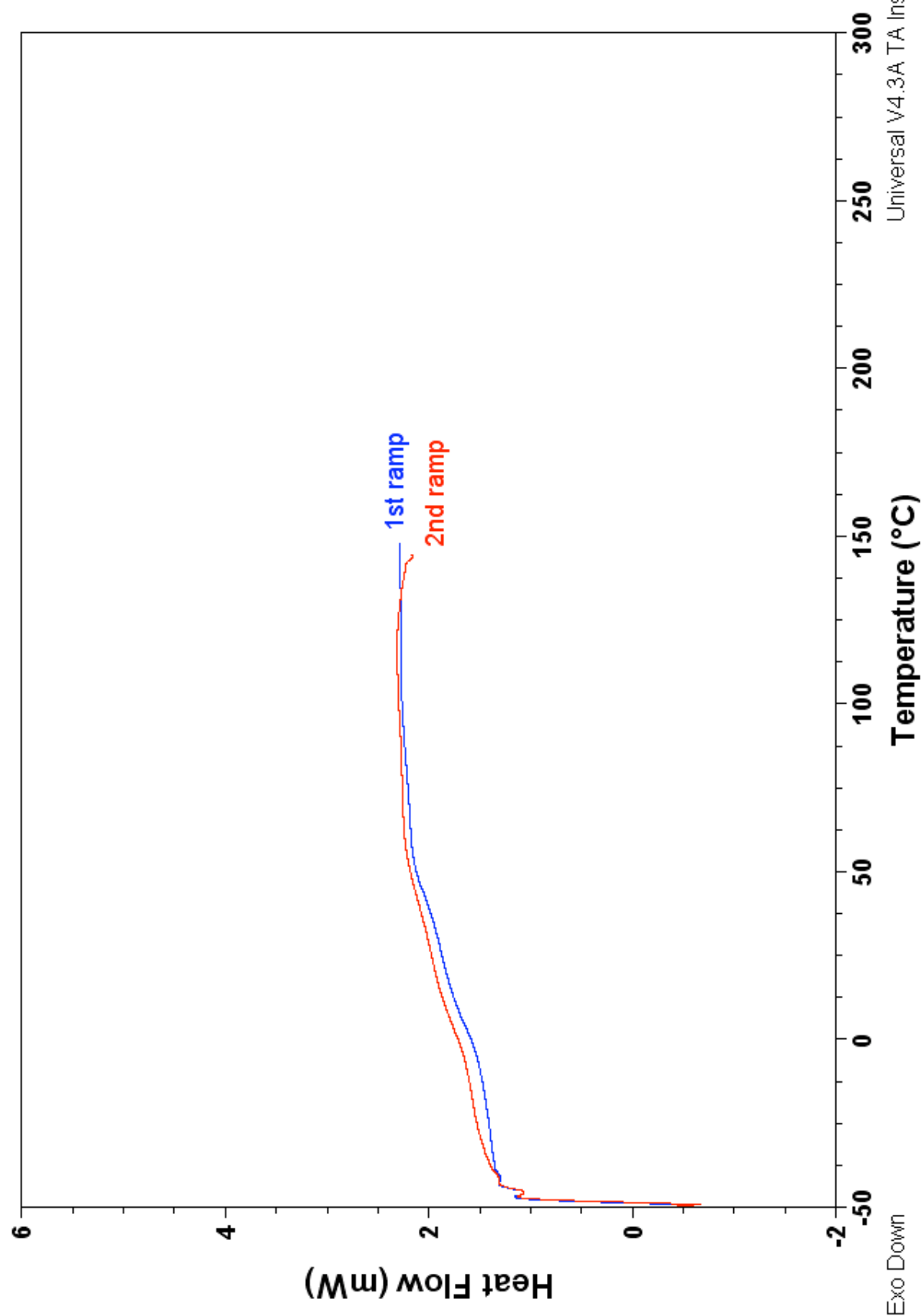
File: C:\...\NTSB-HWY06MH024-EB083-N-3-Smid.00
Operator: Helen
Run Date: 12-Feb-2007 15:31
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB081-S-8-S top (brown)
Size: 4.7000 mg
Method: Helen

DSC

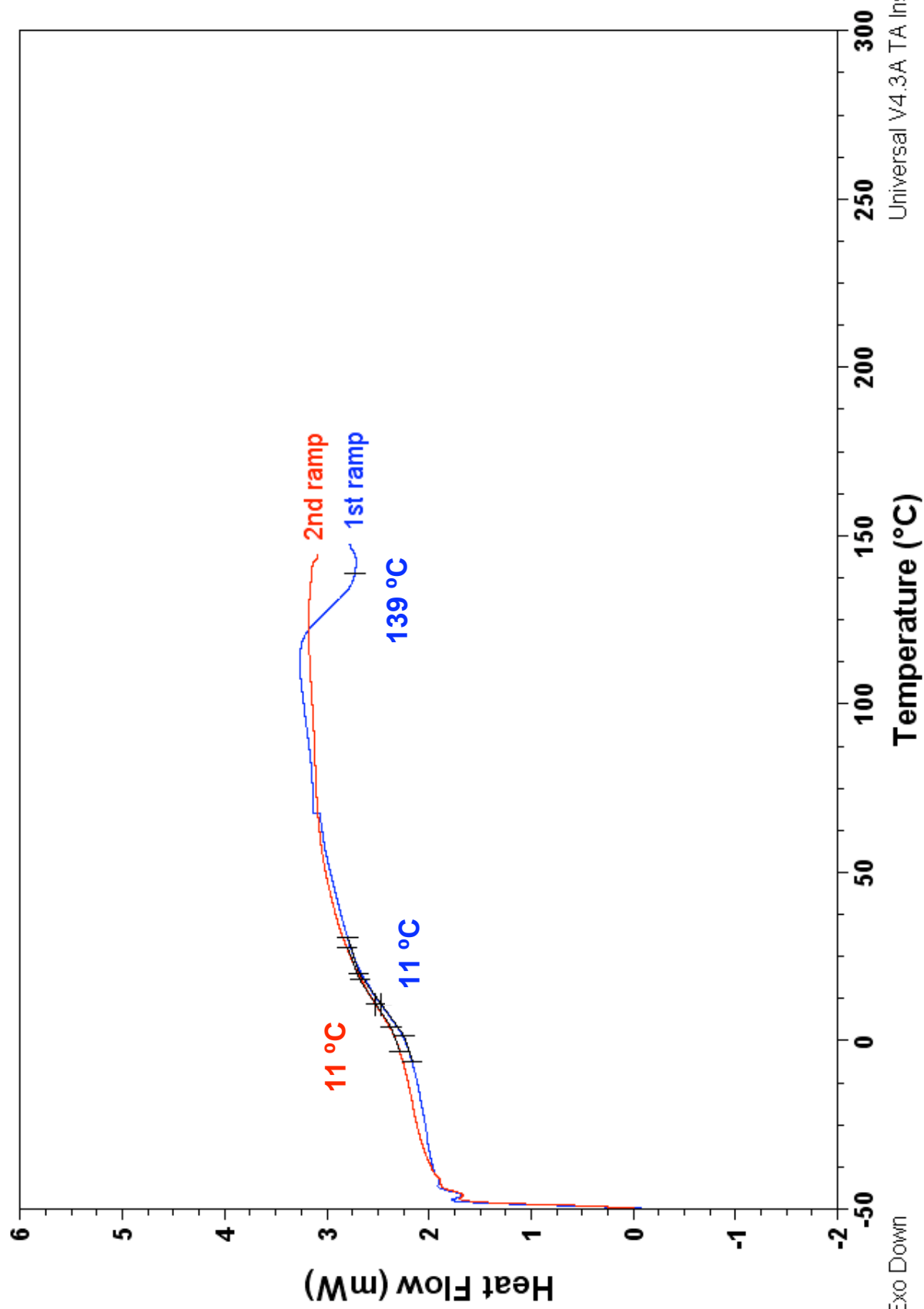
File: ...NTSB-HWY06MH024-EB081-S-8-Stop-2.001
Operator: Helen
Run Date: 13-Feb-2007 13:53
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB083-S-1-S top (brown)
Size: 5.5000 mg
Method: Helen

DSC

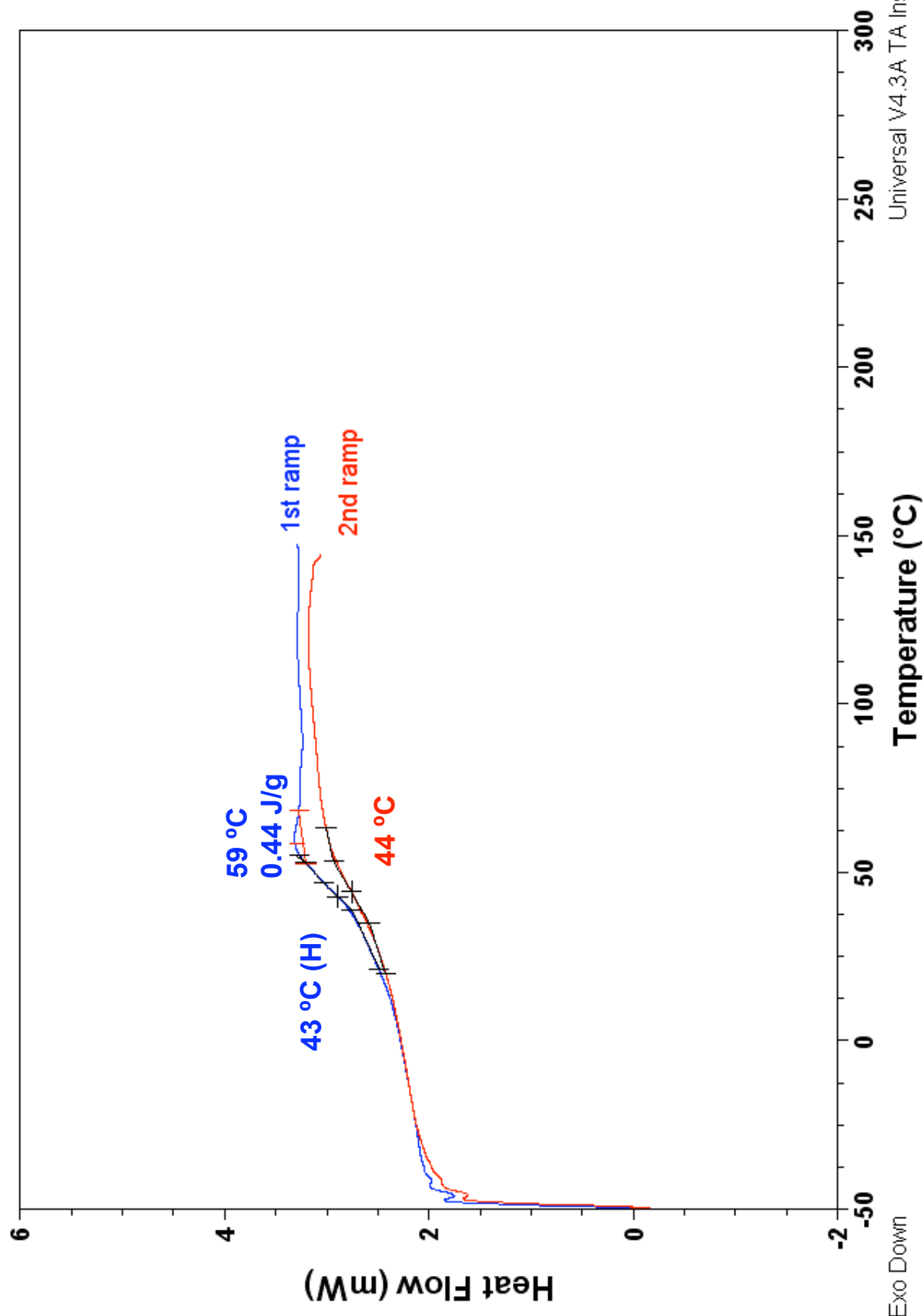
File: C:\NTSB-HWY06MH024\EB083-S-1-Stop.001
Operator: Helen
Run Date: 06-Feb-2007 11:18
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB081-S-10-S top (brown)
Size: 5.4000 mg
Method: Helen

DSC

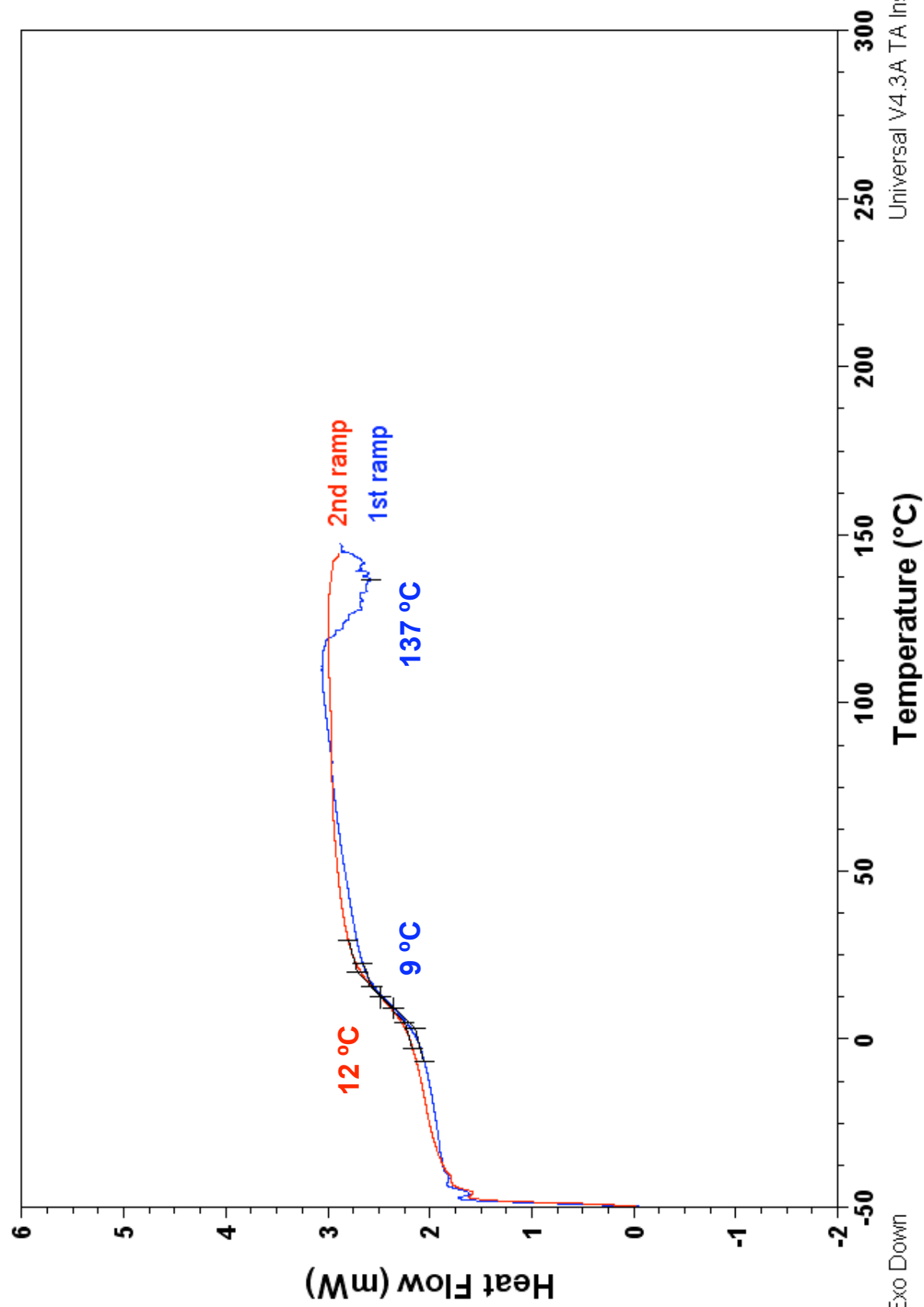
File: ...NTSB-HWY06MH024-EB081-S-10-Stop.001
Operator: Helen
Run Date: 13-Feb-2007 14:44
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB080-N-4/5/6-NE bottom (brown)
Size: 5.2000 mg
Method: Helen

DSC

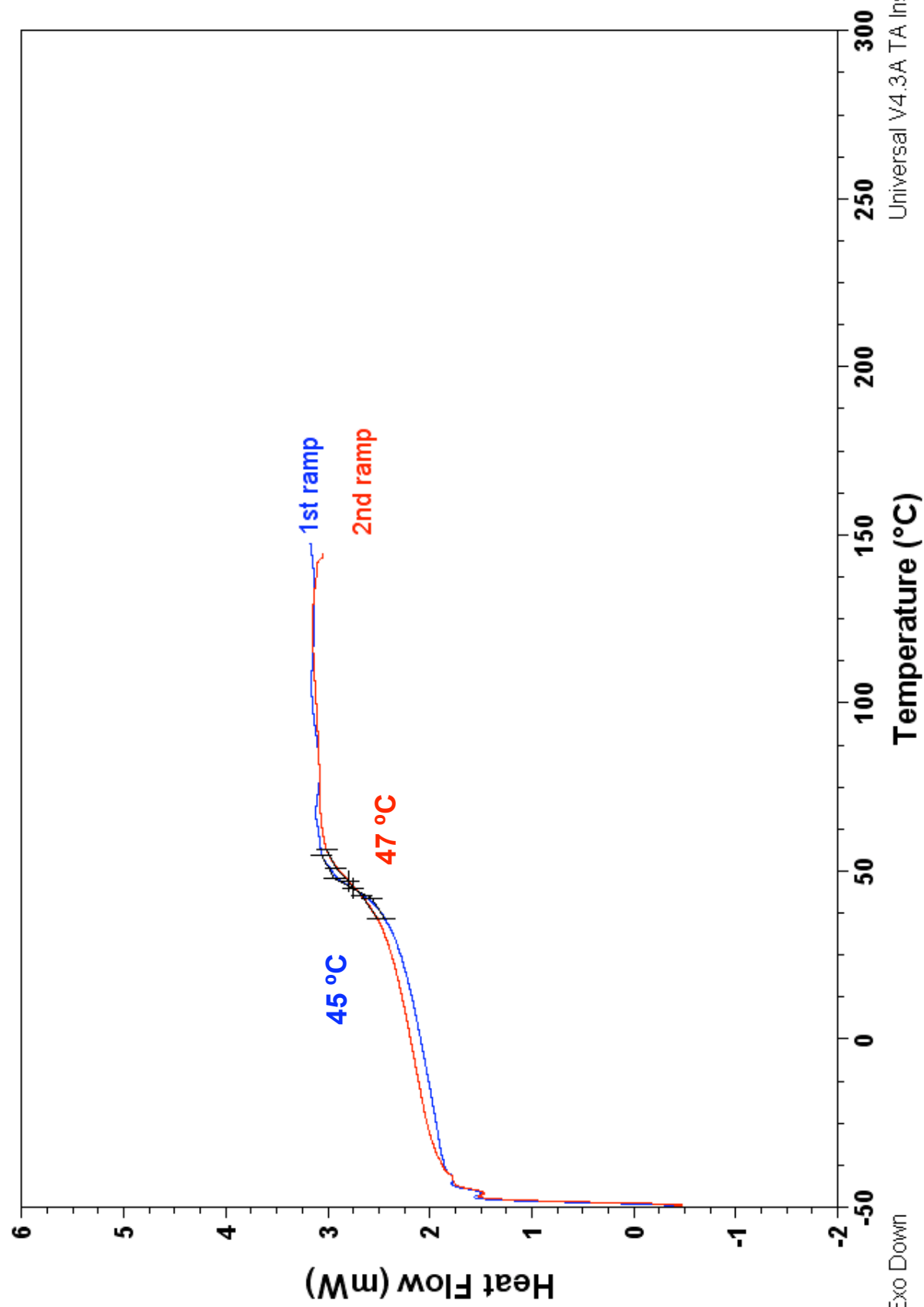
File: NTSB-HWY06MH024-EB080-N-456NEbottom.c
Operator: Helen
Run Date: 06-Feb-2007 10:26
Instrument: DSC Q1000 V9.8 Build 296



Sample: EB083-M2-11-N bottom (ad. fail)
Size: 5.0000 mg
Method: Helen

DSC

File: NTSB-HWY06MH024-EB083-M2-11Nbottom.00
Operator: Helen
Run Date: 06-Feb-2007 13:53
Instrument: DSC Q1000 V9.8 Build 296



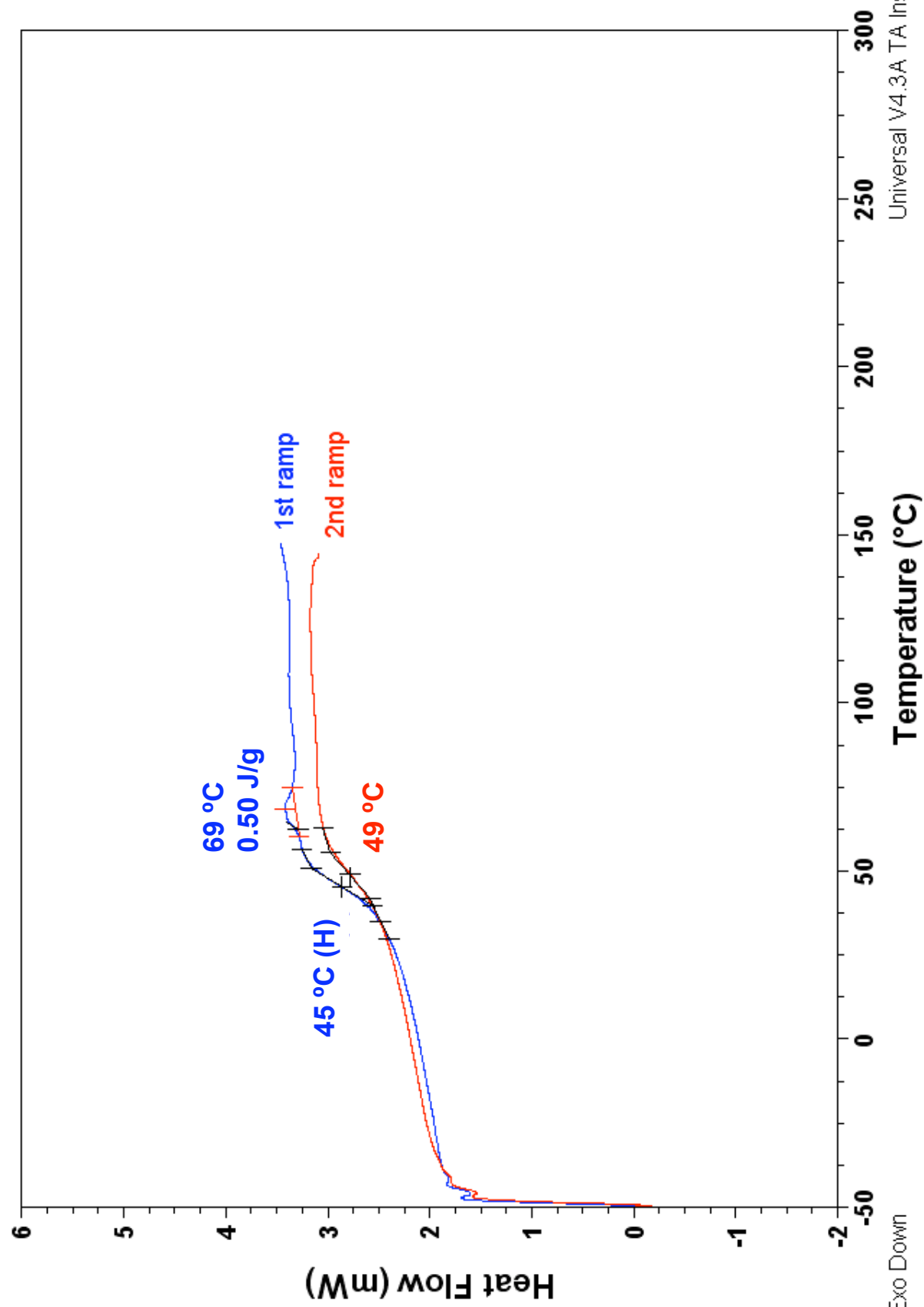
Exo Down

Universal V4.3A TA Instruments

Sample: EB083-M2-7/8/9-SW bot (ad.fail)
Size: 5.1000 mg
Method: Helen

DSC

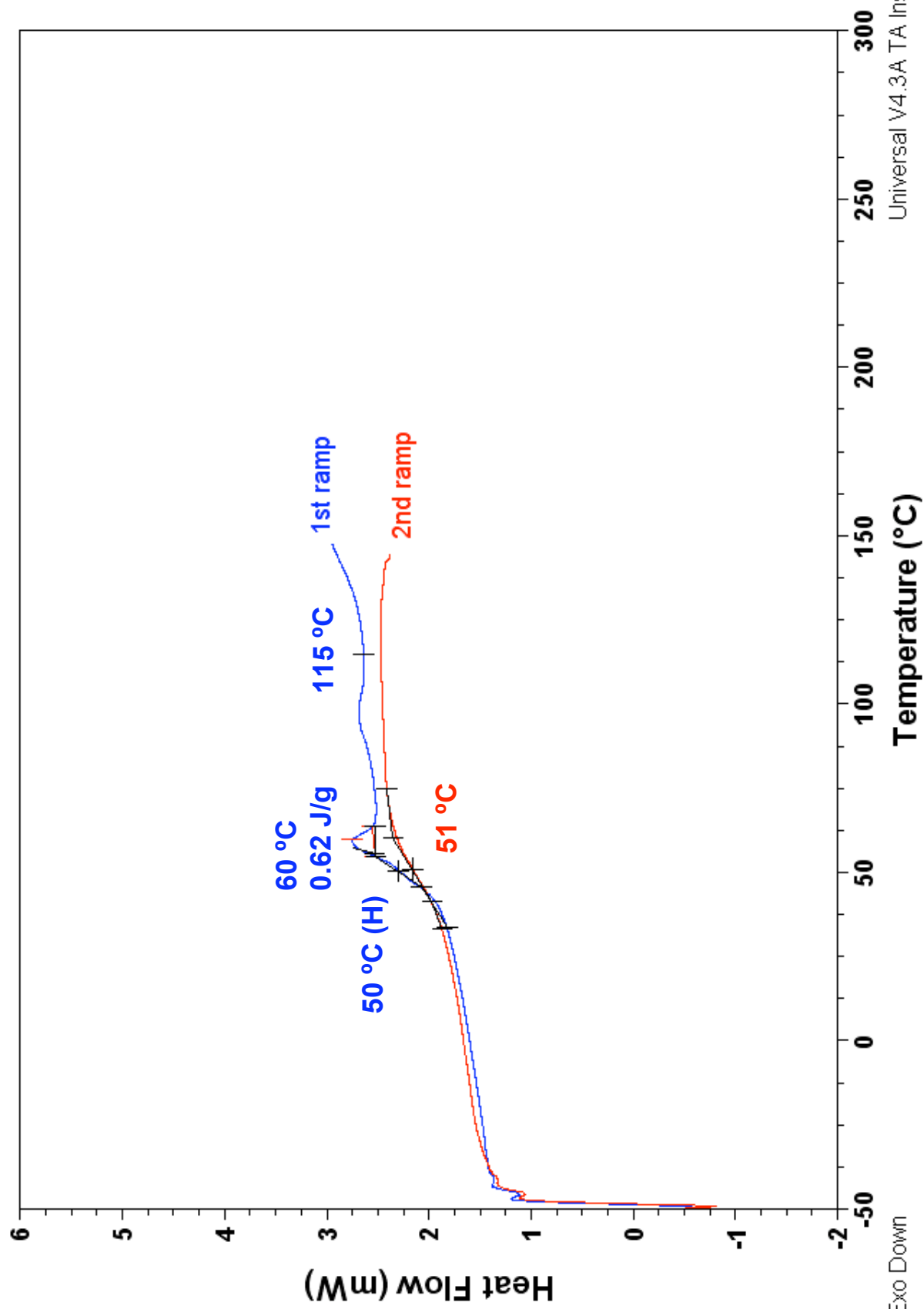
File: NTSB-HWY06MH024-EB083-M2-789SWbottom
Operator: Helen
Run Date: 06-Feb-2007 12:10
Instrument: DSC Q1000 V9.8 Build 296



Sample: FHWA best practice 6
Size: 5.3000 mg
Method: Helen

DSC

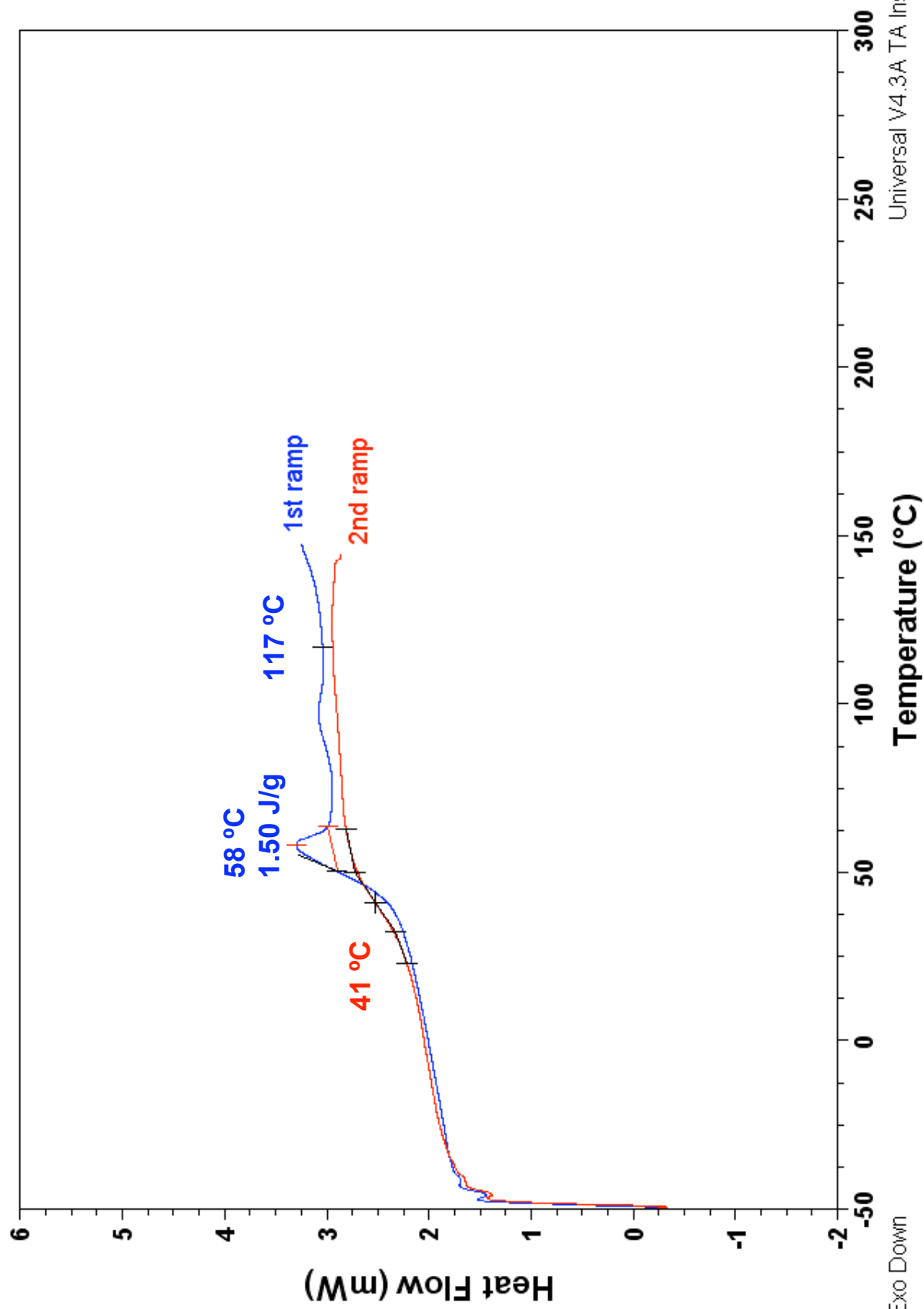
File: ...NTSB-HWY06MH024-fedhwy-bestprac6.001
Operator: Helen
Run Date: 07-Feb-2007 18:41
Instrument: DSC Q1000 V9.8 Build 296



Sample: FHWA bubble w/o twist 9
Size: 5.3000 mg
Method: Helen

DSC

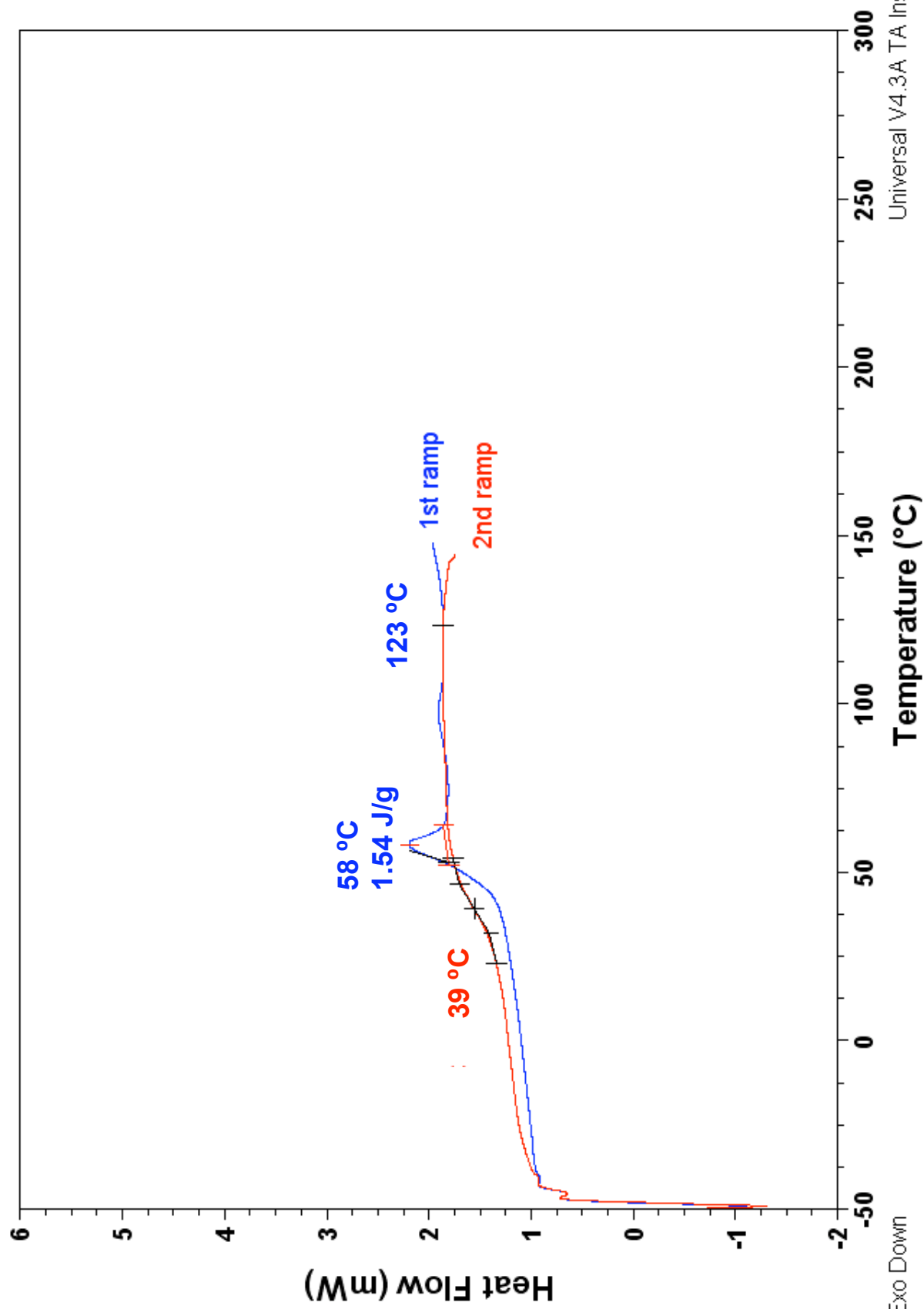
File: NTSB-HWY06MH024-fedhwybubwotwist9.001
Operator: Helen
Run Date: 07-Feb-2007 23:00
Instrument: DSC Q1000 V9.8 Build 296



Sample: FHWA bubble w/o twist 1
Size: 4.5000 mg
Method: Helen

DSC

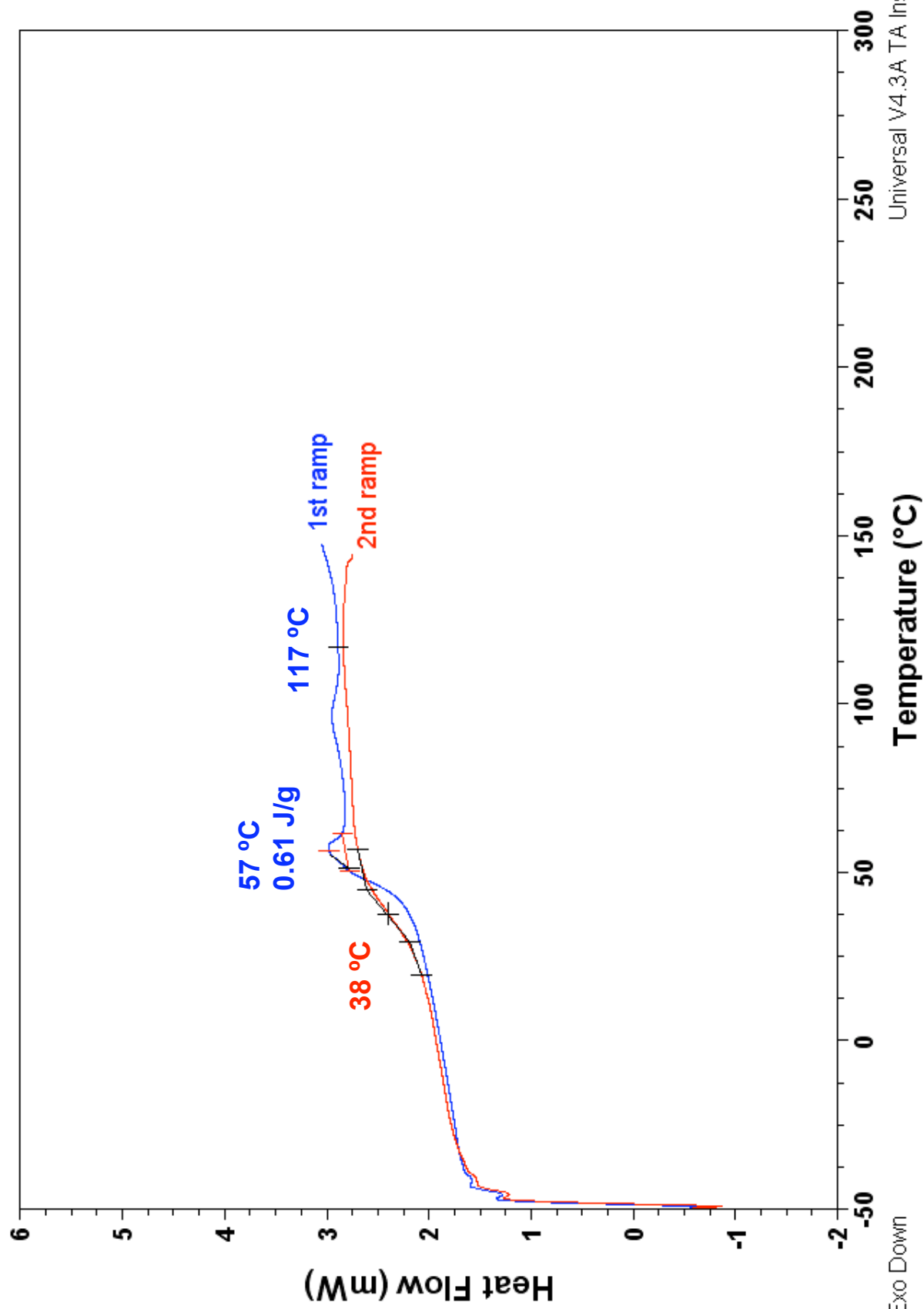
File: NTSB-HWY06MH024-fedhwybubwotwist1.001
Operator: Helen
Run Date: 07-Feb-2007 20:24
Instrument: DSC Q1000 V9.8 Build 296



Sample: FHWA bubble w/ twist 4
Size: 5.4000 mg
Method: Helen

DSC

File: ...NTSB-HWY06MH024-fedhwybubwtwist4.001
Operator: Helen
Run Date: 07-Feb-2007 22:08
Instrument: DSC Q1000 V9.8 Build 296



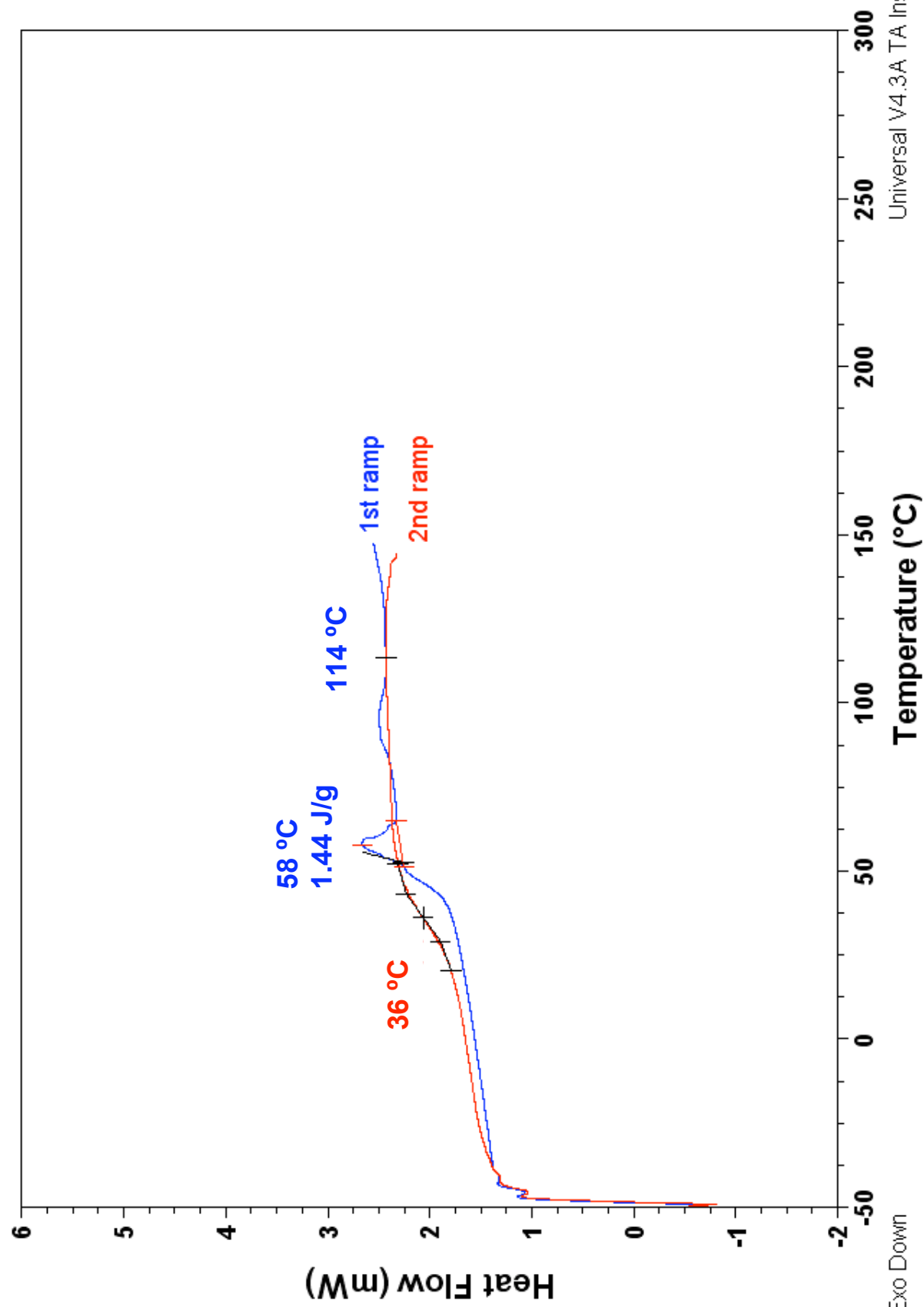
Exo Down

Universal V4.3A TA Instruments

Sample: FHWA unclean 1
Size: 5.3000 mg
Method: Helen

DSC

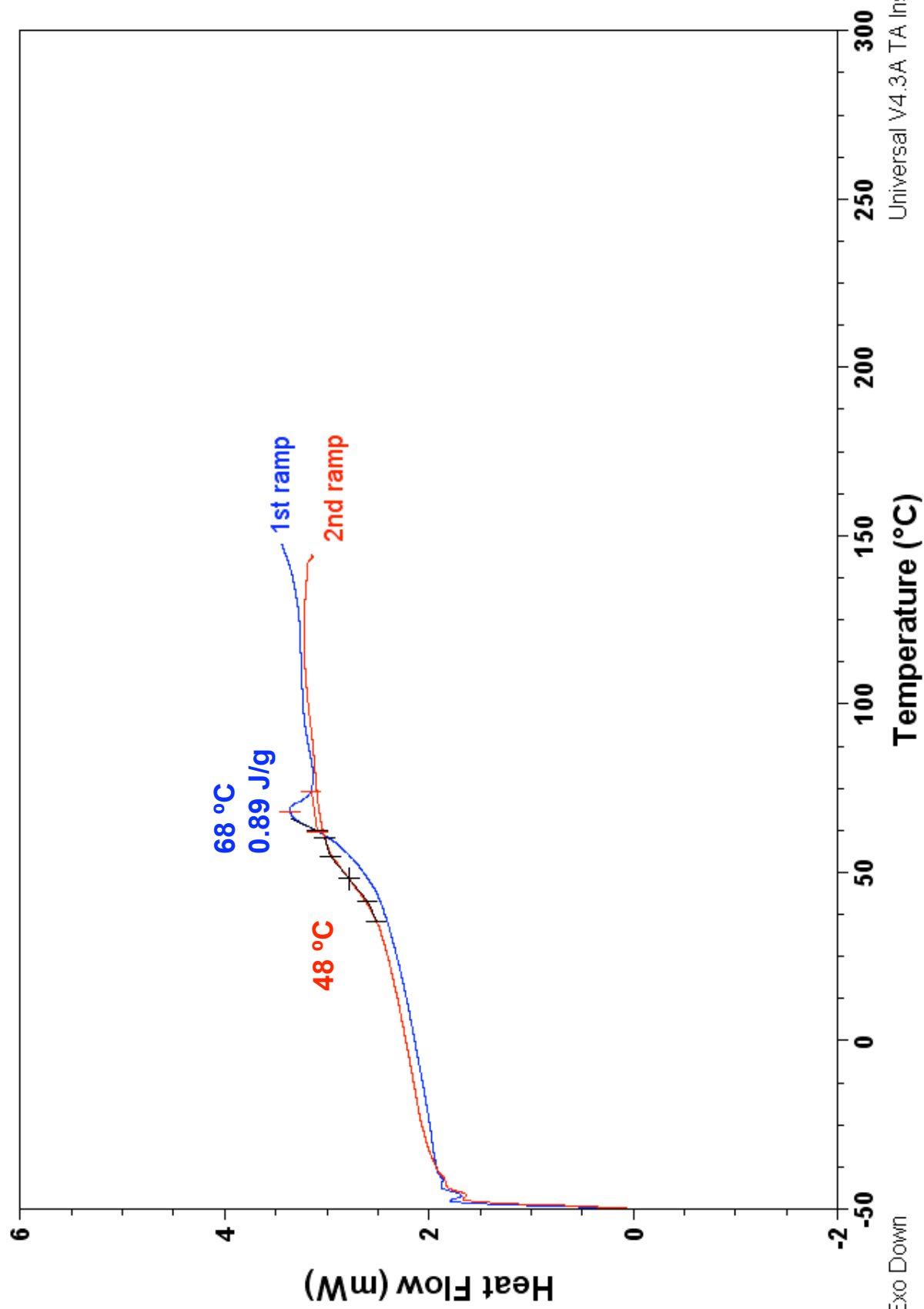
File: C:\NTSB-HWY06MH024-fedhwyunclean1.001
Operator: Helen
Run Date: 07-Feb-2007 19:33
Instrument: DSC Q1000 V9.8 Build 296



Sample: fast set 50/50 reference
Size: 5.6000 mg
Method: Helen

DSC

File: C:\...\NTSB-HWY06MH024.FAST5050-3.001
Operator: Helen
Run Date: 13-Feb-2007 17:19
Instrument: DSC Q1000 V9.8 Build 296



Sample: standard set 50/50 reference
Size: 5.2000 mg
Method: Helen

DSC

File: C:\RZha\NTSB-HWY06MH024-STD5050-2.00
Operator: Helen
Run Date: 02-Feb-2007 19:24
Instrument: DSC Q1000 V9.8 Build 296

